T2K 6.5700×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis

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Abstract

We report the results of the 3-flavour ν_μ-disappearance analysis, performed by the *VaLOR* group, on the combined Run 1+2+3+4 dataset. This analysis is an update of the work presented in Refs. [1], [2], [3], [4] and [5] with improved methods and an enlarged dataset. The Run 1+2+3+4 dataset corresponds to an integrated J-PARC neutrino beam exposure of 6.5700×10^{20} POT.

This analysis predicts 445.98± 23.46(syst) 1 μ-like ring events in SuperK in the absence of any oscillation, but only 120 were observed. The observed deficit has a strong energy dependence; the ratio of observed to expected, under the no-oscillation hypothesis, is ∼26% < 0.5 GeV, ∼10% between 0.5 and 1 GeV and ∼72% > 1 GeV.

A ν_μ-disappearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of 1 μ-like ring events was fitted, and separate fits were made for the normal and the inverted mass hierarchies. In these fits, $\sin^2\theta_{23}$ and either $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy) were allowed to float. The oscillation parameters $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, Δm^2_{21} and δ_{CP} and all 41 systematic parameters considered in this analysis were also allowed to float in the fit.

The 3-flavour ν_μ-disappearance fit of the combined Run 1+2+3+4 dataset, assuming the normal mass hierarchy, gives $|\Delta m^2_{32}| = 2.51\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.514$ as the best-fit values (p-value = 0.93). Assuming the inverted mass hierarchy, the fit yields $|\Delta m^2_{31}| = 2.48\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.511$ as the best-fit values (p-value = 0.93).

Confidence regions in the parameter spaces $\sin^2\theta_{23}$ - $|\Delta m^2_{32}|$ (normal hierarchy) and $\sin^2\theta_{23}$ - $|\Delta m^2_{31}|$ (inverted hierarchy) are shown using both the constant-$\Delta\chi^2$ method and the method of Feldman and Cousins. We use the profile likelihoods obtained from the fit for $\sin^2\theta_{23}$ and either $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy) and the constant-$\Delta\chi^2$ method for one parameter to obtain the 68% (90%) C.L. allowed regions for these parameters. These can be summarized as 0.456 (0.429) < $\sin^2\theta_{23}$ < 0.572 (0.599) and 2.39×10^{-3} (2.32×10^{-3}) eV^2/c^4 < $|\Delta m^2_{32}|$ < 2.63×10^{-3} (2.70×10^{-3}) eV^2/c^4 for the normal mass hierarchy, and 0.454 (0.427) < $\sin^2\theta_{23}$ < 0.569 (0.596) and 2.36×10^{-3} (2.29×10^{-3}) eV^2/c^4 < $|\Delta m^2_{31}|$ < 2.60×10^{-3} (2.67×10^{-3}) eV^2/c^4 for the inverted mass hierarchy.

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1. Introduction

In this technical note, we report the results of a 3-flavour ν_μ-disappearance analysis, performed by the *VaLOR* group, on the combined Run 1+2+3+4 (6.5700×10^{20} POT) dataset. This dataset was collected in four run periods: January - June 2010 (Run 1, 0.323×10^{20} POT), November 2010 - March 2011 (Run 2, 1.108×10^{20} POT), March - June 2012 (Run 3, 1.579×10^{20} POT, following the recovery work from the damage of the 2011 Tohoku earthquake) and October 2012 - 8th May 2013 (Run 4, 3.5600×10^{20} POT). This analysis is an update of the work presented in Refs. [1], [2], [3], [4] and [5] with improved methods and an enlarged dataset.

During our Run 1+2+3 analysis, it was found that the choice of octant of θ_{23} makes a significant difference to the confidence region in the $\sin^2(2\theta_{23})$ - $|\Delta m^2_{32}|$ oscillation parameter space. For this reason, best-fit values were found and confidence regions constructed using $\sin^2\theta_{23}$ as the mixing angle instead of $\sin^2(2\theta_{23})$ [4]. This use of $\sin^2\theta_{23}$ is continued in this Run 1+2+3+4 analysis. There has been a change in the mass-squared differences used in this analysis: we now use $|\Delta m^2_{32}|$ for the normal mass hierarchy and $|\Delta m^2_{31}|$ for the inverted hierarchy, whereas, in our Run 1+2+3 analysis, we used $|\Delta m^2_{32}|$ for both hierarchies. This change has been made because $|\Delta m^2_{32}|$ is the largest mass-squared difference in the inverted hierarchy, but the second largest in the normal hierarchy. The inputs to the calculation of the oscillation probabilities are $+|\Delta m^2_{32}|$ for the normal hierarchy and $-|\Delta m^2_{31}|$ for the inverted hierarchy.

These oscillation parameters are determined by fitting the predicted reconstructed energy spectrum of 1 μ-like ring events at SuperK to the observed spectrum. The method of calculation of this predicted spectrum is presented in Sec. 3. This analysis uses flux tuning based on recent NA61 data [11][26]. It also uses an indirect extrapolation from ND280 to SuperK, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of ν_μ CC-0-pion-like, CC-1-pion-like and CC-other events whose vertices are in FGD1 [21]. Updated nominal predictions of the reconstructed energy spectrum of single μ-like ring events in SuperK are presented for Run 1+2+3+4 in Sec. 3; the changes to the analysis summarized in Sec. 2 are incorporated into these predictions.

We consider a total of 41 energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state rescattering systematics (nuisance parameters). The oscillation parameters $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, Δm^2_{21} and δ_{CP} are also considered as nuisance parameters in this analysis. These nuisance parameters and their effects on the extrapolated single μ-like ring event reconstructed energy spectrum are described in Sec. 4. Further details of them are also given in a separate technical note supplement [7].

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [30]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. In this Run 1+2+3+4 analysis, we allow all 45 nuisance parameters (41 systematic parameters and four non-23-sector oscillation parameters) to float. The oscillation fitting method is presented in detail in Sec. 5.1, while the chosen goodness-of-fit test, suitable for the current statistics, is described in Sec. 5.2. The performance of the fitter was tested in detail for the Run 1+2+3+4 fit using ensembles of toy MC experiments, and a summary of the results of these tests is presented in Sec. 6.

The 2D confidence regions in $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ are constructed using the Feldman-Cousins method [32] as in our previous analyses. Systematic errors are included in the oscillation contours using the Cousins-Highland method (Bayesian integration) [33]. Confidence contours constructed with the constant-$\Delta\chi^2$ method are also shown for both fits for comparison. Both of these methods are summarized in Sec. 5.3, and the exact methodology used for incorporating systematic errors is explained in detail. 1D confidence intervals are also calculated for each of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ separately using a new procedure that is an extension of the Feldman and Cousins method; this procedure is described in Sec. 5.3. The Feldman-Cousins 2D contours and 1D intervals are being prepared, and will be shown in a later version of this note.

This analysis is similar to our Run 1+2+3 analysis with higher statistics, though certain analysis inputs and fitting methods have been updated. For the benefit of readers, all changes with respect to our previously published Run 1+2+3 ν_μ-disappearance analysis described in Ref. [4] are summarized in Sec. 2.

The results of the measurements of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ using the Run 1+2+3+4 dataset are given in Sec. 7.1 of this note, while the goodness-of-fit tests are presented in Sec. 7.2. The best-fit spectra

are shown for each of the normal and inverted mass hierarchy assumptions for the Run 1+2+3+4 dataset in Sec. 7.3. The Feldman-Cousins and constant-$\Delta\chi^2$ confidence regions are shown in Sec. 7.4. Comparisons of our results obtained between different methods for constructing confidence regions are presented in Sec. 7.5. In that section, we also show comparisons of our T2K Run 1+2+3+4 results with results from MINOS and SuperK as well as with the expected T2K sensitivity for an integrated exposure of 6.5700×10^{20} POT.

Finally, a summary is given in Sec. 8.

2. Summary of changes with respect to the *VaLOR* Run 1+2+3 ν_μ-disappearance analysis

This analysis is primarily a statistics update of our Run 1+2+3 ν_μ-disappearance analysis, although certain analysis inputs and the fitting methodology have also been updated. The following changes have been implemented with respect to our Run 1+2+3 analysis described in [4]:

- Added the Run 4 SuperK data. Run 4 adds 3.5600×10^{20} POT to the Run 1+2+3 exposure of 3.010×10^{20} POT. The total Run 1+2+3+4 exposure is 6.5700×10^{20} POT.
- The current analysis uses the most recent official SuperK MC samples (2013a) generated using the 11btunedv3.2 JNUBEAM flux. All MC templates, systematic parameter response functions (cubic splines) and systematic parameter covariance matrices have been regenerated with the new MC samples.
- The following 4 MC templates have been added for completeness.
  - oscillated $\nu_e \to \nu_\mu$ CCQE,
  - oscillated $\nu_e \to \nu_\mu$ CC1π,
  - oscillated $\nu_e \to \nu_\mu$ CC coherent and
  - oscillated $\nu_e \to \nu_\mu$ CC other

  Since a separate official oscillated $\nu_e \to \nu_\mu$ SuperK sample is not available, we emulated it by reweighting the ν_μ SuperK MC sample with the ν_e flux. In the oscillation fit, the contents of those 4 newly-added MC templates are weighted with P($\nu_e \to \nu_\mu$).
- SuperK events with reconstructed energy >30 GeV are now included in the 10-30 GeV bin after discussion with the T2K-SK group (the reconstructed energy binning is unchanged with respect to the Run 1+2+3 analysis).
- The BANFF fit has been updated. This analysis uses version 5 of the Run 1+2+3+4 BANFF2013 fit [21]. Besides adding the Run 4 ND280 data, the BANFF fit method was also updated. For the 2013 analyses, the ND280 ν_μ group updated the division of reconstructed ND280 CC events. These events are divided into CC0π, CC1π and CC other samples, whereas for the Run 1+2+3 analysis they were divided into CCQE and CCnonQE samples.
- The SuperK detector error matrix has been recreated using more SuperK atmospheric data [22].
- The 7 SuperK detector efficiency uncertainties and the 6 FSI+SI uncertainties have been combined into a single 6-parameter covariance matrix. This reduces the number of nuisance parameters in the fit, allowing the fit to run faster.
- We have updated the value of $\sin^2\theta_{12}$ from 0.311 to 0.306 to be consistent with the 2013 T2K ν_e-appearance analyses.
- The response functions (cubic splines) for the uncorrelated cross-section systematic parameters have been regenerated using the following oscillation parameters: $\sin^2(2\theta_{12}) = 0.8495$, $\sin^2(2\theta_{13}) = 0.098$, $\sin^2(2\theta_{23}) = 1.0$, $\Delta m^2_{21} = 7.5 \times 10^{-5}$ eV^2/c^4, $|\Delta m^2_{32}| = 2.44 \times 10^{-3}$ eV^2/c^4, $\delta_{CP} = 0.0$ and the normal mass hierarchy.
- The SuperK energy scale systematic has changed from 2.3% to 2.4% [23].
- The CC $\nu/\bar{\nu}$ cross section normalisation systematic has changed from 40% to 20% [16].
- During the review process of the T2K Run 1+2+3 ν_μ-disappearance PRL [9], it was discovered that the choice of octant ($\theta_{23} \leq \pi/4$ or $\theta_{23} \geq \pi/4$) has a large effect on the confidence contour in $\sin^2(2\theta_{23})$. Therefore, we now fit $\sin^2\theta_{23}$ by default (though the ability to fit $\sin^2(2\theta_{23})$ has been retained).
- The neutrino mass hierarchy assumption also has a large effect on the confidence contour in $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$, and we therefore produce two results: one assumes the normal hierarchy and fits $|\Delta m^2_{32}|$, and the other assumes the inverted hierarchy and fits $|\Delta m^2_{31}|$. The inputs to the calculation of the oscillation probabilities are $+|\Delta m^2_{32}|$ for the normal hierarchy and $-|\Delta m^2_{31}|$ for the inverted hierarchy.

- In this analysis, all four *non-23-sector* oscillation parameters are allowed to float in the fit, whereas they were fixed in our Run 1+2+3 analysis. We use constraints from the the 2012 PDG for Δm^2_{21}, $\sin^2\theta_{13}$ and $\sin^2\theta_{12}$ (see Tab. 10), while δ_{CP} is allowed to float freely in the range [-π, +π]. When generating toy MC experiments, values of Δm^2_{21}, $\sin^2\theta_{13}$ and $\sin^2\theta_{12}$ are thrown from a normal distribution with mean and standard deviation taken from Tab. 10, whereas δ_{CP} is thrown uniformly within its allowed range.

- In this analysis, the spectral function systematic is restricted in the fit to the range [0,1], with a Gaussian constraint that favoured 0 (relativistic Fermi gas). This was also done in our Run 1+2+3 analysis, but, in that analysis, the spectral function systematic was thrown uniformly within its allowed range when generating toy MC experiments. In this analysis, toy MC experiments are generated using a Gaussian constraint, centered at 0, and bounded in the range [0,1].

- We have added 1D confidence intervals in $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ separately using a new procedure that is an extension of the Feldman-Cousins method. These intervals are being prepared, and will be shown in a later version of this note.

3. Predictions of single μ-like ring event reconstructed energy spectrum in SuperK

3.1. Methodology

In this analysis, the oscillation parameters are determined by fitting the reconstructed energy (E_r) spectrum of 1 μ-like ring events observed at SuperK with the predicted spectrum. The predicted number $N_{SK;r}$ of 1 μ-like ring events in the rth reconstructed energy bin is computed as follows:

$$N_{SK;r} = \sum_{m} \sum_{t} \sum_{r\prime} P_{m;t} \cdot T_{r;r\prime;f_{\Delta E}} \cdot S_{m;t;r\prime;\vec{f}} \cdot N^{MC}_{SK;m;r\prime;t}. \quad (1)$$

In Eq. 1, $N^{MC}_{SK;m;r\prime;t}$ is the input SuperK Monte Carlo (MC) template containing the number of events in the 1 μ-like ring MC sample with true reaction mode m, and found in the true energy bin t and the reconstructed energy bin $r\prime$. The construction of the nominal MC templates is discussed in Sec. 3.2. Both flux-tuning weights based on NA61 hadron-production data (see Sec. 3.3) and flux and cross-section tuning weights based on ND280 data (see Sec. 3.4) are applied to the nominal MC templates. $S_{m;t;r\prime;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$ and a vector of nuisance (systematic) parameters $\vec{f}$. A complete description of the nuisance parameters considered in this analysis is given in Sec. 4. $T_{r;r\prime;f_{\Delta E}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the SuperK reconstructed energy scale, expressed here in terms of the nuisance parameter $f_{\Delta E}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the SuperK MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in Sec. 3.5.

3.2. Construction of nominal SuperK Monte Carlo templates

The nominal SuperK 1 μ-like ring MC templates $N^{MC}_{SK;m;r;t}$ are constructed by applying the 1 μ-like ring selection cuts to the official SuperK MC samples. Details of the MC samples are given in Sec. 3.2.1. The calculation of the normalization of each MC sample is described in Sec. 3.2.2. The official 1 μ-like ring selection cuts are listed in Sec. 3.2.3. The list of MC templates used in this analysis is shown in Sec. 3.2.4, while the true and reconstructed energy binning of the templates is discussed in Sec. 3.2.5. The nominal SuperK MC templates used in this analysis are shown in [7].

3.2.1. Input Monte Carlo samples

This analysis uses the official 2013a SuperK MC samples generated using the 11btunedv3.2 JNUBEAM flux. Five MC samples were generated. Four of the samples (ν_μ, $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$) were generated using the nominal JNUBEAM flux without oscillations. A fifth sample of ν_e interactions was also generated with the ν_μ flux. This fifth sample will be denoted the "oscillated ν_e" sample (it corresponds to the "signal-ν_e" sample in the ν_e-appearance analysis). We added four new samples compared with our Run 1+2+3 analysis, and these are oscillated $\nu_e \to \nu_\mu$ CCQE, oscillated $\nu_e \to \nu_\mu$ CC1π, oscillated $\nu_e \to \nu_\mu$ CC coherent and oscillated $\nu_e \to \nu_\mu$ CC other. Since a separate official oscillated $\nu_e \to \nu_\mu$ SuperK sample is not available, we emulated it by reweighting the ν_μ SuperK MC sample with the ν_e flux. All samples were generated including the neutrino flux estimates up to 30 GeV. The flux histograms used for neutrino event generation had 50 MeV bins.

3.2.2. Normalization of Monte Carlo samples

The normalization (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \quad (2)$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section in water for the given neutrino species, I is the beam intensity in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume. The derived POT normalization of each sample is shown in Tab. 1. After being generated from a MC sample corresponding to a calculated integrated beam exposure I, each MC template is normalized to the integrated beam exposure of the Run 1+2+3+4 dataset (6.5700×10^{20} POT) by scaling the bin contents of the template with 6.5700×10^{20}/I.

Sample	POT normalization
ν_μ	4.47598×10^{23}
$\bar{\nu}_\mu$	6.57643×10^{24}
ν_e	2.23833×10^{25}
$\bar{\nu}_e$	1.06961×10^{26}
Oscillated ν_e	2.14913×10^{23}

Table 1: Normalization of input SuperK MC samples.

3.2.3. Single μ-like ring event selection cuts

The SuperK MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated ν_μ samples passing the standard SuperK selection cuts for 1 μ-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is μ-like (`ip[0] = 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay e^- in the event (`nmue <= 1`).

3.2.4. List of Monte Carlo templates

For each SuperK MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template granularity depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3] and 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4]. This analysis uses the following 32 MC templates:

- ν_μ CCQE
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- ν_e CCQE,

- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\nu_e \rightarrow \nu_\mu$ CCQE,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC1π,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC coherent, and
- oscillated $\nu_e \rightarrow \nu_\mu$ CC other.

3.2.5. Binning of Monte Carlo templates

Each of the 32 MC templates has 6132 2-dimensional bins (= 84 true energy bins $\times$ 73 reconstructed energy bins). The 84 true energy bins are arranged as follows:

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV.

The 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,
- 4 250-MeV bins from 3-4 GeV,

- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and
- 1 bin from 10-30 GeV.

The same reconstructed neutrino energy binning is used for the fit of the 1-ring μ-like reconstructed energy spectrum.

A study was performed to decide the optimum binning scheme balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4].

The binning described here is unchanged with respect to the Run 1+2+3 analysis, but, after discussion with T2K-SK, events with reconstructed energy >30 GeV are now included in the 10-30 GeV bin.

3.3. Applying flux tuning based on NA61 data

In this analysis, the nominal MC templates, which were constructed using a MC event sample generated with the 11a flux, are reweighted to the 11bv3.2 flux which was tuned on NA61 hadron production measurements [11][26]. The reweighted MC templates are denoted the *NA61-tuned* MC templates in this document. The applied weights were obtained from [19] and are shown as a function of the true neutrino energy in Figs. 1-4 in Ref. [4]. The effect of this flux tune on the predicted reconstructed energy spectrum of single μ-like ring events is discussed in Sec. 3.6 (see Figs. 4 and 6, and Tabs. 2 and 3)

3.4. Applying flux and cross-section (BANFF) tuning based on ND280 data

This analysis uses the BANFF flux and cross-section parameter tuning to reweight the NA61-tuned MC templates. The MC templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. Updated BANFF weights were used in this analysis with respect to the Run 1+2+3 one.

For the 2013 analyses, the ND280 ν_μ group has updated the division of reconstructed ND280 CC events. These events are now divided into the CC0π, CC1π and CC other samples (whereas for the Run 1+2+3 analysis they were divided into CCQE and CCnonQE samples). Pion tracks are identified using ionisation energy losses in the TPC or FGD, or a pion can be identified by the presence of a Michel electron in FGD1. The CC0π and CC1π samples have exactly 0 and 1 track consistent with a π^+ respectively. The CC other sample consists of events with more than 1 track consistent with a π^+ or at least one negative track or with a track consistent with an electron [20]. The BANFF group has made a fit of these three ND280 samples to find the best-fit values of the flux normalisation parameters and the cross-section parameters whose errors are expected to have some cancellation between the ND280 and SuperK, and the correlations between these parameters [21]. The postfit values of these parameters (taken from version 5 of the 2013 Run 1+2+3+4 BANFF fit in `postfit_banff_2013_disappearance_v5.root`) are used in turn to calculate the nominal predicted single μ-like ring reconstructed energy spectrum in SuperK. A list of BANFF parameters and their best-fit values is shown in Tab. 4.

The effect of the BANFF tune on the predicted reconstructed energy spectrum of single μ-like ring events is discussed in Sec. 3.6 (see Figs. 5 and 7, and Tabs. 2 and 3).

3.5. Applying neutrino oscillations

Oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the VaLOR analysis framework. Details and estimates of the numerical accuracy of this code were first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the VaLOR oscillation probability calculation code and Prob3++ [43], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

Oscillations are applied as a function of true energy to the ν_μ CCQE, ν_μ CC1π, ν_μ CC coherent, ν_μ CC other, $\bar{\nu}_\mu$ CCQE, $\bar{\nu}_\mu$ CC1π, $\bar{\nu}_\mu$ CC coherent, $\bar{\nu}_\mu$ CC other, ν_e CCQE, ν_e CC1π, ν_e CC coherent, ν_e CC other, $\bar{\nu}_e$ CCQE, $\bar{\nu}_e$ CC1π, $\bar{\nu}_e$ CC coherent, $\bar{\nu}_e$ CC other, oscillated $\nu_\mu \to \nu_e$ CCQE, oscillated $\nu_\mu \to \nu_e$ CC1π, oscillated $\nu_\mu \to \nu_e$ CC coherent, oscillated $\nu_\mu \to \nu_e$ CC other, oscillated $\nu_e \to \nu_\mu$ CCQE, oscillated $\nu_e \to \nu_\mu$ CC1π, oscillated $\nu_e \to \nu_\mu$ CC coherent and oscillated $\nu_e \to \nu_\mu$ CC other MC templates.

The MC templates constructed from the unoscillated MC samples are weighted with the corresponding survival probability: The ν_μ templates are weighted with $P(\nu_\mu \to \nu_\mu)$, the $\bar{\nu}_\mu$ templates with $P(\bar{\nu}_\mu \to \bar{\nu}_\mu)$, the ν_e templates with $P(\nu_e \to \nu_e)$ and the $\bar{\nu}_e$ templates with $P(\bar{\nu}_e \to \bar{\nu}_e)$. The MC templates made from the oscillated $\nu_\mu \to \nu_e$ MC sample are weighted with $P(\nu_\mu \to \nu_e)$ (the sample was generated assuming 100% of ν_μ transform to ν_e), while the MC templates made from the oscillated $\nu_e \to \nu_\mu$ MC sample are weighted with $P(\nu_e \to \nu_\mu)$.

In the standard 3-flavour oscillation framework, oscillations of ν_e and ν_μ can yield ν_τ, while oscillations of $\bar{\nu}_e$ and $\bar{\nu}_\mu$ can yield $\bar{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ SuperK MC samples are available).

It should be emphasized here that the ν_μ NC MC templates for a mode m are proxies for the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations (see also Sec. 3.5).

Also it should be noted that there are no explicit NC MC templates made from the oscillated ν_e sample. If they were used, the oscillated ν_e (i.e. ν_e coming from ν_μ oscillations) would be double counted since they are already included in the ν_μ NC MC templates.

The same oscillation parameters are used for ν_μ, $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ oscillations, and these are $\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), $\sin^2\theta_{13}$, $\sin^2\theta_{12}$, Δm^2_{21} and δ_{CP}.

3.6. Predictions of nominal and tuned-MC single μ-like ring spectra

In this section, we present the expected numbers of events and single μ-like ring reconstructed energy spectra in SuperK for various oscillation scenarios. We also present estimates of the effect of the NA61 and BANFF tunings and of assumptions made in the 3-flavour neutrino oscillation framework used in this analysis. Unless explicitly stated otherwise, the normal hierarchy is assumed, $\delta_{CP} = 0$, and the values shown in Tab. 10 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}. All plots were generated for an integrated exposure of 6.5700×10^{20} POT.

The expected number of single μ-like ring events is shown in Fig. 1 as a function of the oscillation parameters $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$. The expected number of events ranges from a minimum of ∼122, to more than ∼146 for oscillation parameter values still within the T2K Run 1+2+3 90% CL region. This illustrates the great sensitivity of the T2K ν_μ-disappearance analysis in measuring $\sin^2\theta_{23}$ with the Run 1+2+3+4 data.

Predicted single μ-like ring SuperK reconstructed energy spectra are shown in Fig. 2 for no oscillations, and in Fig. 3 for oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4.

A MC tune derived using NA61 data is applied to the nominal MC templates, and another tune derived using the ND280 data is applied to the NA61-tuned templates as described in Secs. 3.3 and 3.4. The fractional difference in the expected number of single μ-like ring events when switching from the nominal MC to the NA61-tuned MC templates is shown in Fig. 4 as a function of the oscillation parameters $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$. In Fig. 5 is shown the fractional difference in the expected number of single-ring μ-like events when switching from the NA61-tuned MC templates to the BANFF-tuned MC templates. From these figures it can be seen that, for oscillation parameters in the T2K Run 1+2+3 90% confidence region, ∼12.4-13.0% more single-ring μ-like events are obtained with the NA61-tuned MC than with the nominal MC. In the same part of the parameter space, ∼7.0-8.2% fewer events are expected with the BANFF-tuned MC than with the NA61-tuned MC. The effects of the NA61 and BANFF tunes on the total expected number of single-ring μ-like events and on the numbers in each of the 32 component interaction modes are shown in Tab. 2 for no oscillations, and in Tab. 3 for oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4. Finally, the effects of the NA61 and BANFF tunes on the reconstructed energy spectrum of single μ-like ring events are shown in Figs. 6 and 7 respectively. In each case, spectra are shown for both no oscillations and for oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4. Also shown are the ratio of flux-tuned spectrum to nominal spectrum, and BANFF-tuned spectrum to flux-tuned spectrum in Figs. 6 and 7 respectively.

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like SuperK sample. A SuperK MC study is required to determine whether the SuperK efficiency assumptions made there are valid.

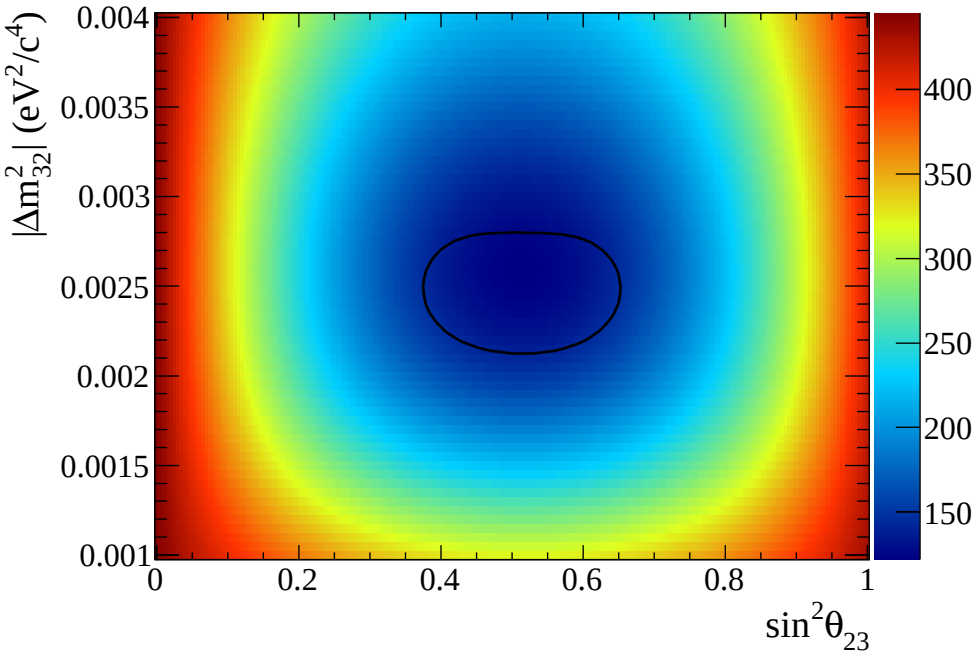


Figure 1: Predicted number of 1 μ-like ring events, as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for an exposure of 6.5700×10^{20} POT. The numbers of events were calculated using the BANFF-tuned MC templates. $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} have the values shown in Tab. 10 and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% C.L. region is superimposed for reference.

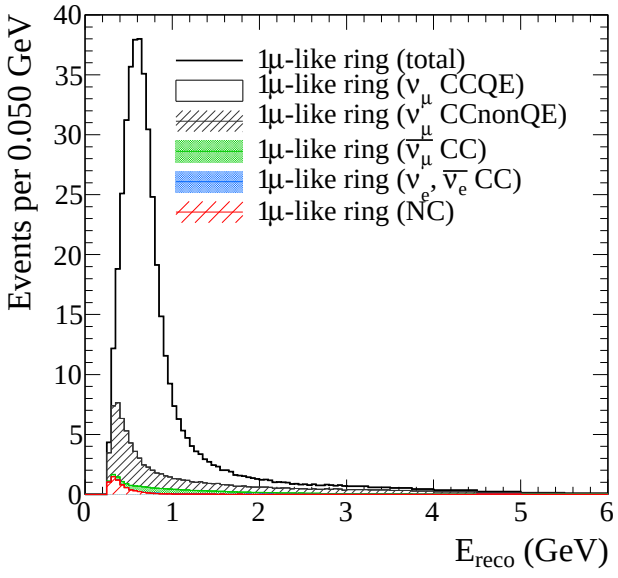


Figure 2: Predicted reconstructed-energy spectrum of 1 μ-like ring events, and contributions from various true neutrino reaction modes, for no oscillations and for an exposure of 6.5700×10^{20} POT. The 32 components of the spectrum are calculated separately in the actual analysis, but, for this plot, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu}_\mu$ CC, $\nu_e/\bar{\nu}_e$ CC and NC. The spectrum was generated using the BANFF-tuned MC templates.

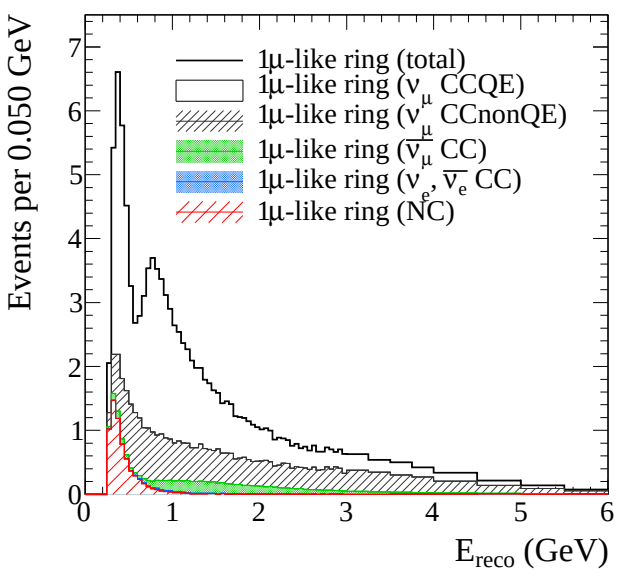


Figure 3: Predicted reconstructed-energy spectrum of 1 μ-like ring events, and contributions from various true neutrino reaction modes, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4 and for an exposure of 6.5700×10^{20} POT. The spectrum was generated using the BANFF-tuned MC templates. $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} have the values shown in Tab. 10 and δ_{CP}=0. The spectrum shown was generated assuming the normal hierarchy. Note that the vertical axis is zoomed in by a factor of 4 compared with Fig. 2.

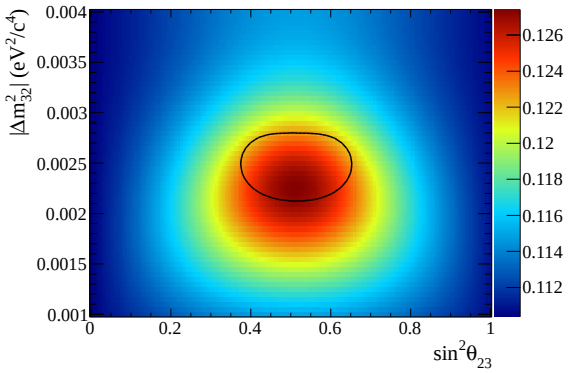


Figure 4: Fractional difference in the predicted number of 1 μ-like ring events as a result of applying the NA61 flux tuning to the nominal MC templates. These numbers are shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for an exposure of 6.5700×10^{20} POT. $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} have the values shown in Tab. 10 and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference.

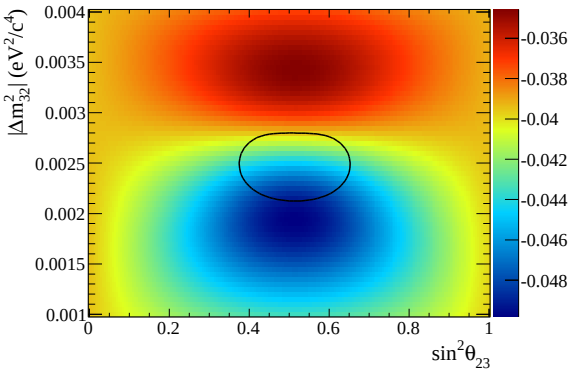


Figure 5: Fractional difference in the predicted number of 1 μ-like ring events as a result of applying the BANFF flux and cross-section tuning to the NA61 flux tuned MC templates. These numbers are shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ and for an exposure of 6.5700×10^{20} POT. $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} have the values shown in Tab. 10 and δ_{CP}=0. The numbers shown were generated assuming the normal hierarchy. The T2K Run 1+2+3 90% CL region is superimposed for reference.

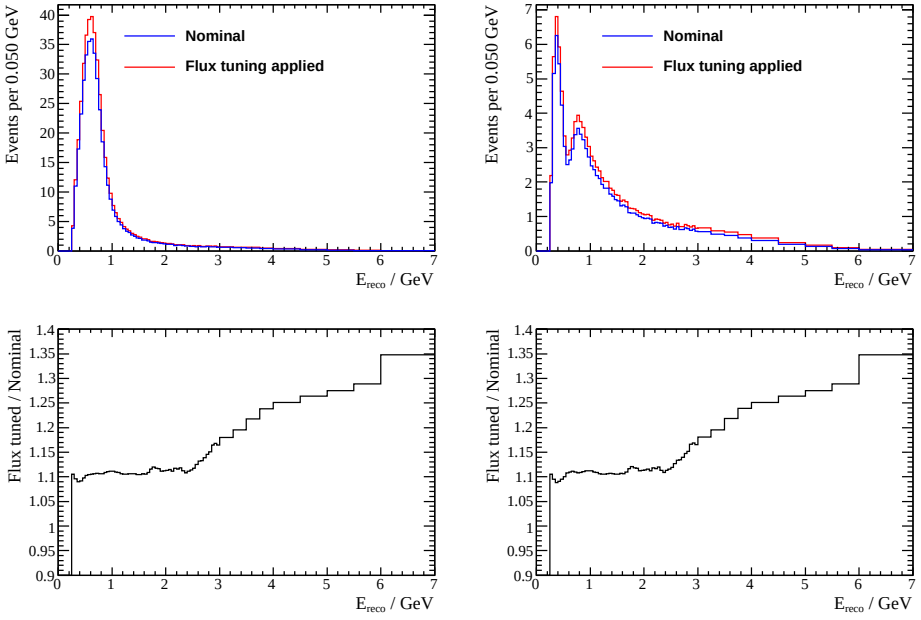


Figure 6: Top: Reconstructed-energy spectrum of 1 μ-like ring events, for an exposure of 6.5700×10^{20} POT, both with and without the effect of the NA61 flux tuning of the nominal MC templates. The spectra are shown both for no oscillations (left) and for oscillations with $\sin^2\theta_{23}$=0.5, $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the 2012 PDG values for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0 (right). Results shown for the normal hierarchy. Bottom: Ratio of flux-tuned spectrum to the nominal MC spectrum.

	N_{SK} (nominal MC)	N_{SK} (NA61-tuned MC)	N_{SK} (BANFF-tuned MC)
Total	418.131220	464.272533	445.980111
ν_μ CCQE	332.238749	367.166655	351.657098
ν_μ CC 1π	56.190220	63.533041	62.227879
ν_μ CC coherent	1.896444	2.136085	2.108162
ν_μ CC other	7.963009	9.600253	9.615971
ν_μ/ν_τ NC $1\pi^{+/-}$	3.403911	3.888131	2.934821
ν_μ/ν_τ NC other	2.912186	3.428374	3.228957
Osc. ν_μ CCQE	0.000000	0.000000	0.000000
Osc. ν_μ CC 1π	0.000000	0.000000	0.000000
Osc. ν_μ CC coherent	0.000000	0.000000	0.000000
Osc. ν_μ CC other	0.000000	0.000000	0.000000
$\bar{\nu_\mu}$ CCQE	9.008088	9.664244	9.521964
$\bar{\nu_\mu}$ CC 1π	2.909354	3.114170	3.061538
$\bar{\nu_\mu}$ CC coherent	0.428081	0.458785	0.466122
$\bar{\nu_\mu}$ CC other	0.519092	0.553722	0.548210
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.188316	0.203257	0.143433
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.176427	0.188866	0.175090
ν_e CCQE	0.035370	0.038166	0.037333
ν_e CC 1π	0.021574	0.023840	0.023994
ν_e CC coherent	0.001057	0.001158	0.001158
ν_e CC other	0.005812	0.006914	0.006795
ν_e NC $1\pi^{+/-}$	0.091432	0.105093	0.075133
ν_e NC other	0.112184	0.131680	0.121606
$\bar{\nu_e}$ CCQE	0.002174	0.002187	0.002102
$\bar{\nu_e}$ CC 1π	0.001376	0.001385	0.001345
$\bar{\nu_e}$ CC coherent	0.000135	0.000137	0.000137
$\bar{\nu_e}$ CC other	0.000442	0.000446	0.000432
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.012254	0.012354	0.008390
$\bar{\nu_e}$ NC other	0.013532	0.013592	0.012439
Osc. ν_e CCQE	0.000000	0.000000	0.000000
Osc. ν_e CC 1π	0.000000	0.000000	0.000000
Osc. ν_e CC coherent	0.000000	0.000000	0.000000
Osc. ν_e CC other	0.000000	0.000000	0.000000

Table 2: Calculated numbers of 1 μ-like ring events without oscillations using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.5700×10^{20} POT.

	N_{SK} (nominal MC)	N_{SK} (NA61-tuned MC)	N_{SK} (BANFF-tuned MC)
Total	116.298039	131.065465	124.997058
ν_μ CCQE	66.141356	73.456438	71.000951
ν_μ CC 1π	27.739731	31.948501	29.802171
ν_μ CC coherent	0.777604	0.896377	0.908341
ν_μ CC other	7.072565	8.586362	8.598179
ν_μ/ν_τ NC $1\pi^{+/-}$	3.403911	3.888131	2.934821
ν_μ/ν_τ NC other	2.912186	3.428374	3.228957
Osc. ν_μ CCQE	0.117390	0.129868	0.127422
Osc. ν_μ CC 1π	0.027377	0.030484	0.030937
Osc. ν_μ CC coherent	0.000821	0.000913	0.000913
Osc. ν_μ CC other	0.002343	0.002647	0.002608
$\bar{\nu_\mu}$ CCQE	4.429601	4.746184	4.622074
$\bar{\nu_\mu}$ CC 1π	2.080385	2.220226	2.133141
$\bar{\nu_\mu}$ CC coherent	0.228793	0.244564	0.248270
$\bar{\nu_\mu}$ CC other	0.460575	0.490922	0.485532
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.188316	0.203257	0.143433
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.176427	0.188866	0.175090
ν_e CCQE	0.032855	0.035470	0.034683
ν_e CC 1π	0.020475	0.022660	0.022716
ν_e CC coherent	0.000987	0.001084	0.001084
ν_e CC other	0.005716	0.006807	0.006690
ν_e NC $1\pi^{+/-}$	0.091432	0.105093	0.075133
ν_e NC other	0.112184	0.131680	0.121606
$\bar{\nu_e}$ CCQE	0.002085	0.002096	0.002014
$\bar{\nu_e}$ CC 1π	0.001337	0.001345	0.001303
$\bar{\nu_e}$ CC coherent	0.000130	0.000132	0.000132
$\bar{\nu_e}$ CC other	0.000438	0.000442	0.000429
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.012254	0.012354	0.008390
$\bar{\nu_e}$ NC other	0.013532	0.013592	0.012439
Osc. ν_e CCQE	0.179706	0.198075	0.189984
Osc. ν_e CC 1π	0.059372	0.065712	0.070965
Osc. ν_e CC coherent	0.004748	0.005234	0.005093
Osc. ν_e CC other	0.001405	0.001577	0.001558

Table 3: Calculated numbers of 1 μ-like ring events using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.5700×10^{20} POT and oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4. Values shown in Tab. 10 are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. The normal mass hierarchy is assumed.

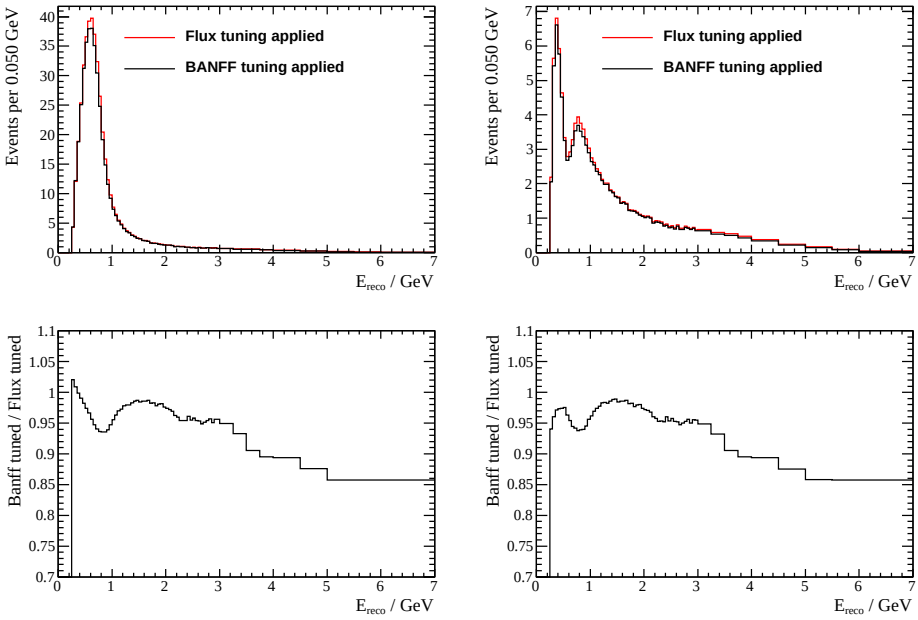


Figure 7: Top: Reconstructed-energy spectrum of 1 μ-like ring events, for an exposure of 6.5700×10^{20} POT, both with and without the effect of BANFF flux and cross-section tuning on the NA61-tuned flux MC templates. The spectra are shown both for no 23-sector oscillations (left) and for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4 (right). In all spectra, the 2012 PDG parameters are used for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} and δ_{CP}=0. Results shown for the normal hierarchy. Bottom: Ratio of BANFF-tuned spectrum to the flux-tuned spectrum.

4. Systematic effects in the prediction of the single μ-like ring event energy spectrum

4.1. Input systematic parameters

This analysis considers 45 systematic parameters which can be grouped into 5 categories:

1. **SuperK energy scale** (1 parameter).
The systematic parameter $f^{SK}_{E;r}$ is included to control the energy scale. Its uncertainty is estimated to be 2.4% [23]. The effect of the uncertainty in the reconstructed energy scale is calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in Tab. 5.

2. **SuperK efficiencies & effects of intranuclear and secondary re-interactions** (6 parameters).
Uncertainties in the efficiencies comes from the uncertainties in the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1412 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples. The NC PID error is evaluated using a NC-enriched control sample, which does not use PID information. Errors are evaluated using a fit to the efficiency, and the fitter errors are taken into account [22].
Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [14]. The NIWG group produced 16 parameter sets that span the 1σ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [14]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the SuperK detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [17]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single μ-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [13]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [14]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.
In this analysis, we consider the SuperK detector efficiency and FSI/SI systematics together, and add their covariance matrices linearly. Each of these covariance matrices has six correlated parameters, corresponding to different modes and reconstructed energy bins, as shown in Tab. 4. These efficiency/FSI/SI systematics are applied to the MC templates as shown in Tab. 5.

3. **Correlated flux and cross-section (BANFF) parameters** (23 parameters).
The postfit uncertainties of all BANFF flux and cross-section parameters and their correlations are taken into account. For the disappearance analysis, we use a version of the BANFF fit which considers only the ν_μ and $\bar{\nu}_\mu$ flux parameters. The effect of the ν_e and $\bar{\nu}_e$ flux parameter uncertainties was studied and it was found to be negligible [4]. Only those cross-section parameters that are expected to have some cancellation between the ND280 and SuperK are included in the BANFF fit. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors is given in Tab. 4.
The BANFF parameters are applied to the MC templates as specified in Tab. 5. All but two of the BANFF cross-section parameters are simple normalization factors. The parameters $f^{banff}_{16;t,r}$ and $f^{banff}_{17;t,r}$ are the axial masses that are parameters in the models of neutrino CC quasi-elastic (M_A^{QE}) and resonance-production (M_A^{RES}) interactions respectively; these have a more complicated, often non-linear, dependence on true and reconstructed neutrino energy. Event reweighting methods were used

to obtain the fractional changes of the number of events, in bins of true and reconstructed neutrino energy, for a range of values (-3.0σ, -2.5σ, -2.0σ, ..., +3.0σ) in the $f^{banff}_{16;t,r}$ and $f^{banff}_{17;t,r}$ parameters. In our p.d.f. calculation, cubic splines[2] are used to interpolate between these event weights to obtain the fractional change for any value of these systematic parameters in the (-3.0σ, +3.0σ) range.

4. **Uncorrelated cross-section parameters**(11 parameters).
In addition to the BANFF cross-section systematics, another 11 cross-section systematics are taken into account; these additional systematics are considered to be uncorrelated between themselves and to have no cancellation between the near and far detectors [14] [15] [16].
Three of these additional systematics relate to the initial-state nuclear environment modelling for CCQE. Neutrino Monte Carlo generators typically use Relativistic Fermi Gas (RFG) models, with or without extensions to take nucleon-nucleon correlation into account, and with some, often simplistic, prescription for the treatment of off-shell kinematics. The RFG model implementation depends on two parameters, the Fermi momentum level (225 MeV for ^{16}O) and the nucleon binding energy (27 MeV for ^{16}O), which are extracted from quasi-elastic electron scattering data. The systematic parameter $f_{pF;t,r}$ takes into account the effect of a 30 MeV 1σ uncertainty in the Fermi momentum level for ^{16}O, while the parameter $f_{bindE;t,r}$ takes into account the effect of a 9 MeV 1σ uncertainty in the nucleon binding energy for for ^{16}O. The implementation details of the nuclear model can have a significant effect and, therefore, the effect of switching between two such models was also included as a systematic uncertainty. This analysis considers the effect of switching between the RFG model and a better-motivated Spectral Function (SF) model. The latter does not assume a fixed binding energy and includes the correlations between nucleon momentum and binding energy. The systematic parameter $f_{SF;t,r}$ includes the effect of switching between the two models.
Simple, energy-independent, generator-level uncertainties are considered for the CC coherent, NC, ν_e CC and $\bar{\nu}$ CC cross sections, as the effect of these processes in the single μ-like ring sample is small. This analysis uses a 100% uncertainty for the CC coherent cross-section, a 30% uncertainty for both the NC1$\pi^{\pm}$ and NC "other" (all but single $\pi^{\pm}$) cross section, a 3% uncertainty on the ν_e / ν_μ CC cross-section ratio and a 20% uncertainty on the $\bar{\nu}$ / ν CC cross-section ratio. The effect of the variation of the above cross sections is included in the analysis through the systematic parameters $f_{CCcoh;t}$, $f_{NC1\pi^{\pm};t}$, $f_{NCoth;t}$, $f_{CC\nu_e;t}$ and $f_{CC\bar{\nu};t}$ respectively.
The treatment of Δ resonances in the nuclear environment has been controversial in the neutrino Monte Carlo generator community. Certain models include a 20%-level of π-less Δ decays in order to include the effect of $\Delta + N \rightarrow N + N$ scattering, in which the Δ interacts before it can decay. In contrast, other models include no π-less Δ decays on the basis that this would amount to double counting since, whatever the size of the effect, it is already included in hadron-nucleus scattering data to which the intranuclear cascade Monte Carlos are tuned. This analysis uses NEUT, in which 20% of Δs have a π-less decay by default, and takes the uncertainty in this branching fraction to be 100%. The systematic parameter $f_{\pi-less\Delta;t,r}$ includes the effect of varying the fraction of π-less Δ decays to the corresponding MC templates.
The $f_{CCothShape;t,r}$ systematic parameterizes the uncertainty in CC multi-pion, CC deep inelastic scattering and CC η, K and photon production. From MINOS data, this is known to be ≈10% at 4 GeV.

[2] In this analysis, the cubic splines are computed for each non-linear systematic and for each MC template, in bins of reconstructed and true energy. To produce these splines, we use the SK13a SuperK MC, the 11bv3.2 flux weights and the following oscillation parameters: $\sin^2(2\theta_{23}) = 1.0$, $\sin^2(2\theta_{13}) = 0.098$, $\sin^2(2\theta_{12}) = 0.8495$, $|\Delta m^2_{32}| = 2.44\times10^{-3}$ eV^2/c^4, $\Delta m^2_{21} = 7.5\times10^{-5}$ eV^2/c^4, $\delta_{CP} = 0$ and the normal mass hierarchy. For all splines, we use 41 bins in reconstructed energy: 1 bin from 0-0.3 GeV, 14 50-MeV bins from 0.3-1.0 GeV, 20 100-MeV bins from 1.0-3.0 GeV, 2 500-MeV bins from 3.0-4.0 GeV 1 bin from 4.0-5.0 GeV, 1 bin from 5.0-7.0 GeV, 1 bin from 7.0-10.0 GeV and 1 bin from 10.0-30.0 GeV. The ν_μ, $\bar{\nu}_\mu$ and oscillated ν_e samples have 11 bins in true energy: 1 bin from 0-0.4 GeV, 3 100-MeV bins from 0.4-0.7 GeV, 1 bin from 0.7-1.0 GeV, 1 bin from 1.0-1.5 GeV, 2 1-GeV bins from 1.5-3.5 GeV, 1 bin from 3.5-5.0 GeV, 1 bin from 5.0-7.0 GeV and 1 bin from 7.0-30.0 GeV. The ν_e and oscillated ν_μ samples have 7 bins in true energy: 1 bin from 0-0.05 GeV, 1 bin from 0.5-0.7 GeV, 1 bin from 0.7-0.8 GeV, 1 bin from 0.8-1.5 GeV, 1 bin from 1.5-2.5 GeV, 1 bin from 2.5-4.0 GeV and 1 bin from 4.0-30.0 GeV. Finally, the $\bar{\nu}_e$ sample have 2 bins in true energy: 1 bin from 0-2.5 GeV and 1 bin from 2.5-30.0 GeV.
The binning of the cubic splines was studied, and the results were presented in Appendix I of [4]. The spline spectra produced using the cubic splines are very similar to those produced using the CPU-intensive event-by-event reweighting, especially for the dominant components of the spectra.

Using this as a reference point, the uncertainty is estimated as $0.4/E_\nu$, where E_ν is in GeV. It might appear that this error could diverge at low energy, but this does not happen since the threshold is ≈0.6 GeV and there are very few interactions below 1 GeV.

The $f_{Wshape;t,r}$ parameter modifies the shape of the initial pion momentum distribution (before final-state interactions) by changing the distribution of the pion/nucleon invariant mass while keeping the total cross section constant. As the W shape parameter decreases, the pion momentum shifts to lower values, and this increases the probability of pion absorption in a final-state interaction. If such an absorption occurs, the event is migrated from CC1π (its true initial state) and appears to be CCQE.

The effect of the $f_{CCcoh;t}$, $f_{NC1\pi^{\pm};t}$, $f_{NCoth;t}$, $f_{CC\nu_e;t}$ and $f_{CC\bar{\nu};t}$ systematic parameters is a simple normalisation change for the corresponding MC templates and thus easily implemented. Systematic parameters $f_{pF;t,r}$, $f_{SF;t,r}$, $f_{bindE;t,r}$, $f_{\pi-less\Delta;t,r}$, $f_{CCothShape;t,r}$ and $f_{Wshape;t,r}$ have a complicated, often non-linear, dependence on true and reconstructed energy. These are treated in a similar way to the BANFF parameters $f^{banff}_{16;t,r}$ and $f^{banff}_{17;t,r}$. Event reweighting methods were used to obtain the fractional changes of the number of events, in bins of true and reconstructed neutrino energy, for a range of values (-3.0σ, -2.5σ, -2.0σ, ..., +3.0σ) of the systematic parameters. Again cross-section splines with polynomials of order 3 were used to interpolate between these event weights to obtain the fractional change for any value of these systematic parameters in the (-3.0σ, +3.0σ) range.

Some of these systematics are special cases:

- $f_{SF;t,r}$ is discrete; it can be either spectral function off (RFG) or spectral function on (SF). Splines are generated with SF off as nominal (0), and SF on as +1σ. This parameter is restricted to the range between 0 and +1σ, and linear interpolation is made between the weights for nominal and +1σ.
- For $f_{Wshape;t,r}$ only ±1σ errors are evaluated using event reweighting. Weights for positive tweaks are interpolated or extrapolated from the nominal and +1σ weights, and weights for negative tweaks are interpolated or extrapolated from the nominal and -1σ weights.
- For $f_{\pi-less\Delta;t,r}$ only ±1.0 sigma errors are evaluated using event reweighting. Weights for positive tweaks are interpolated or extrapolated from the nominal and +1σ weights, and weights for tweaks between nominal and -1σ are interpolated from the nominal and -1σ weights. Weights for tweaks more negative than -1σ are taken to be the same as the -1σ weight; this is done since 20% of Δs have a π-less decay by default in NEUT, and the error is 20%. For this reason, it is unphysical to tweak $f_{\pi-less\Delta;t,r}$ to be more negative than -1σ.

The 11 uncorrelated cross-section systematics considered in this analysis are summarised in Tab. 4. These cross-section systematics are applied to the MC templates as shown in Tab. 5.

5. ***Non-23 sector* oscillation parameters** (4 parameters).

In this analysis, we treat all four non-23-sector oscillation parameters as nuisance parameters in the fit. We use penalty terms for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, with their central values and uncertainties being taken from [27, 41]. The CP-violating phase δ_{CP} can take any value between $-\pi$ and $+\pi$, and is not constrained by a penalty term.

SuperK energy scale systematics (1)

$f^{SK}_{E;r}$	Reconstructed energy scale systematic. The uncertainty is estimated to be 2.4% [23].

SuperK detector efficiency & intranuclear and secondary interaction systematics

SuperK detector efficiency (ring counting, fiducial volume, reduction chain, OD cut, PID cut, momentum cut and decay electron cut efficiency uncertainties), FSI (final state interaction / intranuclear hadron transport) and SI (secondary interaction / detector hadron transport) systematic parameters. Six correlated parameters are included (i=0,...,5) corresponding to different modes and reconstructed energy bins.

$f^{SK+FSI}_{i;r}$

Parameter	Description	1σ fractional error
0	ν_μ & $\bar{\nu}_\mu$ CCQE, $E_{reco} <$ 0.4 GeV	0.025
1	ν_μ & $\bar{\nu}_\mu$ CCQE, 0.4 GeV $\leq E_{reco} \leq$ 1.1 GeV	0.028
2	ν_μ & $\bar{\nu}_\mu$ CCQE, $E_{reco} >$ 1.1 GeV	0.037
3	ν_μ & $\bar{\nu}_\mu$ CCnonQE	0.119
4	ν_e & $\bar{\nu}_e$ CC	1.000
5	All NC	0.600

The covariance matrix is listed in Eq. 3.

Correlated flux and cross-section (BANFF) systematics (23)

Flux and cross-section parameters tuned by BANFF group on ND280 data [21]. Twenty-three correlated parameters are included (i=0,...,22).

$f^{banff}_{i;t,r}$

Parameter	Description	Best-fit	1σ pre/postfit fractional error
0	ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.029	0.121/0.085
1	ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.022	0.130/0.088
2	ν_μ flux normalisation, E = 0.5 - 0.6 GeV	0.995	0.122/0.080
3	ν_μ flux normalisation, E = 0.6 - 0.7 GeV	0.966	0.115/0.076
4	ν_μ flux normalisation, E = 0.7 - 1.0 GeV	0.934	0.129/0.085
5	ν_μ flux normalisation, E = 1.0 - 1.5 GeV	0.992	0.116/0.077
6	ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.037	0.100/0.068
7	ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.054	0.095/0.065
8	ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.035	0.112/0.072
9	ν_μ flux normalisation, E = 5.0 - 7.0 GeV	0.975	0.152/0.073
10	ν_μ flux normalisation, E = 7.0 - 30.0 GeV	0.943	0.187/0.082
11	$\bar{\nu}_\mu$ flux normalisation, E = 0.0 - 0.7 GeV	1.030	0.133/0.102
12	$\bar{\nu}_\mu$ flux normalisation, E = 0.7 - 1.0 GeV	1.011	0.117/0.090
13	$\bar{\nu}_\mu$ flux normalisation, E = 1.0 - 1.5 GeV	1.007	0.119/0.094
14	$\bar{\nu}_\mu$ flux normalisation, E = 1.5 - 2.5 GeV	1.026	0.123/0.104
15	$\bar{\nu}_\mu$ flux normalisation, E = 2.5 -30.0 GeV	1.008	0.122/0.107
16	CCQE axial-mass scaling factor	1.025	0.372/0.059
17	Resonance-production axial-mass scaling factor	0.797	0.183/0.056
18	CCQE normalisation, E = 0.0 - 1.5 GeV	0.966	0.110/0.076
19	CCQE normalisation, E = 1.5 - 3.5 GeV	0.931	0.300/0.103
20	CCQE normalisation, E = 3.5 - 30.0 GeV	0.852	0.300/0.114
21	$CC1\pi$ normalisation, E = 0.0 - 2.5 GeV	1.265	0.317/0.163
22	$CC1\pi$ normalisation, E = 2.5 - 30.0 GeV	1.122	0.400/0.172

The covariance matrix is listed in Eq. 4.1.

Uncorrelated cross-section systematics (11)	
$f_{pF;t,r}$	Uncertainty in Fermi momentum for ^{16}O was taken to be 225±30 MeV [15].
$f_{bindE;t,r}$	Uncertainty in binding energy for ^{16}O was taken to be 27±9 MeV [15].
$f_{SF;t,r}$	Nuclear environment modelling systematic: Switches from Fermi Gas (RFG) to Spectral Function (SF) model (0:RFG - 1:SF) [15].
$f_{\pi-less\Delta;t,r}$	Systematic due to the uncertainty (20% [15]). in the fraction of π-less Δ decays in resonance-production events.
$f_{CCothShape;t,r}$	Systematic due to the uncertainty (40% [15]) in CC multipion, CC DIS and CC resonant η, K and photon production cross section.
$f_{Wshape;t,r}$	Systematic due to the uncertainty (52% [15]) in the shape of the initial pion momentum distribution (before final-state interactions) in resonance interactions.
$f_{CCcoh;t}$	Systematic due to the uncertainty (100% [15]) in the CC coherent cross section.
$f_{NC1\pi^{\pm};t}$	Systematic due to the uncertainty (30% [15]) on the NC1$\pi^{\pm}$ cross section.
$f_{NCoth;t}$	Systematic due to the uncertainty (30% [15]) on the NC other (all but single $\pi^{\pm}$) cross section.
$f_{CC\nu_e;t}$	Systematic due to the uncertainty (3% [14]) on the ν_e / ν_μ CC cross-section ratio.
$f_{CC\bar{\nu};t}$	Systematic due to the uncertainty (20% [16]) on the $\bar{\nu}$ / ν CC cross-section ratio.
***Non-23 sector* oscillation parameters** (4)	
$\sin^2\theta_{13}$	Uncertainty in $\sin^2\theta_{13}$ of 13.9%, taken from the 2012 PDG [27].
$\sin^2\theta_{12}$	Uncertainty in $\sin^2\theta_{12}$ of 5.9%, taken from a global fit [41].
Δm^2_{21}	Uncertainty in Δm^2_{21} of 2.7%, taken from the 2012 PDG [27].
δ_{CP}	Allowed to vary between -π and +π with no constraint.

Table 4: Summary of systematics included the VaLOR Run 1+2+3+4 ν_μ-disappearance analysis.

$$C^{SK+FSI} = \begin{array}{c} \\ f^{SK+FSI}_{0;r} \\ f^{SK+FSI}_{1;r} \\ f^{SK+FSI}_{2;r} \\ f^{SK+FSI}_{3;r} \\ f^{SK+FSI}_{4;r} \\ f^{SK+FSI}_{5;r} \end{array} \begin{array}{c} \begin{array}{cccccc} f^{SK+FSI}_{0;r} & f^{SK+FSI}_{1;r} & f^{SK+FSI}_{2;r} & f^{SK+FSI}_{3;r} & f^{SK+FSI}_{4;r} & f^{SK+FSI}_{5;r} \end{array} \\ \begin{pmatrix} 0.00065 & 0.00066 & 0.00062 & 0.00031 & -0.00000 & 0.00030 \\ 0.00066 & 0.00080 & 0.00079 & 0.00037 & 0.00001 & 0.00030 \\ 0.00062 & 0.00079 & 0.00137 & 0.00057 & 0.00004 & 0.00030 \\ 0.00031 & 0.00037 & 0.00057 & 0.01412 & 0.00106 & -0.00074 \\ -0.00000 & 0.00001 & 0.00004 & 0.00106 & 1.00023 & 0.00041 \\ 0.00030 & 0.00030 & 0.00030 & -0.00074 & 0.00041 & 0.35949 \end{pmatrix} \end{array} \quad (3)$$

$$C^{banff} = \begin{array}{c} \\ f^{banff}_{0;t,r} \\ f^{banff}_{1;t,r} \\ f^{banff}_{2;t,r} \\ f^{banff}_{3;t,r} \\ f^{banff}_{4;t,r} \\ f^{banff}_{5;t,r} \\ f^{banff}_{6;t,r} \\ f^{banff}_{7;t,r} \\ f^{banff}_{8;t,r} \\ f^{banff}_{9;t,r} \\ f^{banff}_{10;t,r} \\ f^{banff}_{11;t,r} \\ f^{banff}_{12;t,r} \\ f^{banff}_{13;t,r} \\ f^{banff}_{14;t,r} \\ f^{banff}_{15;t,r} \\ f^{banff}_{16;t,r} \\ f^{banff}_{17;t,r} \\ f^{banff}_{18;t,r} \\ f^{banff}_{19;t,r} \\ f^{banff}_{20;t,r} \\ f^{banff}_{21;t,r} \\ f^{banff}_{22;t,r} \end{array} \begin{array}{c} f^{banff}_{0;t,r} \\ f^{banff}_{1;t,r} \\ f^{banff}_{2;t,r} \\ f^{banff}_{3;t,r} \\ f^{banff}_{4;t,r} \\ f^{banff}_{5;t,r} \\ f^{banff}_{6;t,r} \\ f^{banff}_{7;t,r} \\ f^{banff}_{8;t,r} \\ f^{banff}_{9;t,r} \\ f^{banff}_{10;t,r} \\ f^{banff}_{11;t,r} \\ f^{banff}_{12;t,r} \\ f^{banff}_{13;t,r} \\ f^{banff}_{14;t,r} \\ f^{banff}_{15;t,r} \\ f^{banff}_{16;t,r} \\ f^{banff}_{17;t,r} \\ f^{banff}_{18;t,r} \\ f^{banff}_{19;t,r} \\ f^{banff}_{20;t,r} \\ f^{banff}_{21;t,r} \\ f^{banff}_{22;t,r} \end{array} \begin{pmatrix} 0.007 & 0.007 & 0.006 & 0.005 & 0.004 & 0.004 & 0.004 & 0.003 & 0.002 & 0.002 & 0.001 & 0.008 & 0.006 & 0.005 & 0.005 & 0.003 & -0.001 & -0.000 & -0.004 & -0.002 & -0.001 & -0.005 & -0.002 \\ 0.007 & 0.008 & 0.007 & 0.005 & 0.005 & 0.005 & 0.004 & 0.003 & 0.002 & 0.002 & 0.001 & 0.006 & 0.005 & 0.004 & 0.004 & 0.002 & -0.001 & -0.000 & -0.005 & -0.002 & -0.001 & -0.006 & -0.001 \\ 0.006 & 0.007 & 0.006 & 0.006 & 0.005 & 0.004 & 0.004 & 0.003 & 0.002 & 0.002 & 0.001 & 0.005 & 0.005 & 0.004 & 0.004 & 0.002 & -0.001 & -0.000 & -0.005 & -0.003 & -0.001 & -0.005 & -0.001 \\ 0.005 & 0.005 & 0.006 & 0.006 & 0.005 & 0.005 & 0.004 & 0.003 & 0.002 & 0.002 & 0.001 & 0.005 & 0.005 & 0.005 & 0.004 & 0.002 & -0.001 & -0.000 & -0.005 & -0.003 & -0.001 & -0.006 & -0.001 \\ 0.004 & 0.005 & 0.005 & 0.005 & 0.007 & 0.006 & 0.004 & 0.003 & 0.002 & 0.002 & 0.001 & 0.004 & 0.004 & 0.004 & 0.003 & 0.001 & -0.001 & -0.000 & -0.005 & -0.002 & -0.001 & -0.007 & -0.001 \\ 0.004 & 0.005 & 0.004 & 0.005 & 0.006 & 0.006 & 0.005 & 0.003 & 0.002 & 0.002 & 0.001 & 0.005 & 0.005 & 0.005 & 0.004 & 0.002 & -0.000 & -0.000 & -0.004 & -0.003 & -0.001 & -0.007 & -0.001 \\ 0.004 & 0.004 & 0.004 & 0.004 & 0.004 & 0.005 & 0.005 & 0.003 & 0.002 & 0.002 & 0.002 & 0.005 & 0.005 & 0.004 & 0.005 & 0.003 & -0.000 & -0.000 & -0.003 & -0.003 & -0.001 & -0.006 & -0.002 \\ 0.003 & 0.003 & 0.003 & 0.003 & 0.003 & 0.003 & 0.003 & 0.004 & 0.003 & 0.002 & 0.001 & 0.004 & 0.003 & 0.003 & 0.003 & 0.001 & -0.000 & -0.000 & -0.002 & -0.002 & -0.001 & -0.003 & -0.003 \\ 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & 0.003 & 0.005 & 0.004 & 0.003 & 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & -0.000 & -0.000 & -0.002 & -0.001 & -0.002 & -0.002 & -0.005 \\ 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & 0.002 & 0.004 & 0.005 & 0.005 & 0.002 & 0.002 & 0.002 & 0.003 & 0.004 & -0.000 & 0.000 & -0.001 & -0.000 & -0.002 & -0.001 & -0.006 \\ 0.001 & 0.001 & 0.001 & 0.001 & 0.001 & 0.001 & 0.002 & 0.001 & 0.003 & 0.005 & 0.007 & 0.001 & 0.002 & 0.001 & 0.003 & 0.005 & 0.000 & 0.000 & -0.001 & 0.000 & -0.002 & -0.001 & -0.005 \\ 0.008 & 0.006 & 0.005 & 0.005 & 0.004 & 0.005 & 0.005 & 0.004 & 0.002 & 0.002 & 0.001 & 0.010 & 0.008 & 0.008 & 0.007 & 0.004 & -0.001 & -0.000 & -0.004 & -0.003 & -0.000 & -0.006 & -0.001 \\ 0.006 & 0.005 & 0.005 & 0.005 & 0.004 & 0.005 & 0.005 & 0.003 & 0.002 & 0.002 & 0.002 & 0.008 & 0.008 & 0.008 & 0.007 & 0.004 & -0.001 & -0.000 & -0.004 & -0.003 & -0.001 & -0.006 & -0.001 \\ 0.005 & 0.004 & 0.004 & 0.005 & 0.004 & 0.005 & 0.004 & 0.003 & 0.002 & 0.002 & 0.001 & 0.008 & 0.008 & 0.009 & 0.007 & 0.004 & -0.001 & -0.000 & -0.004 & -0.003 & -0.001 & -0.006 & -0.001 \\ 0.005 & 0.004 & 0.004 & 0.004 & 0.003 & 0.004 & 0.005 & 0.003 & 0.002 & 0.003 & 0.003 & 0.007 & 0.007 & 0.007 & 0.011 & 0.009 & -0.000 & -0.000 & -0.003 & -0.003 & -0.000 & -0.005 & -0.002 \\ 0.003 & 0.002 & 0.002 & 0.002 & 0.001 & 0.002 & 0.003 & 0.001 & 0.002 & 0.004 & 0.005 & 0.004 & 0.004 & 0.004 & 0.009 & 0.011 & -0.000 & -0.000 & -0.001 & -0.002 & -0.001 & -0.002 & -0.003 \\ -0.001 & -0.001 & -0.001 & -0.001 & -0.001 & -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & 0.000 & -0.001 & -0.001 & -0.001 & -0.000 & -0.000 & 0.004 & -0.001 & -0.001 & -0.002 & -0.002 & 0.001 & 0.002 \\ -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & 0.000 & 0.000 & -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & -0.001 & 0.003 & 0.000 & 0.000 & 0.000 & -0.004 & -0.004 \\ -0.004 & -0.005 & -0.005 & -0.005 & -0.005 & -0.004 & -0.003 & -0.002 & -0.002 & -0.001 & -0.001 & -0.004 & -0.004 & -0.004 & -0.003 & -0.001 & -0.001 & 0.000 & 0.006 & 0.004 & 0.002 & 0.005 & 0.000 \\ -0.002 & -0.002 & -0.003 & -0.003 & -0.002 & -0.003 & -0.003 & -0.002 & -0.001 & -0.000 & 0.000 & -0.003 & -0.003 & -0.003 & -0.003 & -0.002 & -0.002 & 0.000 & 0.004 & 0.011 & 0.005 & 0.001 & -0.005 \\ -0.001 & -0.001 & -0.001 & -0.001 & -0.001 & -0.001 & -0.001 & -0.001 & -0.002 & -0.002 & -0.002 & -0.000 & -0.001 & -0.001 & -0.000 & -0.001 & -0.002 & 0.000 & 0.002 & 0.005 & 0.013 & -0.003 & -0.006 \\ -0.005 & -0.006 & -0.005 & -0.006 & -0.007 & -0.007 & -0.006 & -0.003 & -0.002 & -0.001 & -0.001 & -0.006 & -0.006 & -0.006 & -0.005 & -0.002 & 0.001 & -0.004 & 0.005 & 0.001 & -0.003 & 0.027 & 0.011 \\ -0.002 & -0.001 & -0.001 & -0.001 & -0.001 & -0.001 & -0.002 & -0.003 & -0.005 & -0.006 & -0.005 & -0.001 & -0.001 & -0.001 & -0.002 & -0.003 & 0.002 & -0.004 & 0.000 & -0.005 & -0.006 & 0.011 & 0.029 \end{pmatrix} \quad (4)$$

	ν_μ CCQE	ν_μ CC1π	ν_μ CC coherent	ν_μ CC other	ν_μ NC1$\pi^\pm$	ν_μ NC other	$\bar{\nu_\mu}$ CCQE	$\bar{\nu_\mu}$ CC1π	$\bar{\nu_\mu}$ CC coherent	$\bar{\nu_\mu}$ CC other	$\bar{\nu_\mu}$ NC1$\pi^\pm$	$\bar{\nu_\mu}$ NC other	ν_e CCQE	ν_e CC1π	ν_e CC coherent	ν_e CC other	ν_e NC1$\pi^\pm$	ν_e NC other	$\bar{\nu_e}$ CCQE	$\bar{\nu_e}$ CC1π	$\bar{\nu_e}$ CC coherent	$\bar{\nu_e}$ CC other	$\bar{\nu_e}$ NC1$\pi^\pm$	$\bar{\nu_e}$ NC other	Osc-ν_e CCQE	Osc-ν_e CC1π	Osc-ν_e CC coherent	Osc-ν_e CC other	Osc-ν_μ CCQE	Osc-ν_μ CC1π	Osc-ν_μ CC coherent	Osc-ν_μ CC other
$\sin^2\theta_{13}$	✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
$\sin^2\theta_{12}$	✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
Δm^2_{21}	✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
δ_{CP}	✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{0;r}$	✓						✓																						✓			
$f^{SK+FSI}_{1;r}$	✓						✓																						✓			
$f^{SK+FSI}_{2;r}$	✓						✓																						✓			
$f^{SK+FSI}_{3;r}$		✓	✓	✓				✓	✓	✓																				✓	✓	✓
$f^{SK+FSI}_{4;r}$													✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓				
$f^{SK+FSI}_{5;r}$					✓	✓					✓	✓					✓	✓					✓	✓								
$f^{banff}_{0;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{1;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{2;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{3;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{4;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{5;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{6;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{7;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{8;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{9;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{10;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓				
$f^{banff}_{11;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{12;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{13;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{14;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{15;t,r}$							✓	✓	✓	✓	✓	✓																				
$f^{banff}_{16;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{17;t,r}$		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓		✓		✓
$f^{banff}_{18;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{19;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{20;t,r}$	✓						✓						✓						✓						✓				✓			
$f^{banff}_{21;t,r}$		✓						✓						✓						✓						✓				✓		
$f^{banff}_{22;t,r}$		✓						✓						✓						✓						✓				✓		
$f_{pF;t,r}$	✓						✓						✓						✓						✓				✓			
$f_{SF;t,r}$	✓						✓						✓						✓						✓				✓			
$f_{\pi-less\Delta;t,r}$		✓			✓	✓		✓			✓	✓		✓			✓	✓		✓			✓	✓		✓				✓		
$f_{bindE;t,r}$	✓						✓						✓						✓						✓				✓			
$f_{CCothShape;t,r}$				✓						✓						✓						✓						✓				✓
$f_{Wshape;t,r}$					✓	✓					✓	✓					✓	✓					✓	✓								
$f_{CCcoh;t}$			✓						✓						✓						✓						✓				✓	
$f_{NC1\pi^\pm;t}$					✓						✓						✓						✓									
$f_{NCoth;t}$						✓						✓						✓						✓								
$f_{CC\nu_e;t}$													✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓				
$f_{CC\bar{\nu};t}$							✓	✓	✓	✓									✓	✓	✓	✓										

Table 5: Each row of this table represents one of the 45 systematic parameters considered in this analysis and each column represents one of the 32 MC templates used to construct the single μ-like ring reconstructed energy spectrum p.d.f. A ✓symbol denotes that the given MC template is modified when the given systematic parameter is tweaked.

4.2. Evaluation of effects of systematic parameters on the predicted single μ-like ring reconstructed energy spectrum

In this section, we present the effects of the systematics described in Sec. 4.1 on the predicted number of single μ-like ring events and the predicted reconstructed energy spectrum in SuperK. These effects are shown both without oscillations and with oscillations with $\sin^2\theta_{23} = 0.5$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, the values shown in Tab. 10 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. All results are generated for an integrated exposure of 6.5700×10^{20} POT and the BANFF-tuned MC templates are used.

The effects of systematic uncertainties on the predicted number of events are summarised in Tab. 6, both for no oscillations and for oscillations with the typical parameter values given above. In this table, systematic parameters are grouped into five categories and all parameter correlations are taken into account. Percentage uncertainties are given for both the BANFF prefit and postfit errors. The individual effect of each parameter is also presented in Tab. 7, in which the correlations between parameters are necessarily ignored.

The effects of the combined systematic uncertainties on the reconstructed energy spectrum of single μ-like ring events is shown in Fig. 8 for the typical oscillation scenario used. This plot shows the total error envelope for the 1 μ-like ring event reconstructed energy spectrum; this is calculated as the $\pm1\sigma$ spread of bin contents from 100k toy MC experiments generated with randomized systematic parameters. In Fig. 8, all systematic parameters were considered and their correlations were taken into account.

Figs. 9 and 11 show the effect of the systematic error, relative to the statistical error at the Run 1-4 exposure, for the no oscillation and oscillation cases respectively. It can be seen that the statistical error dominates at all energies.

Figs. 10 and 12 show the effect of the four categories of systematic error, relative to the total systematic error, for the no oscillation and oscillation cases respectively. It can be seen that the SuperK detector + FSI + SI and uncorrelated cross section systematics dominate the oscillation dip region.

Detailed studies showing the effects of individual systematics on the expected number of single μ-like ring events are shown in [7].

The individual effect of each systematic parameter on the best-fit values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for the normal hierarchy was quantified by making a toy MC experiment with no statistical fluctuations and a $+1\sigma$ tweak of a single systematic parameter. A fit for only $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, with systematics fixed are their nominal values, was made of that experiment, and the $+1\sigma$ effect of that systematic parameter on the best-fit values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ was taken to be the difference between the best-fit values from the experiment with the $+1\sigma$ tweak and an experiment with nominal values of the systematics. Similarly, the -1σ effect of that parameter was the difference between the best-fit values from an experiment with a -1σ tweak and an experiment with nominal values of the systematics. This procedure was repeated for each systematic parameter in turn. For systematic parameters which are correlated, this procedure was repeated twice, both respecting and ignoring correlations. The study is repeated at multiple input points in the $\sin^2\theta_{23}$-$|\Delta m^2_{32}|$ plane.

An estimate of the order of the effect of systematics is given in Tabs. 8 and 9, ordered by the effect on $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ respectively. To make this estimate, the systematic effect is compared with the statistical error for the specific values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$. The statistical error was evaluated using 10k toy MC datasets with no systematic fluctuations at each input point, with each dataset being fitted for only $\sin^2_{23}$ and $|\Delta m^2_{32}|$. For each systematic, the largest effect seen in the study is given, whether that is from $+1\sigma$ or -1σ, including or neglecting correlations, and from any true values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ in the study.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (no oscillations)	$\delta N_{SK}/N_{SK}$ (oscillations with typical parameter values)
SuperK detector + FSI + SI	3.39%	5.60%
BANFF (prefit)	26.20%	21.92%
BANFF (postfit)	2.67%	2.74%
Uncorrelated XSec	3.00%	5.01%
Non-23-sector oscillation parameters	0.00%	0.22%
Total (BANFF prefit)	26.59%	23.17%
Total (BANFF postfit)	5.26%	8.00%

Table 6: Effect of 1σ systematic parameter variation on the number of 1 μ-like ring events, computed for no oscillations and for oscillations with $\sin^2\theta_{23}$=0.5, $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the values shown in Tab. 10 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.5700×10^{20} POT and BANFF-tuned MC templates were used.

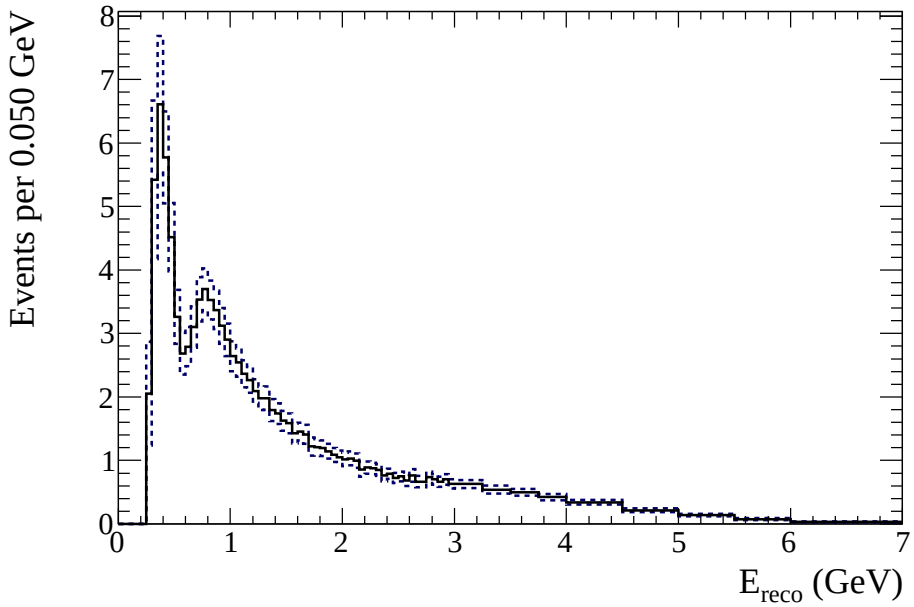


Figure 8: Total error envelope of the 1 μ-like ring event reconstructed energy spectrum, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the values shown in Tab. 10 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.5700×10^{20} POT. BANFF-tuned MC templates were used. The error envelope was calculated as the $\pm1\sigma$ spread of bin contents using an ensemble of 100k toy MC experiments generated with randomized systematic parameters. All systematic parameters were considered and their correlations were taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (no oscillation)		$\delta N_{SK}/N_{SK}$ (oscillations with typical params)	
	BANFF prefit	BANFF postfit	BANFF prefit	BANFF postfit
$f^{SK}_{E;r}$	0	0	0	0
$f^{SK+FSI}_{0;r}$	0.09	0.098	0.15	0.18
$f^{SK+FSI}_{1;r}$	1.8	1.9	0.68	0.74
$f^{SK+FSI}_{2;r}$	0.4	0.43	0.92	1.0
$f^{SK+FSI}_{3;r}$	2.4	2.1	4.6	4.0
$f^{SK+FSI}_{4;r}$	0.017	0.016	0.25	0.27
$f^{SK+FSI}_{5;r}$	1.1	0.9	3.8	3.2
$f^{banff}_{0;t,r}$	0.36	0.27	0.85	0.66
$f^{banff}_{1;t,r}$	1.0	0.74	0.67	0.5
$f^{banff}_{2;t,r}$	1.9	1.3	0.12	0.085
$f^{banff}_{3;t,r}$	2.3	1.6	0.3	0.22
$f^{banff}_{4;t,r}$	3.4	2.4	2.2	1.6
$f^{banff}_{5;t,r}$	1.0	0.66	1.8	1.2
$f^{banff}_{6;t,r}$	0.63	0.39	1.7	1.1
$f^{banff}_{7;t,r}$	0.33	0.2	1.0	0.65
$f^{banff}_{8;t,r}$	0.41	0.22	1.3	0.74
$f^{banff}_{9;t,r}$	0.19	0.077	0.65	0.27
$f^{banff}_{10;t,r}$	0.051	0.019	0.17	0.066
$f^{banff}_{11;t,r}$	0.061	0.05	0.035	0.03
$f^{banff}_{12;t,r}$	0.061	0.049	0.049	0.04
$f^{banff}_{13;t,r}$	0.085	0.067	0.16	0.13
$f^{banff}_{14;t,r}$	0.1	0.08	0.27	0.22
$f^{banff}_{15;t,r}$	0.078	0.06	0.25	0.2
$f^{banff}_{16;t,r}$	25.0	3.4	18.0	2.8
$f^{banff}_{17;t,r}$	2.8	1.1	6.7	2.4
$f^{banff}_{18;t,r}$	7.9	5.9	4.2	3.3
$f^{banff}_{19;t,r}$	1.4	0.55	3.9	1.6
$f^{banff}_{20;t,r}$	0.36	0.16	1.2	0.53
$f^{banff}_{21;t,r}$	3.7	1.5	4.9	2.0
$f^{banff}_{22;t,r}$	1.7	0.47	5.4	1.6
$f_{pF;t,r}$	0.062	0.065	0.13	0.13
$f_{SF;t,r}$	2.9	2.9	0.85	0.9
$f_{Wshape;t,r}$	0.12	0.073	0.4	0.26
$f_{CCothShape;t,r}$	0.28	0.3	0.78	0.88
$f_{\pi-less\Delta;t,r}$	3.2	2.6	5.6	4.5
$f_{bindE;t,r}$	0.5	0.43	0.24	0.2
$f_{CCcoh;t}$	0.53	0.58	0.81	0.93
$f_{NC1\pi^{\pm};t}$	0.31	0.21	1.1	0.76
$f_{NCoth;t}$	0.25	0.24	0.84	0.85
$f_{CC\nu_e;t}$	0.0005	0.00049	0.0076	0.0081
$f_{CC\bar{\nu};t}$	0.61	0.61	1.2	1.2
$\sin^2(\theta_{13})$	0	0	0.069	0.078
$\sin^2(\theta_{12})$	0	0	0.04	0.037
Δm^2_{21}	0	0	0.027	0.025
δ_{CP}	0	0	0.21	0.19

Table 7: Effects of 1σ systematic parameter variations on the number of 1 μ-like ring events, computed for no oscillations and for oscillations with $\sin^2\theta_{23}$=0.5, $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the values shown in Tab. 10 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. Correlations are ignored for this table. The normal mass hierarchy is assumed. The numbers shown were calculated for an exposure of 6.5700×10^{20} POT and BANFF-tuned MC templates were used.

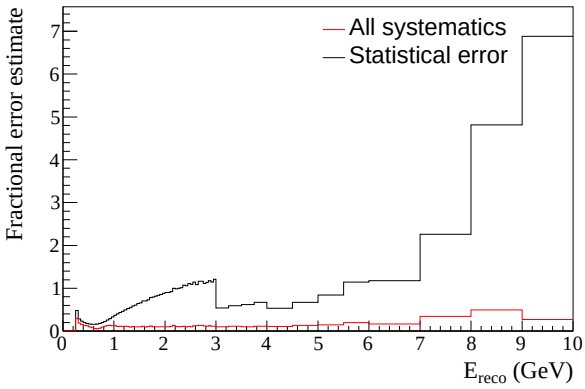


Figure 9: Comparative size of systematic and statistical error effects per bin. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. The fractional systematic error is estimated as the 1σ spread divided by the expected number of events in each bin. The fractional statistical error is estimated as the inverse square root of the number of events in each bin at 6.5700×10^{20} POT. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillation was applied.

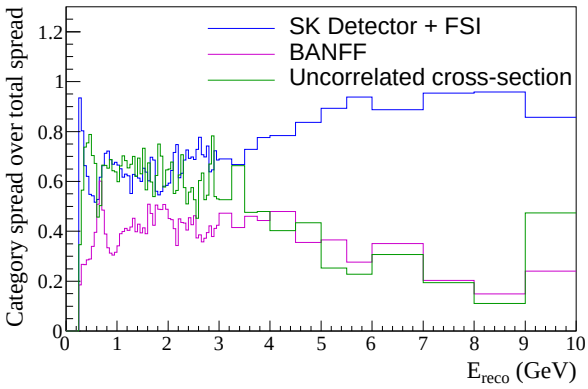


Figure 10: Ratio of event-per-bin 1σ spread with individual systematic categories randomized, to the equivalent spread with all systematics randomized. 100k toy MC experiments each were generated with different groups of systematics randomized, and with all systematics randomized, and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. No oscillation was applied.

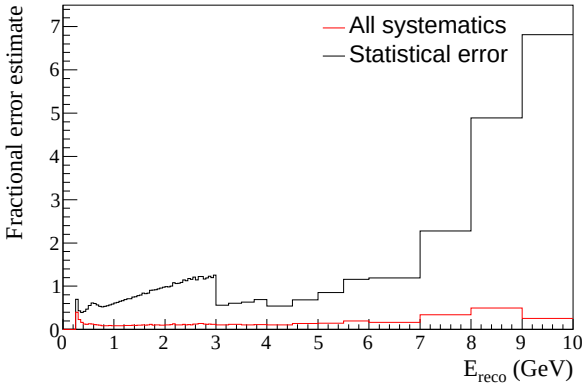


Figure 11: As in Fig. 9, but with oscillations applied for $\sin^2\theta_{23}$=0.5, $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the values shown in Tab. 10 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed.

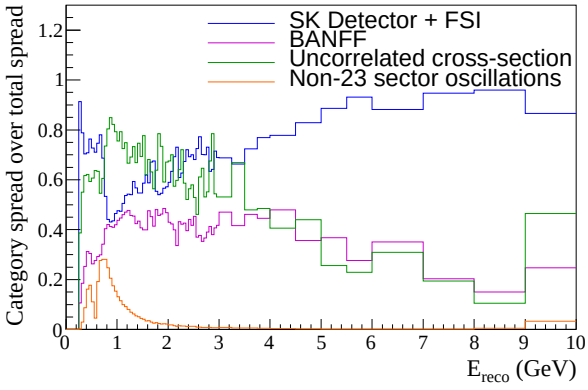


Figure 12: As in Fig. 10, but with oscillations applied for $\sin^2\theta_{23}$=0.5, $|\Delta m^2_{32}|$=2.4×10^{-3} eV^2/c^4, the values shown in Tab. 10 for $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21}, and δ_{CP}=0. The normal mass hierarchy is assumed.

Parameter	$\frac{\delta_{syst} sin^2\theta_{23}}{\delta_{stat} sin^2\theta_{23}}$	$\frac{\delta_{syst}\lvert\Delta m^2_{32}\rvert(eV^2/c^4)}{\delta_{stat}\lvert\Delta m^2_{32}\rvert(eV^2/c^4)}$
SK+FSI NC	1.1773	0.3993
Spectral Func	1.0255	0.3641
π-less Δ-decay	0.9775	0.2689
CCQE 0.0-1.5 GeV	0.9734	0.3171
SK+FSI $\nu_\mu/\bar{\nu_\mu}$ CCnQE	0.7611	0.1014
CC1π 0.0-2.5 GeV	0.7223	0.1277
Binding energy	0.7211	0.4678
ν_μ flux 0.7-1.0 GeV	0.6886	0.2540
M_AQE	0.6812	0.2096
SK energy scale	0.6529	0.2262
SK+FSI $\nu_\mu/\bar{\nu_\mu}$CCQE 0.4-1.1 GeV	0.6299	0.0681
SK+FSI $\nu_\mu/\bar{\nu_\mu}$CCQE <0.4 GeV	0.6095	0.0573
NC other norm	0.5898	0.1048
M_ARES	0.5850	0.0810
NC $1\pi^\pm$ norm	0.5830	0.1036
SK+FSI $\nu_\mu/\bar{\nu_\mu}$CCQE >1.1 GeV	0.5701	0.0674
ν_μ flux 1.0-1.5 GeV	0.5277	0.0987
ν_μ flux 0.4-0.5 GeV	0.5266	0.1880
ν_μ flux 0.0-0.4 GeV	0.4829	0.1113
ν_μ flux 0.5-0.6 GeV	0.4372	0.0991
W-shape	0.4074	0.0549
$\bar{\nu_\mu}$ flux 0.0-0.7 GeV	0.3947	0.0722
ν_μ flux 0.6-0.7 GeV	0.3907	0.0684
ν_μ flux 5.0-7.0 GeV	0.3691	0.0373
$\bar{\nu_\mu}$ flux 1.0-1.5 GeV	0.3495	0.0333
ν_μ flux 2.5-3.5 GeV	0.3484	0.0291
ν_μ flux 1.5-2.5 GeV	0.3475	0.0471
$\bar{\nu_\mu}$ flux 1.5-2.5 GeV	0.3469	0.0238
ν_μ flux 3.5-5.0 GeV	0.3450	0.0267
$\bar{\nu_\mu}$ flux 0.7-1.0 GeV	0.3438	0.0333
CC $\bar{\nu}/\nu$ norm	0.3421	0.0505
CCQE 1.5-3.5 GeV	0.3364	0.0418
ν_μ flux 7.0-30.0 GeV	0.3297	0.0312
CC Other shape	0.3283	0.0281
CC coherent norm	0.3257	0.0460
SK+FSI $\nu_e/\bar{\nu_e}$/sigν_e CC	0.3046	0.0215
CCQE 3.5-30 GeV	0.2914	0.0380
Fermi Momentum	0.2873	0.0732
$\bar{\nu_\mu}$ flux 2.5-30.0 GeV	0.2683	0.0178
CC1π 2.5-30 GeV	0.1797	0.0269
$\sin^2(\theta_{13})$	0.1339	0.0084
$\sin^2(\theta_{12})$	0.0568	0.0165
Δm^2_{21}	0.0544	0.0102
δ_{CP}	0.0383	0.1308
CC ν_e/ν_μ norm	0.0302	0.0006

Table 8: Maximum deviations, as a fraction of statistical error at the oscillation parameter grid point, from the input values of fits for only $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ to datasets with ±1 sigma individual systematic variations at a range of oscillation points. These maximum deviations usually occur at maximum disappearance, where the statistical error is smaller, as explained in Sec. 6. The values are sorted by the size of the deviation in $\sin^2\theta_{23}$.

Parameter	$\frac{\delta_{syst} sin^2\theta_{23}}{\delta_{stat} sin^2\theta_{23}}$	$\frac{\delta_{syst}\lvert\Delta m^2_{32}\rvert(eV^2/c^4)}{\delta_{stat}\lvert\Delta m^2_{32}\rvert(eV^2/c^4)}$
Binding energy	0.7211	0.4678
SK+FSI NC	1.1773	0.3993
Spectral Func	1.0255	0.3641
CCQE 0.0-1.5 GeV	0.9734	0.3171
π-less Δ-decay	0.9775	0.2689
ν_μ flux 0.7-1.0 GeV	0.6886	0.2540
SK energy scale	0.6529	0.2262
M_AQE	0.6812	0.2096
ν_μ flux 0.4-0.5 GeV	0.5266	0.1880
δ_{CP}	0.0383	0.1308
CC1π 0.0-2.5 GeV	0.7223	0.1277
ν_μ flux 0.0-0.4 GeV	0.4829	0.1113
NC other norm	0.5898	0.1048
NC $1\pi^\pm$ norm	0.5830	0.1036
SK+FSI $\nu_\mu/\bar{\nu_\mu}$ CCnQE	0.7611	0.1014
ν_μ flux 0.5-0.6 GeV	0.4372	0.0991
ν_μ flux 1.0-1.5 GeV	0.5277	0.0987
M_ARES	0.5850	0.0810
Fermi Momentum	0.2873	0.0732
$\bar{\nu_\mu}$ flux 0.0-0.7 GeV	0.3947	0.0722
ν_μ flux 0.6-0.7 GeV	0.3907	0.0684
SK+FSI $\nu_\mu/\bar{\nu_\mu}$CCQE 0.4-1.1 GeV	0.6299	0.0681
SK+FSI $\nu_\mu/\bar{\nu_\mu}$CCQE >1.1 GeV	0.5701	0.0674
SK+FSI $\nu_\mu/\bar{\nu_\mu}$CCQE <0.4 GeV	0.6095	0.0573
W-shape	0.4074	0.0549
CC $\bar{\nu}/\nu$ norm	0.3421	0.0505
ν_μ flux 1.5-2.5 GeV	0.3475	0.0471
CC coherent norm	0.3257	0.0460
CCQE 1.5-3.5 GeV	0.3364	0.0418
CCQE 3.5-30 GeV	0.2914	0.0380
ν_μ flux 5.0-7.0 GeV	0.3691	0.0373
$\bar{\nu_\mu}$ flux 1.0-1.5 GeV	0.3495	0.0333
$\bar{\nu_\mu}$ flux 0.7-1.0 GeV	0.3438	0.0333
ν_μ flux 7.0-30.0 GeV	0.3297	0.0312
ν_μ flux 2.5-3.5 GeV	0.3484	0.0291
CC Other shape	0.3283	0.0281
CC1π 2.5-30 GeV	0.1797	0.0269
ν_μ flux 3.5-5.0 GeV	0.3450	0.0267
$\bar{\nu_\mu}$ flux 1.5-2.5 GeV	0.3469	0.0238
SK+FSI $\nu_e/\bar{\nu_e}$/sigν_e CC	0.3046	0.0215
$\bar{\nu_\mu}$ flux 2.5-30.0 GeV	0.2683	0.0178
$\sin^2(\theta_{12})$	0.0568	0.0165
Δm^2_{21}	0.0544	0.0102
$\sin^2(\theta_{13})$	0.1339	0.0084
CC ν_e/ν_μ norm	0.0302	0.0006

Table 9: Maximum deviations, as a fraction of statistical error at the oscillation parameter grid point, from the input values of fits for only $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ to datasets with ±1 sigma individual systematic variations at a range of oscillation points. These maximum deviations usually occur at maximum disappearance, where the statistical error is smaller, as explained in Sec. 6. The values are sorted by the size of the deviation in $|\Delta m^2_{32}|$.

5. Neutrino oscillation fitting

5.1. Fit method

Several choices exist for the 23-sector oscillation parameters that could be fitted, and these choices have been presented in the context of the VaLOR joint 3-flavour fit of the 1 μ-like ring and 1 e-like ring reconstructed SuperK spectra [28]. During the review process of the T2K Run 1+2+3 ν_μ-disappearance PRL [9], it was discovered that the choice of octant ($\theta_{23} \leq \pi/4$ or $\theta_{23} \geq \pi/4$) has a large effect on the confidence contour in $\sin^2(2\theta_{23})$. Therefore, we now fit $\sin^2\theta_{23}$ by default. This is a sensible choice since the value of $\sin^2\theta_{23}$ giving maximal disappearance is not the same as the value giving maximal mixing ($\sin^2\theta_{23} = 0.5$); this difference is due to the non-zero value of θ_{13}. There are also different choices for the mass-squared splitting parameter: typically $|\Delta m^2_{32}|$ is used in ν_μ-disappearance analyses, but this is the largest mass-squared splitting for the inverted mass hierarchy and the second largest for the normal mass hierarchy. It is therefore not correct to compare results obtained under the different mass hierarchy hypotheses. Other choices include fitting a mass-squared difference whose absolute value is the same for both mass hierarchies, such as $\Delta m^2 = \left(m^2_3 - \frac{m^2_1+m^2_2}{2}\right)$ [41], or fitting $|\Delta m^2_{32}|$ for one hierarchy and $|\Delta m^2_{31}|$ for the other [42]. All these choices are implemented in the VaLOR oscillation analysis framework, and our analysis can switch between them. For the present analysis, it was decided to fit $|\Delta m^2_{32}|$ for the normal hierarchy and $|\Delta m^2_{31}|$ for the inverted hierarchy. The actual inputs to the calculation of the oscillation probabilities are $+|\Delta m^2_{32}|$ for the normal hierarchy and $-|\Delta m^2_{31}|$ for the inverted hierarchy. Fits of $|\Delta m^2_{32}|$ for the inverted hierarchy were also performed for validation.

The VaLOR disappearance analysis uses a binned likelihood-ratio method. Measurements of the oscillation parameters $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ are obtained by comparing the observed and predicted SuperK reconstructed energy spectra for 1-ring μ-like events.

Let N be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the SuperK reconstructed energy spectrum prediction (e.g parameters that control the beam, neutrino interaction simulations and the detector response). Best-fit values are obtained by minimizing:

$$-2\ln\lambda(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp} - n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \quad (5)$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematics parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp} = n_i^{exp}(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a})$ is the corresponding expected number of events.

The likelihood-ratio method used in this analysis is equivalent to the extended maximum likelihood method used in the alternative T2K oscillation analysis (see [1, 30] for more details). The advantage of the likelihood ratio method, compared with the extended maximum-likelihood method, is that in the large-sample limit, the quantity $-2\ln\lambda(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a})$ in Eq.5 has a χ^2 distribution and it can therefore be used as a goodness-of-fit test. In this note, $-2\ln\lambda(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a})$ and $\chi^2(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a})$ are used interchangeably.

The minimization is performed with Minuit [31], using the Migrad method. If Migrad does not converge, the starting values of the oscillation parameters are re-initialised: the new initial values are randomly drawn from uniform distributions in the ranges 1.5×10^{-3} eV^2/c^4 $\leq |\Delta m^2_{32}|$ or $|\Delta m^2_{31}| \leq 3.5 \times 10^{-3}$ eV^2/c^4 and either $0.35 \leq \sin^2\theta_{23} \leq 0.50$ or $0.53 \leq \sin^2\theta_{23} \leq 0.68$ (the new initial value of $\sin^2\theta_{23}$ is in the same octant as the previous initial value). The fit is retried with the new initial values, and this is repeated up to 10 times. If the fit still fails to converge after 10 attempts, the Simplex algorithm is used. The search for the minimum value of $-2\ln\lambda(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a})$ (see Eq. 5) is performed in the range $0 \leq \sin^2\theta_{23} \leq 1$ and 1×10^{-3} eV^2/c^4 $\leq |\Delta m^2_{32}|$ or $|\Delta m^2_{31} \leq 6 \times 10^{-3}$ eV^2/c^4. After a successful fit, The Hesse algorithm is called to improve Minuit's estimates of the errors on the fitted parameters.

Each fit is repeated twice: in the first fit, the initial values are $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}| = 2.4 \times 10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.6$, while the second fit uses as initial values the best fit of $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ and the mirror point in the other octant of the best fit of $\sin^2\theta_{23}$ from the first fit.

Parameter	Best-fit	1σ range
Δm^2_{21}	7.5×10^{-5} eV^2/c^4	$\pm \sim$2.7%
$\sin^2(\theta_{12})$	0.306	$\pm \sim$5.9%
$\sin^2(\theta_{13})$	0.0251	$\pm \sim$13.9%
δ_{CP}	-	[-π,+π]

Table 10: The best-fit values and 1σ range for the measured *non-23-sector* oscillation parameters [27, 41].

All the systematic parameters are allowed to float in the fit, and are restricted to the range [-$5\sigma_s$, +$5\sigma_s$] where σ_s is the one standard deviation error assigned to each systematic parameter[3]. Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin are not allowed. If this scenario arises, the number of events is changed to +1E-8 in that bin. This is done for the sum of all the interaction modes rather than for each mode separately.

In a 3-flavour ν_μ-disappearance fit, it is also necessary to make a choice for the treatment of the non-23-sector oscillation parameters. In the current analysis, we allow each of $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{12})$, Δm^2_{21} and δ_{CP} to float. We use penalty terms for $\sin^2(2\theta_{13})$, $\sin^2(2\theta_{12})$, Δm^2_{21}, and their central values and uncertainties are shown in Tab. 10; these are taken from the 2012 PDG [27] for $\sin^2\theta_{13}$ and Δm^2_{21} and from a global fit [41] for $\sin^2\theta_{12}$ in order to be consistent with the latest ν_e-appearance fits [25, 24]. There is no penalty term for δ_{CP}, and it is allowed to float freely in the range $[-\pi, +\pi]$[4].

5.2. Goodness-of-fit tests

In our ν_μ-disappearance analysis of the Run1+2 datasets, p-values were computed using both a) the rate of single μ-like ring events and b) the rate and reconstructed energy distribution of single μ-like ring events using the likelihood-ratio method [34]. The choice is not unique.

D. Karlen has pointed out that, if the statistics are low, "the use of the distribution is not helpful in forming the p-value. This is directly related to the well known problem that is difficult to define a general goodness-of-fit for sparsely populated distributions. Large bins (or in the extreme a single bin = rate-only) are necessary for small statistics samples" [35].

It was decided by the oscillation analysis group that, for all ν_μ-disappearance analyses, p-values will be calculated using the rate and the reconstructed energy distribution, and that the distribution will use a very coarse binning scheme to ensure that there are sufficient numbers of events in each bin: this scheme has 5 bins from 0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV.

5.3. Construction of confidence regions

In this analysis, uncertainties in the jointly measured parameters are quoted using both the constant-$\Delta\chi^2$ [27] (pp. 390-401) and the Feldman-Cousins [32] methods. A definite recommendation was made in favour of the Feldman and Cousins method for constructing confidence regions by the T2K ν_μ contour committee [36]. The work of Feldman and Cousins uses the freedom which is inherent in the Neyman construction of confidence regions to address cases (in particular, cases with Poisson processes with background and Gaussian errors near a physical boundary) where the classical construction of confidence regions produces overcoverage, undercoverage, or in extreme cases, an empty region.

The two methods have been described in detail in past T2K technical notes and presentations. For completeness, we summarise both of these methods here and provide definite statements on the exact methodology used for incorporating systematic errors.

The methods differ in the calculation of the critical values of $\Delta\chi^2$, which are used to identify the areas of the parameter space to be included in the allowed region for a given confidence level. The constant-$\Delta\chi^2$

[3] Such an error is defined for all but one of the systematic parameters considered in this analysis. The exception is the systematic parameters f_{SF} which parameterizes the uncertainty on nuclear modelling, switching between the Relativistic Fermi Gas (RFG) and Spectral Function (SF) models. The allowed range of the parameter f_{SF} in the oscillation fit is [0,1].

[4] The *non-23-sector* oscillation parameters are allowed to float in the fit in the following ranges: $\sin^2\theta_{12}$ and $\sin^2\theta_{13}$ in the range [0,1]; Δm^2_{21} in the range [6,9] $\times 10^{-3} eV^2/c^4$; and δ_{CP} in the range $[-\pi, +\pi]$.

method is a fully frequentist treatment using the Gaussian approximation. One finds the best-fit point and calculates the confidence contours as lines of *constant* $\Delta\chi^2$ from this point[5].

The method of Feldman and Cousins method finds the critical values of $\Delta\chi^2$ as follows: many toy MC experiment fits are performed for each point of the ($\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$) 2-dimensional grid. The value of $\Delta\chi^2_{MC}$ for each toy MC experiment is calculated as

$$\Delta\chi^2_{MC} = \chi^2(\text{MC, true}) - \chi^2(\text{MC, best fit}) \tag{6}$$

where "true" means that the χ^2 was computed using the true values of the oscillation parameters at that grid point, and "best-fit" means that the χ^2 was computed using the actual best-fit values for that MC experiment. The χ^2 values are calculated using Eq. 5. For each grid point, the critical value of $\Delta\chi^2$ ($\Delta\chi^2_{critical}$) is found such that α of the toy MC experiments have $\Delta\chi^2_{MC} < \Delta\chi^2_{critical}$, where α is the desired confidence level. The principal difference between the Feldman and Cousins and the constant $\Delta\chi^2$ methods is that the former determines the value of $\Delta\chi^2_{critical}$ individually for each grid point, whereas the latter uses the same value for all grid points.

The procedure for constructing the $\Delta\chi^2(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|)$ surface itself from the T2K Run 1+2+3+4 data is common to both methods. Assuming that N nuisance parameters are allowed to float in the oscillation fit, the values of $\chi^2(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a})$ are calculated via a fit at *each* point of the ($\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$) 2-dimensional grid, where only the nuisance parameters are allowed to float and $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ are fixed to the true values for the given grid point. The following grid spacing is used in all confidence region construction methods: $\delta\sin^2\theta_{23} = 5\times10^{-3}$ and $\delta|\Delta m^2_{32}|$ or $\delta|\Delta m^2_{31}| = 5\times10^{-5}$ eV^2/c^4.

The allowed region in the 2D oscillation parameter space is then given by

$$\Delta\chi^2(\sin^2\theta_{23}, |\Delta m^2_{32}|/|\Delta m^2_{31}|) = \chi^2(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a}) - \chi^2_{min} < \Delta\chi^2{}_{critical} \tag{7}$$

where χ^2_{min} is the minimum value of $\chi^2(\sin^2\theta_{23}, |\Delta m^2_{32}| \text{ or } |\Delta m^2_{31}|; \mathbf{a})$ from the fit of the T2K Run 1+2+3+4 data over the oscillation parameter space considered.

The Feldman and Cousins method does not specify how systematics are to be included in the confidence region construction and different choices exist. To construct a 2-dimensional ($\sin^2 2\theta_{23}$, $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$) confidence region, where N systematic parameters are included in the p.d.f. and all of these N systematics are allowed to float in the fit, this analysis uses the following methodology: the value of $\Delta\chi^2_{critical}$ at each true ($\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$) grid point is calculated from an ensemble of toy MC experiments, each one of which is generated with the true oscillation parameters of that point and with all N systematics randomised. For each toy MC experiment, both $\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ and all N nuisance parameters are fitted simultaneously to obtain the value of χ^2(MC, best fit) used in Eq. 6. A separate fit, with $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ fixed to their true values and with all N nuisance parameters allowed to float, is performed to obtain the value of χ^2(MC, true) (i.e. χ^2 is *independently* minimised, for *each* toy MC experiment, with respect to the nuisance parameters).

Confidence intervals are also given for each of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ separately (this is done only for the normal mass hierarchy). These can be found in a straightforward way by calculating the 1D profile likelihood for each parameter, and finding the values of each parameter corresponding to the constant $\Delta\chi^2$ for one parameter and the desired confidence level (e.g. $\Delta\chi^2 = 1.0$ for 68% C.L.). Unfortunately, however, there is no standard procedure in the literature for constructing 1D Feldman-Cousins confidence intervals from 2D confidence regions. To find the 1D intervals, values of $\Delta\chi^2{}_{critical}$ need to be computed, as a function of the corresponding oscillation parameter, using an ensemble of toy MC experiments. One issue concerns the choice of the true input value(s) of the "other" oscillation parameter in those experiments. For example, the true input values of $|\Delta m^2_{32}|$ must be chosen for the toy MC experiment ensemble used for constructing the confidence interval in $\sin^2\theta_{23}$. One option in this case would be to use the T2K Run 1+2+3+4 best-fit value of $|\Delta m^2_{32}|$ as the true input value for all the toy MC experiments, but this fails to account for the uncertainty

[5] For 1 fit parameter, the critical values of $\Delta\chi^2$ used are 1.00 (68% CL), 2.71 (90% CL), 3.84 (95% CL), 6.63 (99% CL). For 2 parameters fitted simultaneously, the critical values of $\Delta\chi^2$ used are 2.30 (68% CL), 4.61 (90% CL), 5.99 (95% CL), 9.21 (99% CL).

in $|\Delta m^2_{32}|$. Alternatively, a uniform distribution of true input values of $|\Delta m^2_{32}|$ could be used in some range, say from 2-3$\times 10^{-3}$ eV^2/c^4, but this ignores the fact that some values of $|\Delta m^2_{32}|$ are preferred by the T2K data to others. In this analysis, the true input values of $|\Delta m^2_{32}|$ are chosen using the likelihood from the fit of the T2K Run 1+2+3+4 data. At each true value of $\sin^2\theta_{23}$, the numbers of toy MC experiments with a given true input value of $|\Delta m^2_{32}|$ are proportional to $e^{-\frac{\Delta\chi^2}{2}}$, where $\Delta\chi^2$ is the value of the 2-dimensional $\Delta\chi^2$ surface at those values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ from the fit of the T2K Run 1+2+3+4 data. Each toy MC experiment is fitted twice; in one fit, $\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ and all the nuisance parameters are allowed to float, and, in the other, $|\Delta m^2_{32}|$ and the nuisance parameters are allowed to float while $\sin^2\theta_{23}$ is fixed to its true value. For each experiment, $\Delta\chi^2_{MC}$ is calculated as the difference between the minimum values of χ^2 from the two fits. The critical value of $\Delta\chi^2$ at each true value of $\sin^2\theta_{23}$ is found by combining all the toy MC experiments with that input value of $\sin^2\theta_{23}$ and different true input values of $|\Delta m^2_{32}|$. It is the value of $\Delta\chi^2$ such that α of the toy experiments have $\Delta\chi^2_{MC} < \Delta\chi^2_{critical}$, where α is the desired confidence level. The 1-dimensional Feldman and Cousins confidence intervals in $|\Delta m^2_{32}|$ are constructed in an analogous way.

The Feldman-Cousins 2D contours and 1D intervals are being prepared, and will be shown in a later version of this note.

6. Validation of the *VaLOR* Run 1+2+3+4 ν_μ-disappearance analysis

The performance of the fitter was evaluated using ensembles of 5.1k toy MC experiments, and these were generated with statistical fluctuations and variations of all 41 systematic parameters and the oscillation parameters $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, Δm^2_{21} and δ_{CP}. Ensembles of experiments were generated for different true input values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, and for both 6.5700×10^{20} POT (Run 1+2+3+4) and 7.8×10^{21} POT (the ultimate T2K goal) These toy MC experiments were fitted using MINUIT and the MINUIT HESSE errors [6] were computed.

Distributions of fitter residuals for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ are shown in Figs. 13 and 14 for true input values $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.650$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4 respectively, for both 6.5700×10^{20} POT and 7.8×10^{21} POT. A clear three-peak structure is seen at Run 1+2+3+4 POT in the distribution of residuals of $\sin^2\theta_{23}$, with peaks occurring at the input value, the mirror of the input value in the other octant, and the maximum disappearance value. The peak at the mirror value in the other octant occurs because $P(\nu_\mu \rightarrow \nu_\mu)$ is almost identical for the input and mirror values (note that the centre of symmetry of $\sin^2\theta_{23}$ is at $\approx \frac{1}{2(1-\sin^2(\theta_{13}))} = 0.513$ for $\sin^2\theta_{13} = 0.0251$ for reasons given in Appendix E in [6]). The peak at maximal disappearance (which is also at $\sin^2\theta_{23} \approx 0.513$) is due to downward statistical fluctuations in the numbers of events in the bins around the oscillation maximum in some of the toy datasets. When these datasets are fitted, the best fit of $\sin^2\theta_{23}$ is the value giving the maximum physically-allowed disappearance, i.e. 0.513. This peak is not seen for 7.8×10^{21} POT since downward statistical fluctuations have a smaller effect on the toy datasets at higher statistics.

Distributions of fitter residuals for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ are shown in Fig. 15 for true input values $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4 at 6.5700×10^{20} POT and 7.8×10^{21} POT. The distribution for $\sin^2\theta_{23}$ has a large peak at 0.513 as expected, but also two smaller peaks with one peak in each octant. This can be explained by considering the effects of statistical fluctuations in the bins near the oscillation maximum in the toy datasets. If there are no statistical fluctuations in these bins, the fitted value of $\sin^2\theta_{23}$ is the same as the input value of 0.513, and downward statistical fluctuations also result in a fitted value of 0.513. However upward statistical fluctuations in these bins in some datasets mean that ν_μ-disappearance is less than maximal, and this results in a fitted value of $\sin^2\theta_{23}$ that is different from the input value of 0.513.

The non-maximal values of $\sin^2\theta_{23}$ returned by the fitter can be quantified by considering the ratio of the number of events in a single bin at the lowest part of the oscillation dip with oscillations to that without oscillations. Examples of this ratio are shown in Tab. 11 for 6.5700×10^{20} POT (run 1+2+3+4) and 7.8×10^{21} POT (the ultimate T2K goal). As shown in the table, the ratio is 2.672 / 37.93 = 0.070 at 6.5700×10^{20} POT for a dataset without statistical fluctuations. The smallest upward statistical fluctuation that can be made is to 3 events, and this increases the ratio to 0.079; this is the ratio that would be obtained in a dataset made with an input value of $\sin^2\theta_{23}$ of 0.463 (first octant) or 0.565 (second octant). These input values are close to the values of the non-maximal peaks in the fitted values of $\sin^2\theta_{23}$ as shown by the Gaussian fits in Fig. 16. At 7.8×10^{21} POT, it is necessary to consider upward statistical fluctuations from 31.545 to 32, 33 and 34 events in order to obtain a ratio of number of events with oscillations to that without oscillations corresponding to the values of the non-maximal peaks in the fitted values of $\sin^2\theta_{23}$. Such upward fluctuations are to be expected as shown by the distribution of 100K Poisson fluctuations from a mean of 31.545 in the right plot in Fig. 18.

Distributions of pulls of the 41 systematic parameters and four non-23 oscillation parameters are shown in Figs. 19, 22 and 25, for true input values of $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, $\sin^2\theta_{23} = 0.650$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, and $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4 respectively. In these figures, the pulls are calculated as $(f_{bestfit} - f_{true})/\sigma_{fit}$. For most systematics, the distributions are approximately Gaussian with mean 0, width 1.

A summary of the pull mean, RMS, skewness, kurtosis, and the p-value from a Kolmogorov-Smirnov (KS) test comparing the pull distribution with a Gaussian of mean 0 and width 1 is shown in Figs. 20, 23 and 26,

[6] MINOS errors were computed for the SK energy scale systematic because the number of events in a bin is not a smooth function of $f^{SK}_{E;r}$. Tweaking this systematic involves migrating events to and from neighbouring bins, and, although the number of events itself is a continuous function of $f^{SK}_{E;r}$, there is a discontinuity in the differential when moving from positive to negative values of $f^{SK}_{E;r}$. There is also a discontinuity at large values of $f^{SK}_{E;r}$ since events are also migrated to and from the next-to-neighbouring bins. For these reasons, HESSE has a problem evaluating errors for this systematic parameter.

for input values of $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4, $\sin^2\theta_{23} = 0.650$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4, and $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4 respectively. Most systematic distributions are Gaussian, with mean ≈ 0, RMS ≈ 1, and skewness ≈ 0. The kurtosis has some spread, due to peaks forming from the limited statistics of the study. The KS test also suffers from limited statistics.

No problems were identified with the fitter, which performed as expected. There are some small biases in the pull distributions for some systematic parameters, but these biases are understood.

- **δ_{CP}.** This systematic is thrown uniformly between $-\pi$ and $+\pi$ (rather than with a Gaussian distribution), is constrained between $-\pi$ and $+\pi$ in the fit, and has no penalty term. Therefore a Gaussian distribution is not expected.

- **Spectral function.** This systematic is limited to the range [0,1], both for toy experiment generation, and in the fit. It is constrained with a Gaussian constraint. Therefore a Gaussian distribution with mean 0, width 1 is not expected.

- **π-less Δ decay.** The bias is due to the method of calculation of event weights when this parameter is varied. By default, 20% of Δs decay in this way in NEUT, and the uncertainty is 20%. This means that it is unphysical to tweak this parameter to be more negative than -1σ, and consequently there is an asymmetry in weights between positive tweaks (which change continuously as the tweak increases) and negative tweaks (for which all tweaks more negative than -1σ are assigned the same weight as that for a -1σ tweak). This leads to a small negative bias.

- **SuperK energy scale.** This systematic is different from the others in that it migrates events from bin to bin instead of increasing or decreasing the numbers of events in the bins of reconstructed energy. Its pull is not expected to be a Gaussian with zero mean since the shape of the predicted 1 μ-like ring energy spectrum is not symmetric, and, consequently, a positive tweak of this parameter has a different effect on that spectrum from a negative tweak (please see figure 46 in the supplement to this technical note).

- **NC SuperK detector efficiency + FSI + SI.** The bias comes from the low statistics at current POT. When Poisson fluctuations are made of the bin contents of the E_{reco} distributions, a downward fluctuation is more likely than an upward fluctuation. For example, a Poisson distribution with mean 3 has P(1) = 0.149 and P(2) = 0.224, whereas P(4) = 0.168 and P(5) = 0.101. The NC events are mainly around the oscillation maximum where there are few events, and downward Poisson fluctuations are more likely than upward ones. Fitting these downward fluctuations involves moving this parameter down from zero, and this explains the negative bias in its pull. When moving away from the point of maximal disappearance as input (and so reducing the depth of the oscillation dip), the NC bin in the SuperK detector efficiency + FSI + SI matrix bias is reduced.

The pull distributions for these 5 systematics are shown in Figs. 21, 24 and 27, for input values of $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4, $\sin^2\theta_{23} = 0.650$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4, and $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4 respectively.

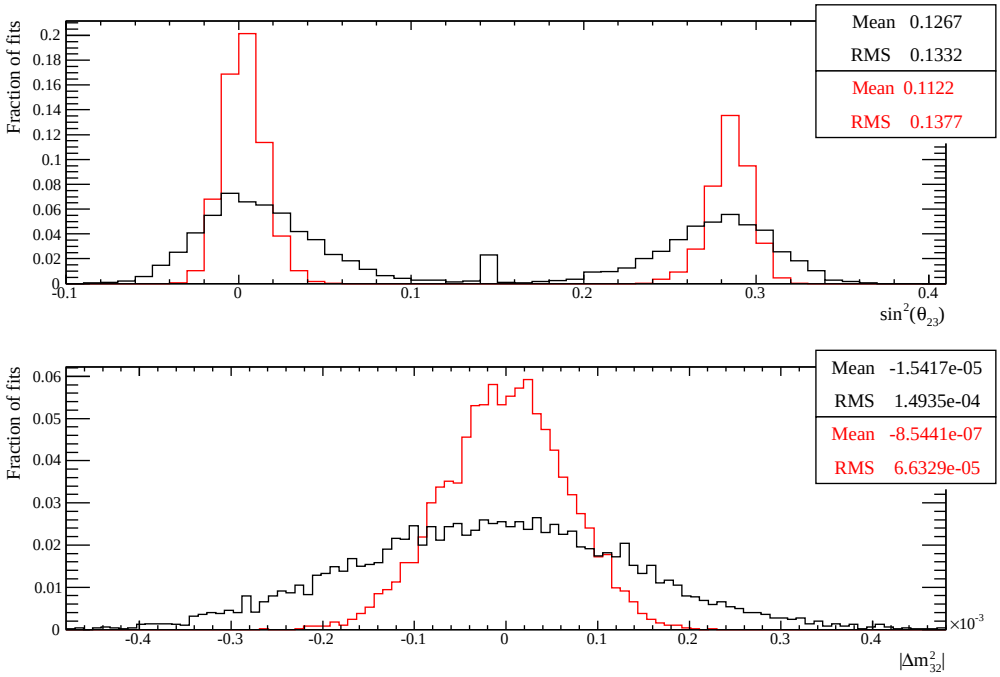


Figure 13: Distributions of fitter residuals for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for 5.1k toy MC experiments, generated at $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic and non-23 oscillation parameters. The distribution is shown for both 6.5700×10^{20} POT (Run 1+2+3+4, black histogram) and 7.8×10^{21} POT (the T2K ultimate goal, red histogram).

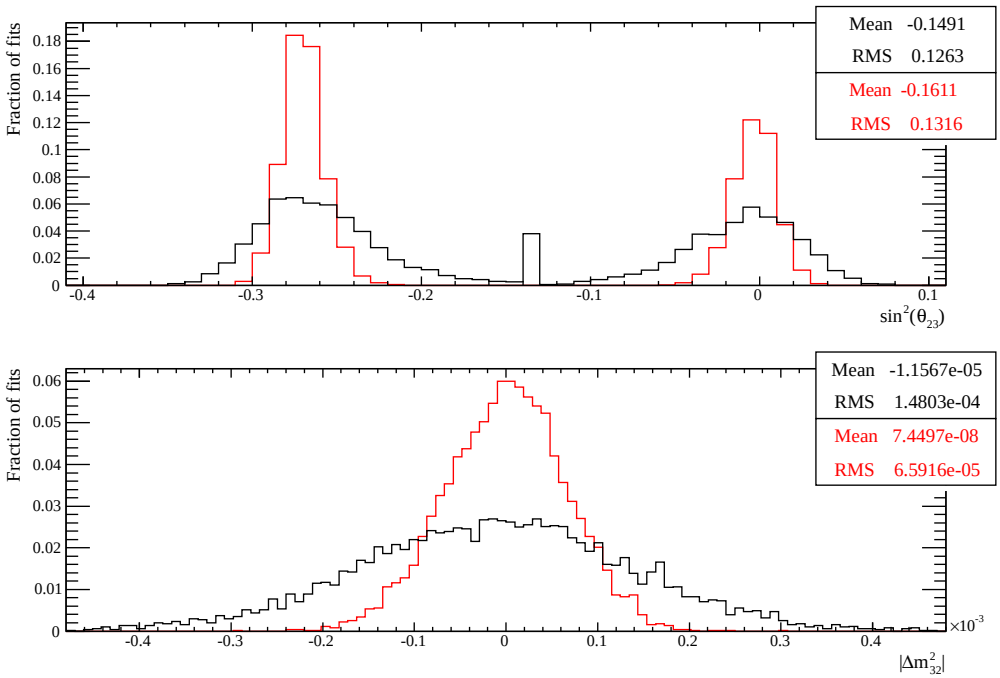


Figure 14: Distributions of fitter residuals for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for 5.1k toy MC experiments, generated at $\sin^2\theta_{23} = 0.650$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic and non-23 oscillation parameters. The distribution is shown for both 6.5700×10^{20} POT (Run 1+2+3+4, black histogram) and 7.8×10^{21} POT (the T2K ultimate goal, red histogram).

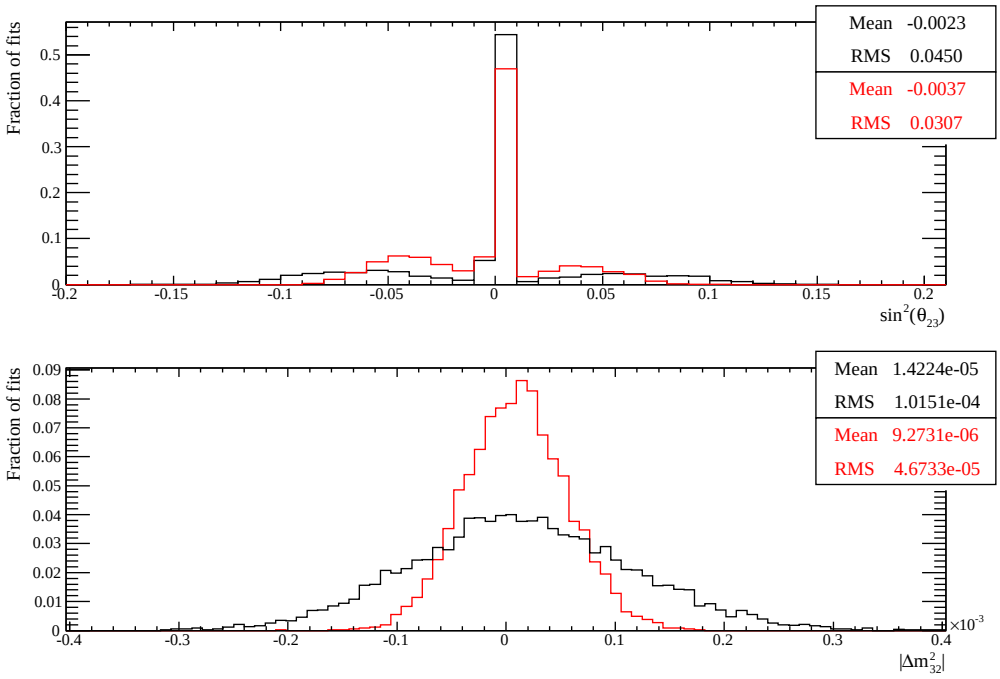


Figure 15: Distributions of fitter residuals for $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for 5.1k toy MC experiments, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic and non-23 oscillation parameters. The distribution is shown for both 6.5700×10^{20} POT (Run 1+2+3+4, black histogram) and 7.8×10^{21} POT (the T2K ultimate goal, red histogram).

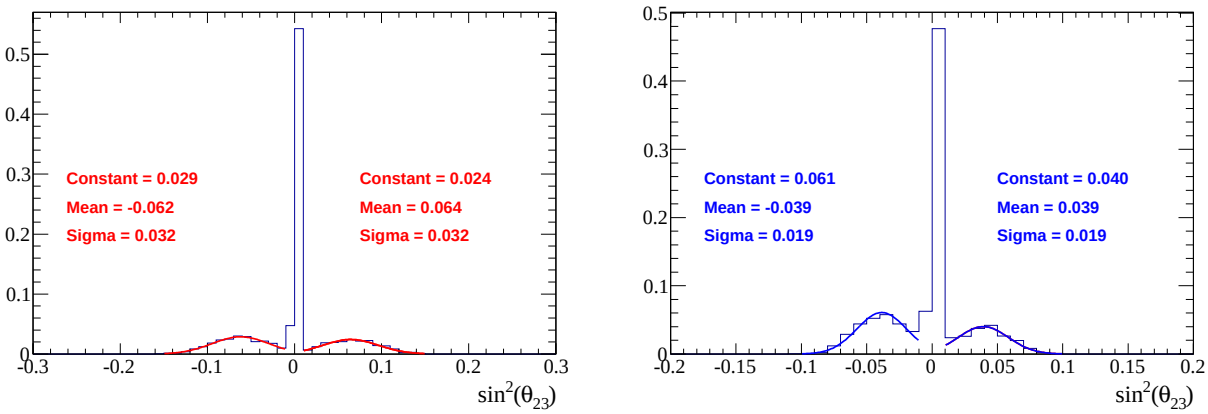


Figure 16: Distributions of fitter residuals for $\sin^2\theta_{23}$ for 5.1k toy MC experiments, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic and non-23 oscillation parameters, stressing the three-peak structure and including a Gaussian fit for the extra peaks at non-maximal disappearance. The distribution is shown for both 6.5700×10^{20} POT (Run 1+2+3+4, left plot) and 7.8×10^{21} POT (the T2K ultimate goal, right plot).

POT		Number of events	Ratio osc/ no osc	Input value of $\sin^2\theta_{23}$ giving this ratio osc/no osc		Peak of non-maximal fitted values of $\sin^2\theta_{23}$	
				(Octant 1)	(Octant 2)	(Octant 1)	(Octant 2)
6.57×10^{20}	No oscillations	37.934	1			0.451	0.577
	$\sin^2\theta_{23} = 0.513$ (no fluctuation)	2.672	0.070				
	$\sin^2\theta_{23} = 0.513$ (fluctuation 1)	3	0.079	0.463	0.565		
7.8×10^{21}	No oscillations	450.352	1			0.474	0.552
	$\sin^2\theta_{23} = 0.513$ (no fluctuation)	31.545	0.070				
	$\sin^2\theta_{23} = 0.513$ (fluctuation 1)	32	0.071	0.497	0.531		
	$\sin^2\theta_{23} = 0.513$ (fluctuation 2)	33	0.073	0.484	0.544		
	$\sin^2\theta_{23} = 0.513$ (fluctuation 3)	34	0.075	0.474	0.553		
	$\sin^2\theta_{23} = 0.513$ (fluctuation 4)	35	0.078	0.467	0.561		
	$\sin^2\theta_{23} = 0.513$ (fluctuation 5)	36	0.080	0.461	0.566		

Table 11: Numbers of events in a single bin at the lowest part of the oscillation dip with oscillations and ratio to number without oscillations for a true input value $\sin^2\theta_{23} = 0.513$ at 6.57×10^{20} POT (Run 1+2+3+4) and 7.8×10^{21} POT (the ultimate T2K goal). These numbers are shown without statistical fluctuations and with different upward fluctuations. Also shown are the input values of $\sin^2\theta_{23}$ to toy datasets from Fig. 17 matching the ratio of number of events with oscillations to number without oscillations. These values of $\sin^2\theta_{23}$ can be compared with the peaks of the non-maximal fitted values of $\sin^2\theta_{23}$ from the Gaussian fits in Fig. 16.

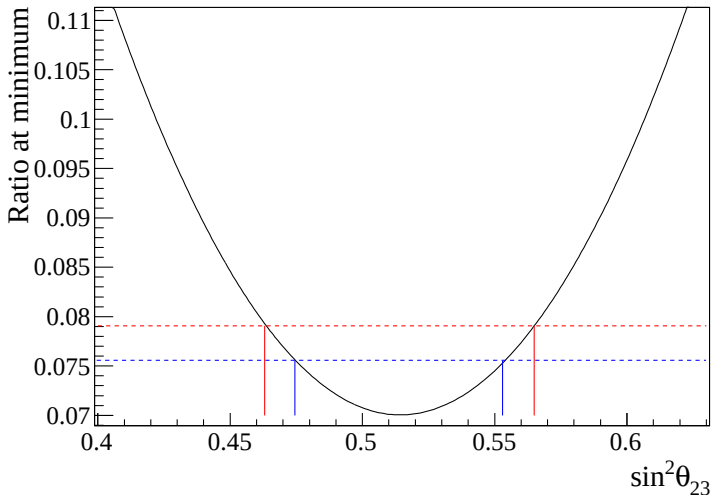


Figure 17: Ratio of number of events in a single bin at the lowest part of the oscillation dip for toy datasets created with different input values of $\sin^2\theta_{23}$ to number without oscillations against the input value of $\sin^2\theta_{23}$. Red (blue) lines show the value corresponding to the ratio found at 6.5700×10^{20} POT (7.8×10^{21} POT, the T2K ultimate POT goal) in Fig. 16.

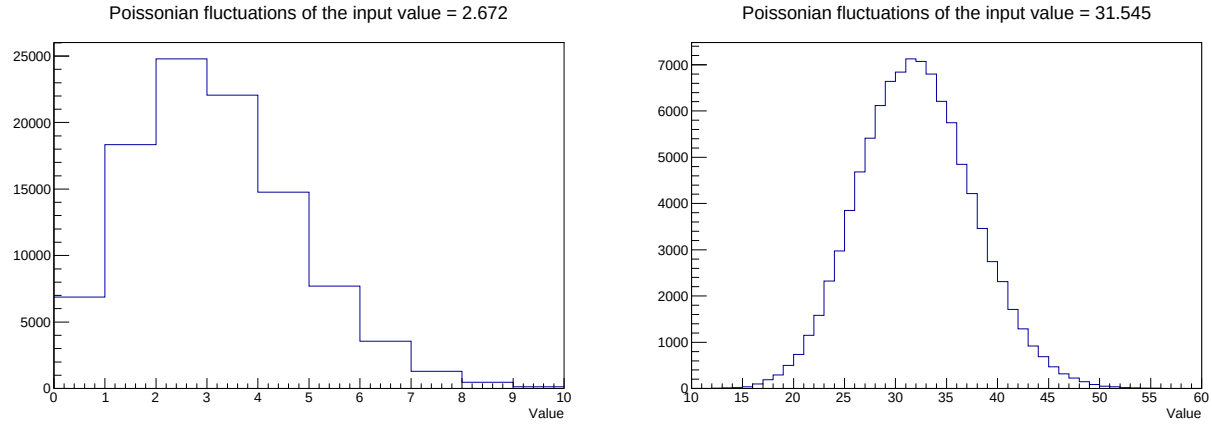


Figure 18: Distributions of 100k Poisson fluctuations of the number of events in a single bin at the lowest part of the oscillation dip for an input value of $\sin^2\theta_{23} = 0.513$ These distributions are shown for means of 2.672 events (for 6.5700×10^{20} POT, left plot) and 31.545 events (for 7.8×10^{21} POT, the T2K ultimate POT goal, right plot).

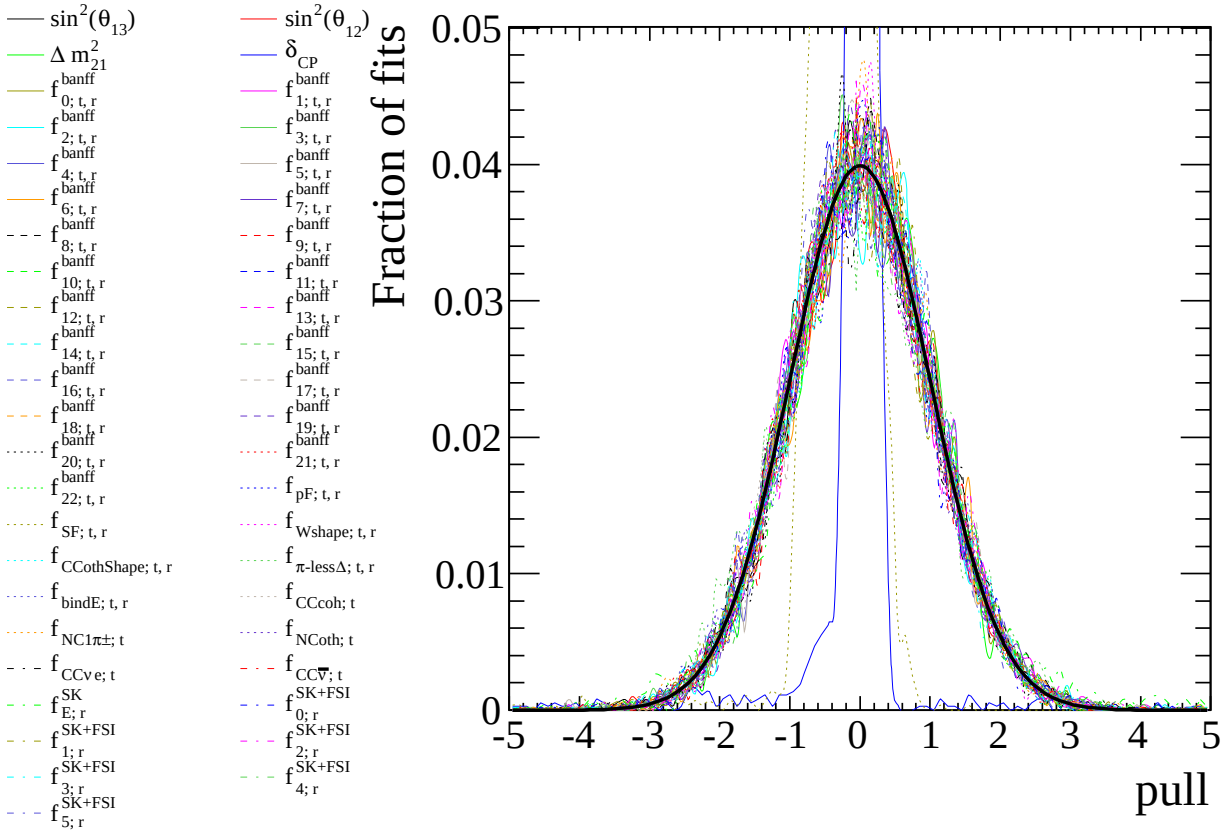


Figure 19: Systematic pull distributions for all systematic parameters for 5.1k toy MC experiments at 6.5700×10^{20} POT generated at $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. A Gaussian distribution of mean 0, width 1 is shown for reference.

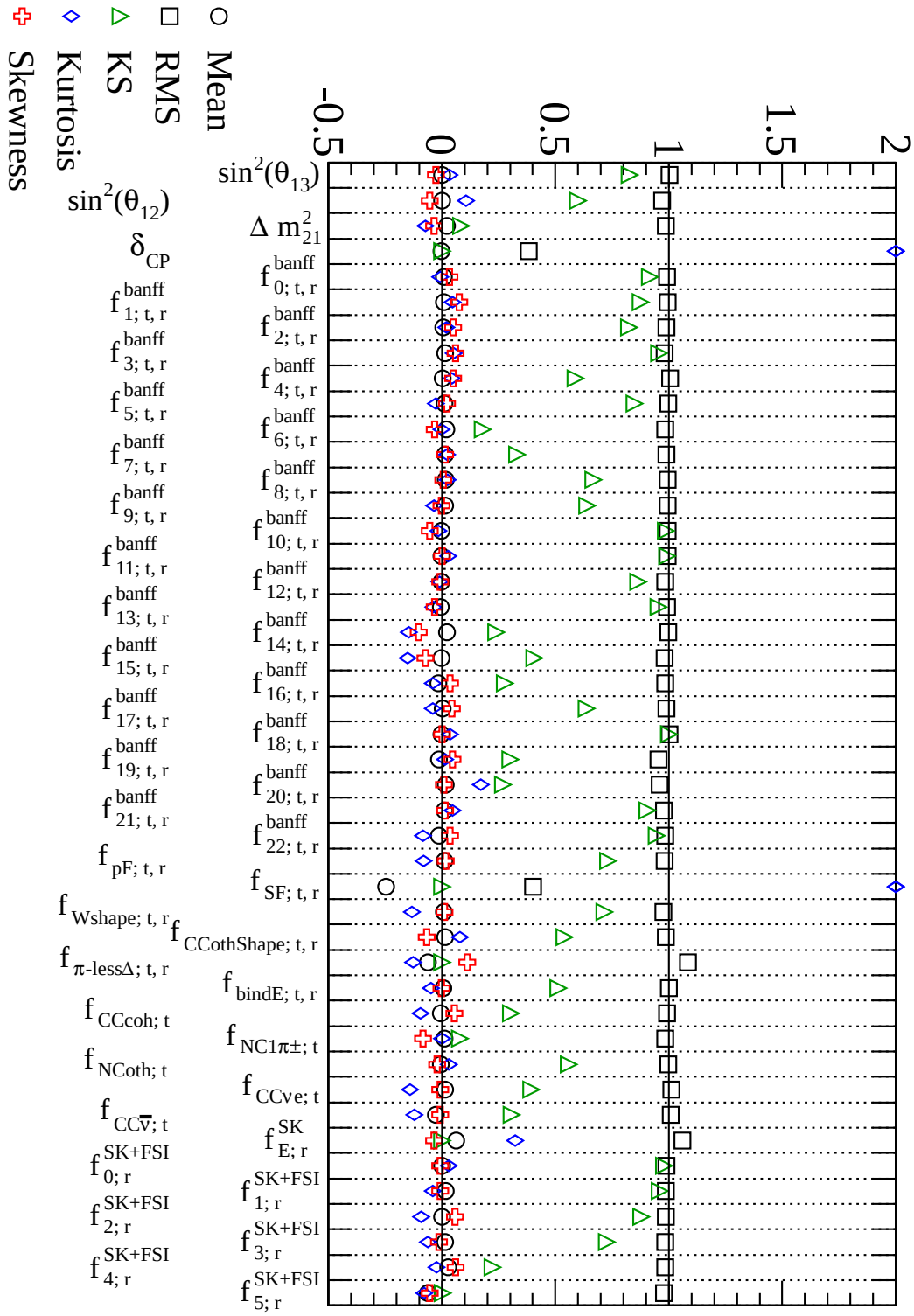


Figure 20: Summary of systematic pull distributions for all systematic parameters for 5.1k toy MC experiments at 6.5700×10^{20} POT generated at $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV$^2/c^4$, including statistical fluctuations and with randomized systematic parameters. The tests shown are the mean, RMS, skewness, kurtosis and the p-value from a Kolmogorov-Smirnov test comparing the pull distribution with a normal distribution of mean 0, width 1.

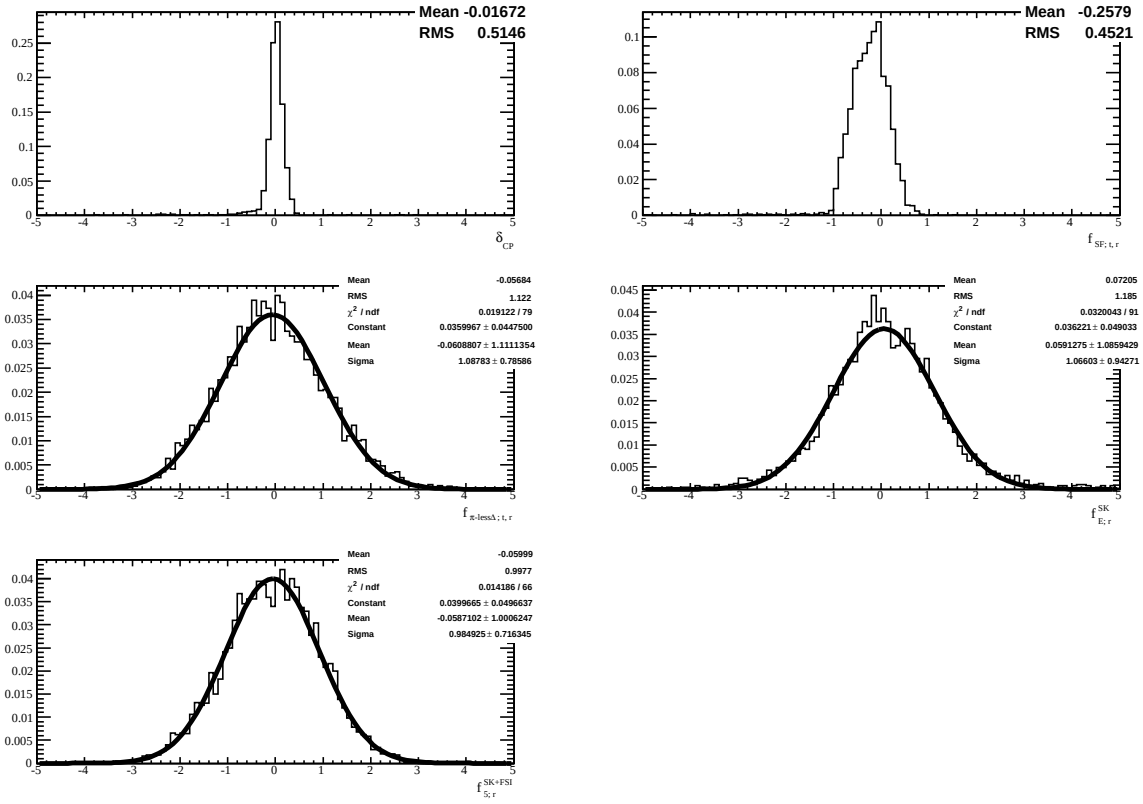


Figure 21: Pull distributions for δ_{CP}, $f_{SF;t,r}$, $f_{\pi-less\Delta;t,r}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{i;r}$, computed from fits of 5.1k toy MC experiments at $6.5700{\times}10^{20}$ POT, generated at $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

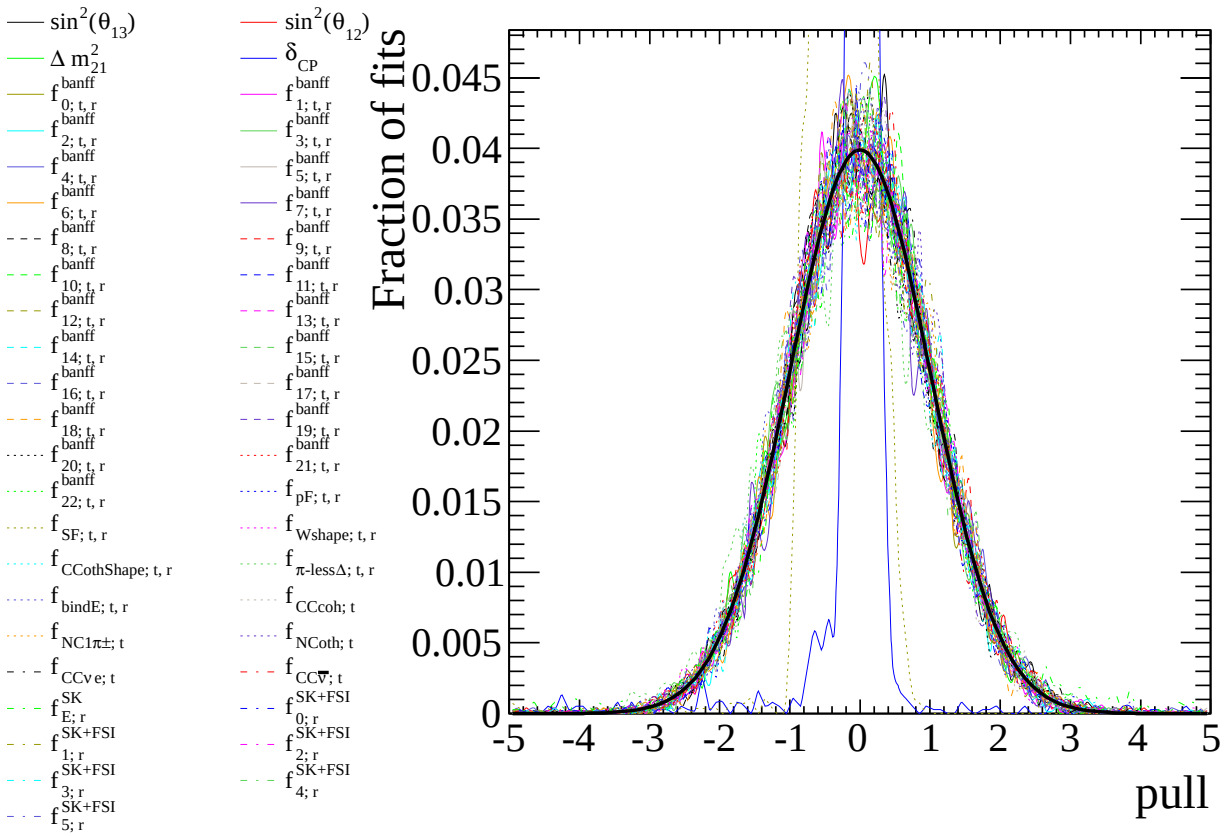


Figure 22: Systematic pull distributions for all systematic parameters for 5.1k toy MC experiments at 6.5700×10^{20} POT generated at $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. A Gaussian distribution of mean 0, width 1 is shown for reference.

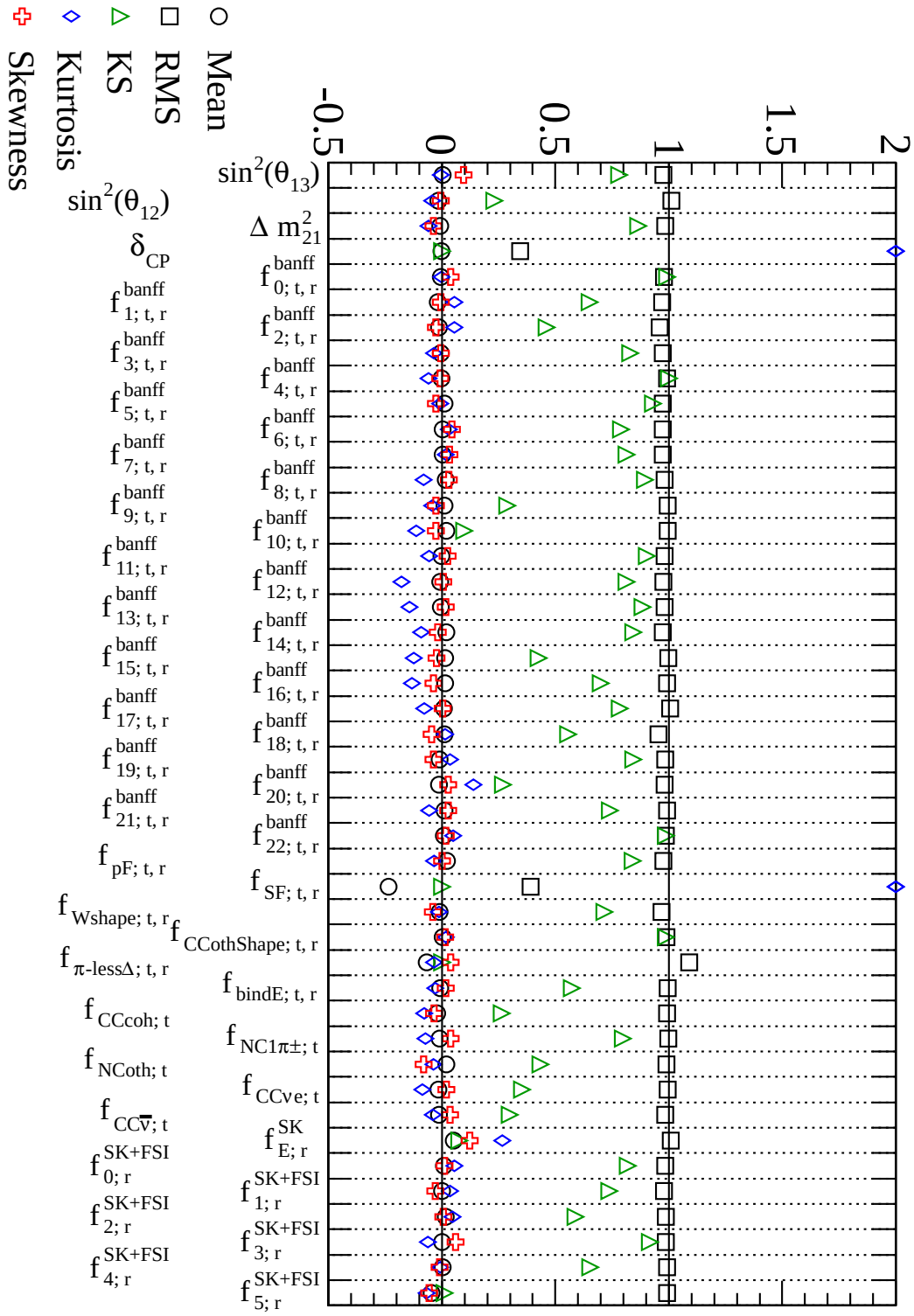


Figure 23: Summary of systematic pull distributions for all systematic parameters for 5.1k toy MC experiments at 6.5700×10^{20} POT generated at $\sin^2\theta_{23} = 0.650$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV$^2/c^4$, including statistical fluctuations and with randomized systematic parameters. The tests shown are the mean, RMS, skewness, kurtosis and the p-value from a Kolmogorov-Smirnov test comparing the pull distribution with a normal distribution of mean 0, width 1.

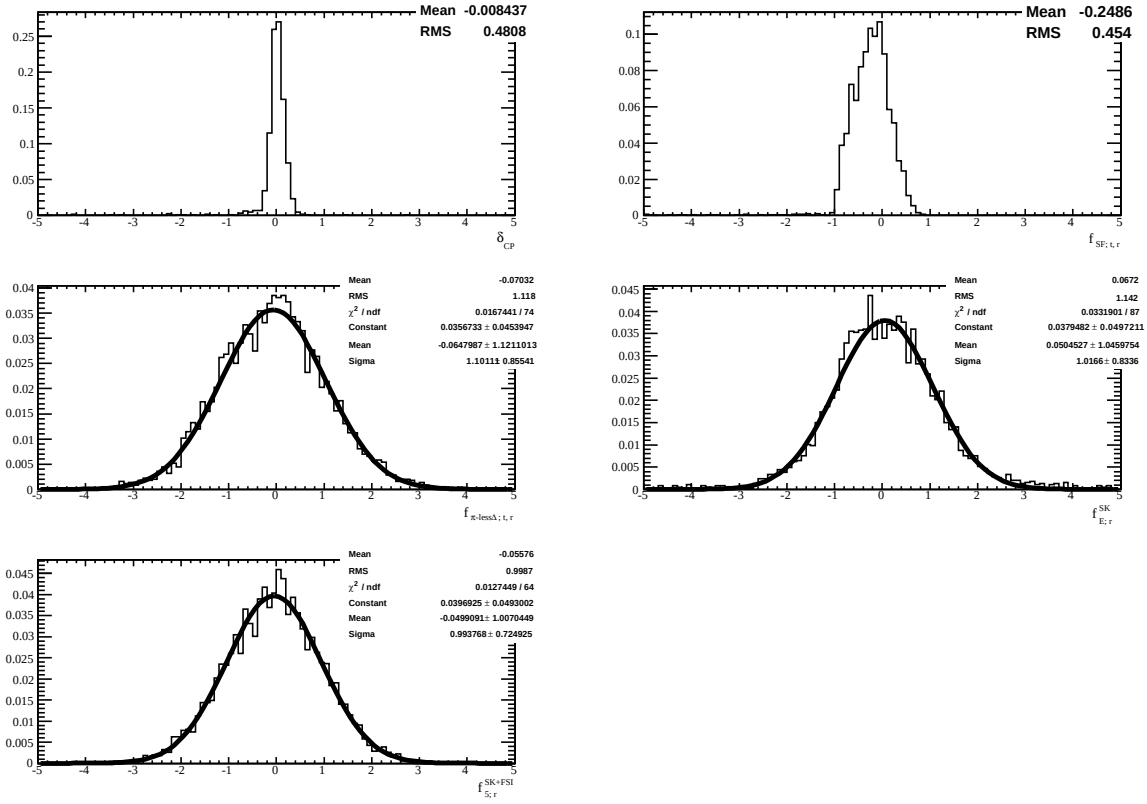


Figure 24: Pull distributions for δ_{CP}, $f_{SF;t,r}$, $f_{\pi-less\Delta;t,r}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{i;r}$, computed from fits of 5.1k toy MC experiments at 6.5700×10^{20} POT, generated at $\sin^2\theta_{23} = 0.650$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

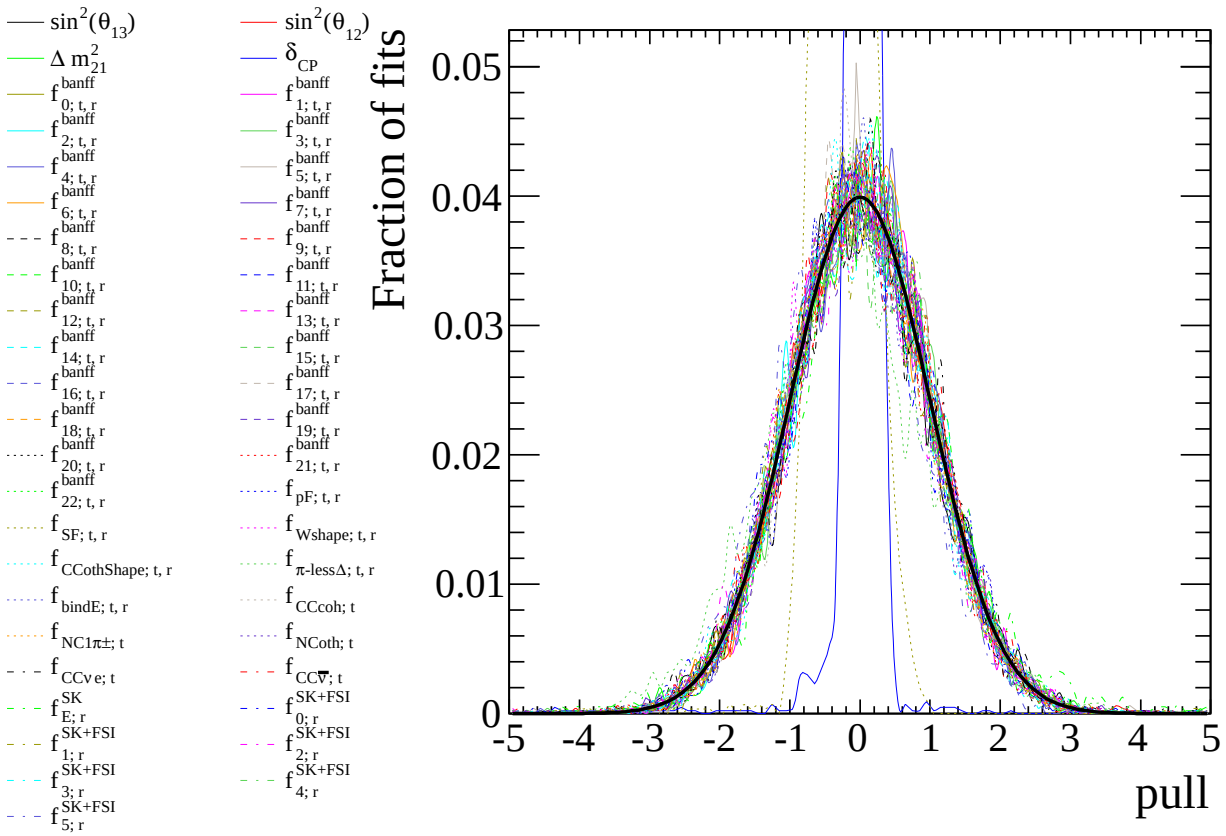


Figure 25: Systematic pull distributions for all systematic parameters for 5.1k toy MC experiments at 6.5700×10^{20} POT generated at $\sin^2\theta_{23} = 0.370$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. A Gaussian distribution of mean 0, width 1 is shown for reference.

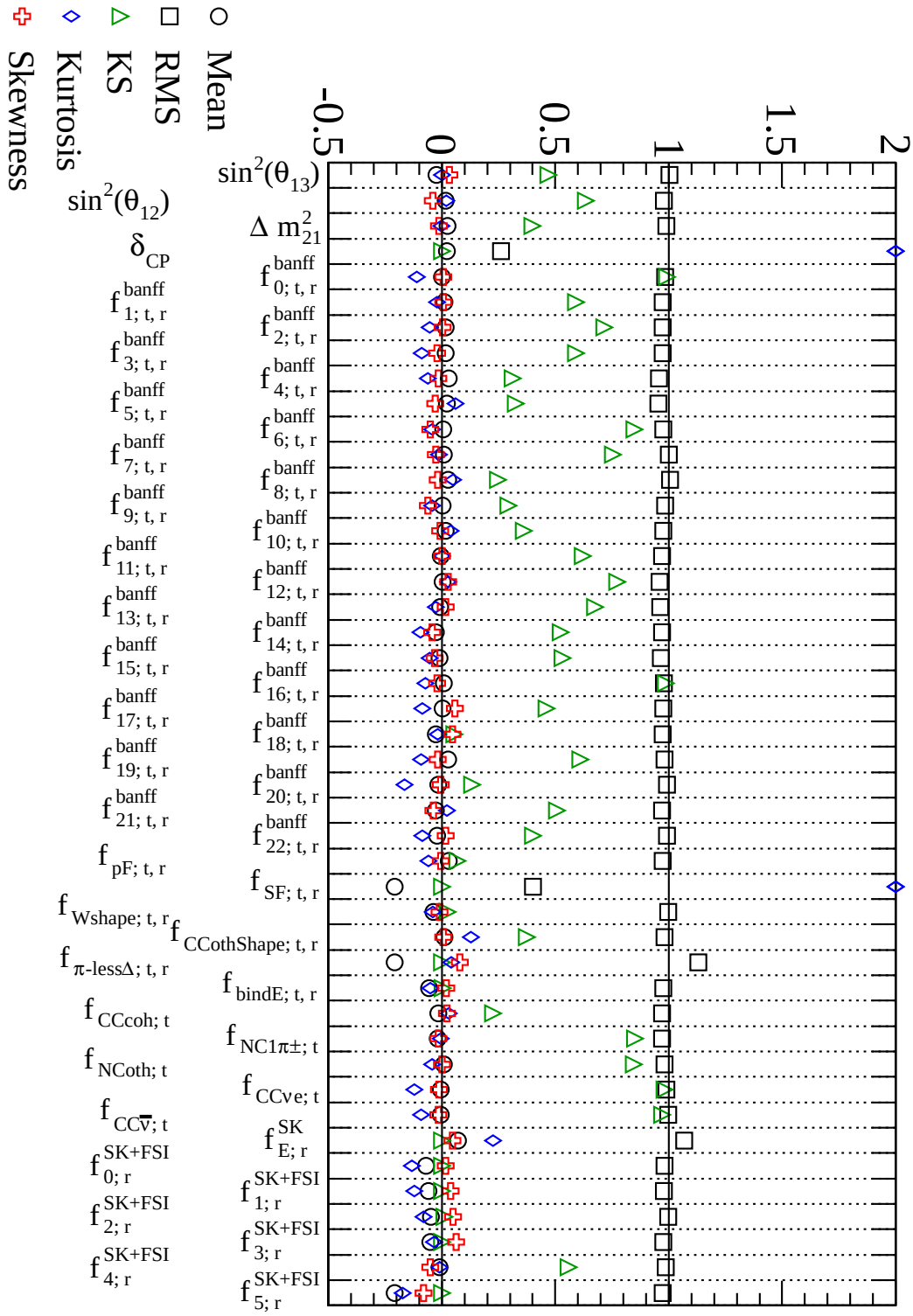


Figure 26: Summary of systematic pull distributions for all systematic parameters for 5.1k toy MC experiments at 6.5700×10^{20} POT generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4\times10^{-3}$ eV$^2/c^4$, including statistical fluctuations and with randomized systematic parameters. The tests shown are the mean, RMS, skewness, kurtosis and the p-value from a Kolmogorov-Smirnov test comparing the pull distribution with a normal distribution of mean 0, width 1.

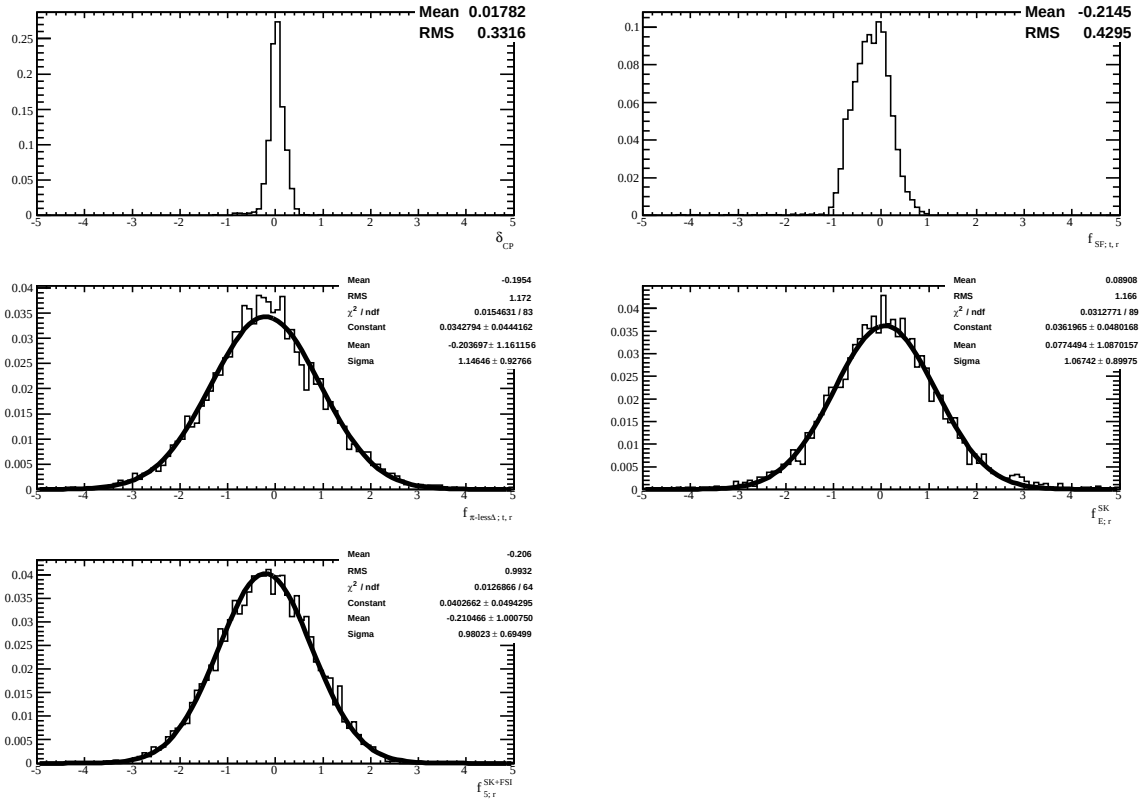


Figure 27: Pull distributions for δ_{CP}, $f_{SF;t,r}$, $f_{\pi-less\Delta;t,r}$, $f^{SK}_{E;r}$, and $f^{SK+FSI}_{i;r}$, computed from fits of 5.1k toy MC experiments at $6.5700{\times}10^{20}$ POT, generated at $\sin^2\theta_{23} = 0.513$ and $|\Delta m^2_{32}| = 2.4{\times}10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters.

7. Results of the ν_μ-disappearance analysis with the Run 1+2+3+4 dataset

7.1. Best-fit oscillation parameters and systematic parameter pulls

Two 3-flavour ν_μ-disappearance fits were performed on the combined 6.5700×10^{20} POT Run 1+2+3+4 dataset, using the method described in Sec. 5.1. In the first fit, the normal mass hierarchy was assumed, and $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ were allowed to float, while, in the second fit, the inverted mass hierarchy was assumed and $\sin^2\theta_{23}$ and $|\Delta m^2_{31}|$ were allowed to float. In both fits, all 41 systematic parameters considered in this analysis and the oscillation parameters $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, Δm^2_{21} and δ_{CP} were also allowed to float. The best-fit values were $\sin^2\theta_{23} = 0.514$ and $|\Delta m^2_{32}| = 2.51\times10^{-3}$ eV^2/c^4 assuming the normal mass hierarchy, and $|\Delta m^2_{31}| = 2.48\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.511$ assuming the inverted mass hierarchy.

The results of these fits are summarized in Tab. 12. In this table are the best-fit oscillation parameters, the observed and expected numbers of single μ-like ring events, and the values of χ^2 per degree of freedom at the best fit.

The pull for each systematic parameter f included in the fit was calculated as

$$\frac{f_{bestfit} - f_{nom}}{\sigma}, \tag{8}$$

where $f_{bestfit}$ is the best-fit value of the systematic parameter, f_{nom} is the nominal value of the parameter (corresponding to no systematic variation), and σ is either σ_{input} (the prefit error)[7] or $\sigma_{bestfit}$ (the Hesse error output from Minuit).

The pulls of all 45 nuisance parameters allowed to float in these fits are shown in Figs. 28 and 29 assuming the normal and inverted mass hierarchies respectively. From these figures, it is apparent that most systematic parameters barely move from their nominal values. We note that, although all 45 nuisance parameters are allowed to float in the fit, only 4(5) of them move by more than $\pm0.1\sigma_{bestfit}$ for the normal(inverted) hierarchy fit. When a systematic parameter that is free to move stays close to its nominal value, it means that the reduction in χ^2 from moving that parameter is less than the corresponding increase in the penalty term defined in Eq. 5 in Sec. 5.1. Only certain systematics are able to change the predicted reconstructed energy spectrum to a sufficient extent to offset the penalty term.

Analysis	Run period	$\sin^2\theta_{23}$	Δm^2	N_{obs}	N_{exp}	χ^2_{bf}/ndf	p-value
2013 3-flavour Normal	Run 1+2+3+4	0.514	$2.51\cdot10^{-3}$	120	121.46	62.34/70	0.93
2013 3-flavour Inverted	Run 1+2+3+4	0.511	$2.48\cdot10^{-3}$	120	121.42	62.34/70	0.93

Table 12: Summary of best-fit parameters from the fits of the Run 1+2+3+4 dataset. The quoted best-fit χ^2 values (χ^2_{bf}) were computed from Eq. 5 in Sec. 5.1. using the reconstructed energy binning scheme given in Sec. 3.2.5. The table also shows the p-values obtained from the goodness-of-fit test described in Sec. 5.2 for all fits.

7.2. Goodness-of-fit tests

Goodness-of-fit tests were performed as described in Sec. 5.2, and the p-values from the fits of the Run 1+2+3+4 dataset are shown in Tab. 12. The χ^2 distributions from which these p-values were calculated are shown for the Run 1+2+3+4 data in Figs. 32 and 33 for the normal and inverted mass hierarchies respectively. Both fits have a high degree of plausibility.

7.3. Best-fit single-ring μ-like reconstructed energy spectra

The reconstructed neutrino energy distribution of the single μ-like ring events in the Run 1+2+3+4 dataset is shown in Fig. 34 along with the best-fit predicted spectrum assuming the normal mass hierarchy and the predicted spectrum under the no-oscillation hypothesis. Also shown is the ratio of the data to the no-oscillation spectrum and the ratio of the best-fit spectrum to the no-oscillation spectrum; in both of these ratios, the characteristic energy-dependent deficit can be clearly seen. The corresponding spectra for the fit assuming the inverted mass hierarchy are shown in Fig. 35. All these distributions are shown in the 73-bin scheme in reconstructed neutrino energy that was used in the fitting procedure (see Sec. 3.2.5).

[7] For δ_{CP}, σ_{input} is taken to be π.

Since the reconstructed energy binning used in the fit is relatively fine, it can be beneficial to see the above distributions using a coarser binning. The best-fit spectra obtained from the fits of the Run 1+2+3+4 dataset is shown using a coarser binning scheme[8] in Figs. 36 and 37 for the normal and inverted mass hierarchies respectively. It is also interesting to see the components of the best-fit predicted spectrum, and these are shown in Figs. 38 and 39 for the normal and inverted mass hierarchies respectively. In these plots, the 32 components of the predicted spectrum are grouped into 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu}_\mu$ CC, $\nu_e/\bar{\nu}_e$ CC and NC (though each of the 32 components is calculated separately in the actual analysis).

Finally, the best-fit predicted spectra for the normal and inverted mass hierarchies are compared for the Run 1+2+3+4 dataset in Fig. 40, using the 73 reconstructed energy bins used in the fit, and in Fig. 41, using the coarser binning described above. The two best-fit spectra are nearly identical.

[8] 16 reconstructed energy bins arranged as follows: 1 0.3-GeV bin from 0 - 0.3 GeV, 8 0.1-GeV bins from 0.3 - 1.1 GeV, 1 0.15-GeV bin from 1.1 GeV - 1.25 GeV, 1 0.25-GeV bin from 1.25 - 1.5 GeV GeV, 1 0.5-GeV bin from 1.5 - 2 GeV, 2 1-GeV bin from 2-4 GeV, 1 2-GeV bin from 4-6 GeV and 1 bin above 6 GeV.

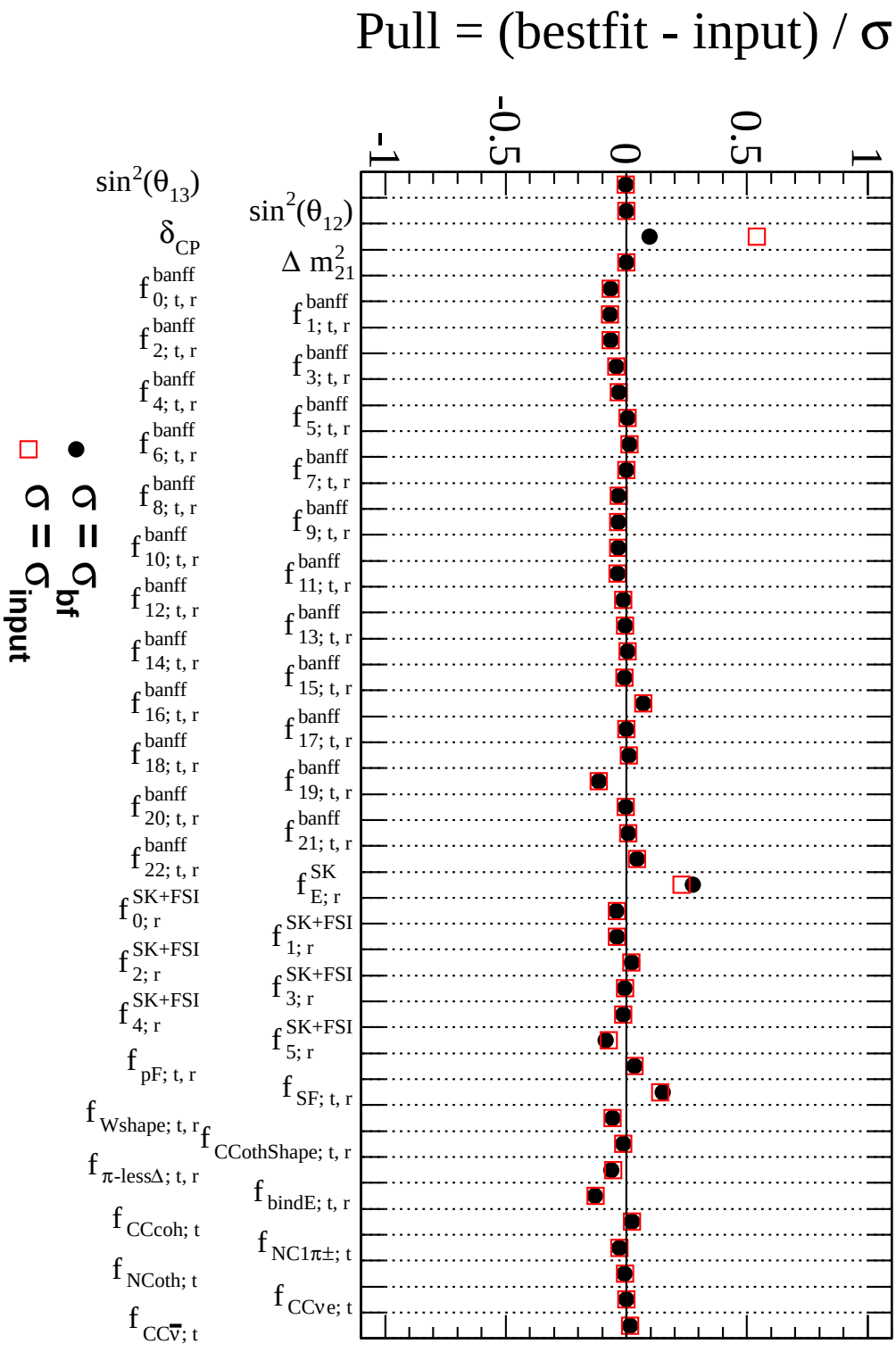


Figure 28: Systematic parameter pulls from the fit of the Run 1+2+3+4 dataset. The normal mass hierarchy is assumed.

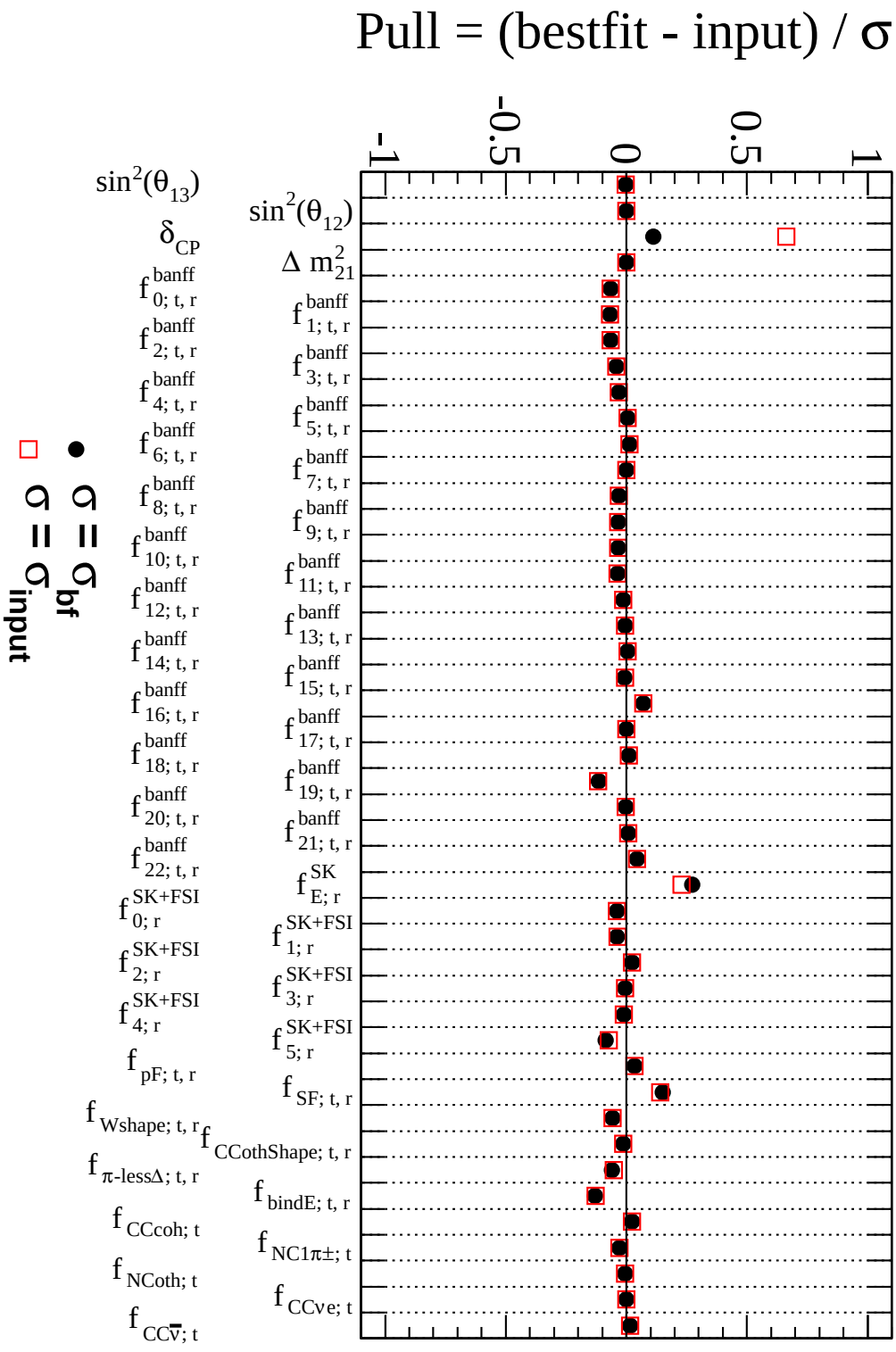


Figure 29: Systematic parameter pulls from the fit of the Run 1+2+3+4 dataset. The inverted mass hierarchy is assumed.

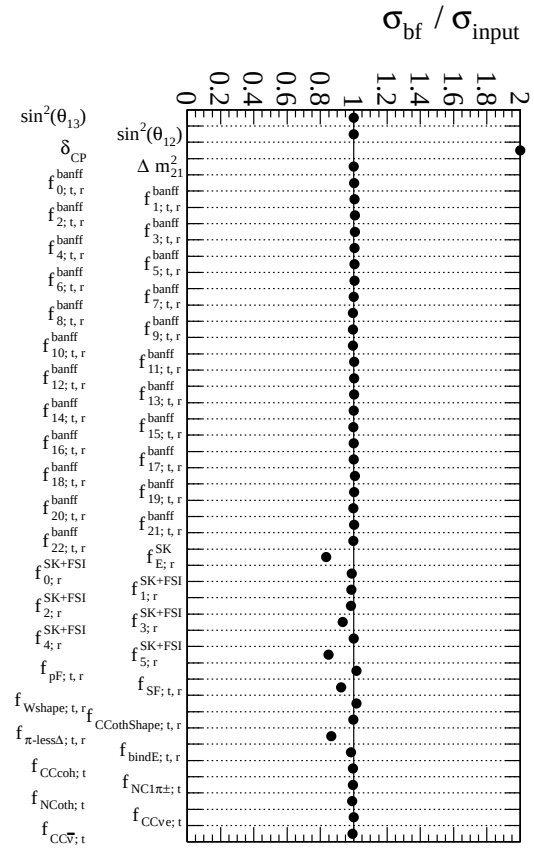


Figure 30: Ratio of $\sigma_{bestfit}$ / σ_{input} from the fit of the Run 1+2+3+4 dataset. The normal mass hierarchy is assumed.

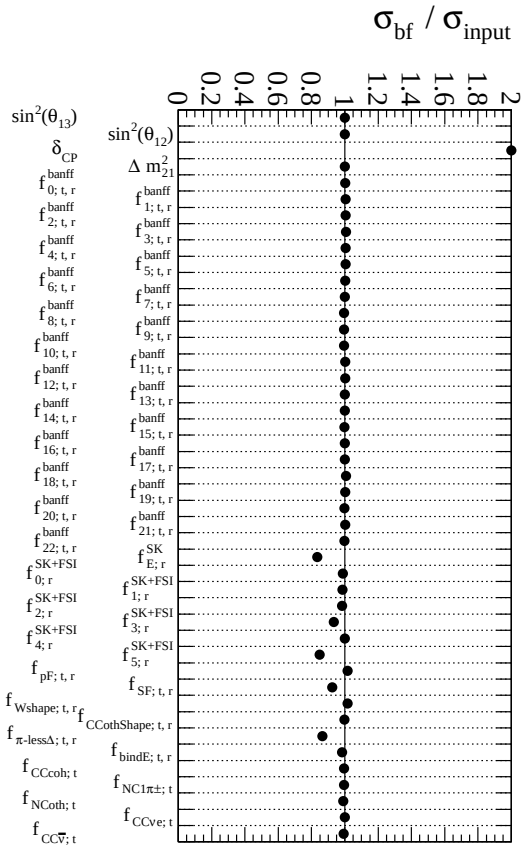


Figure 31: Ratio of $\sigma_{bestfit}$ / σ_{input} from the fit of the Run 1+2+3+4 dataset. The inverted mass hierarchy is assumed.

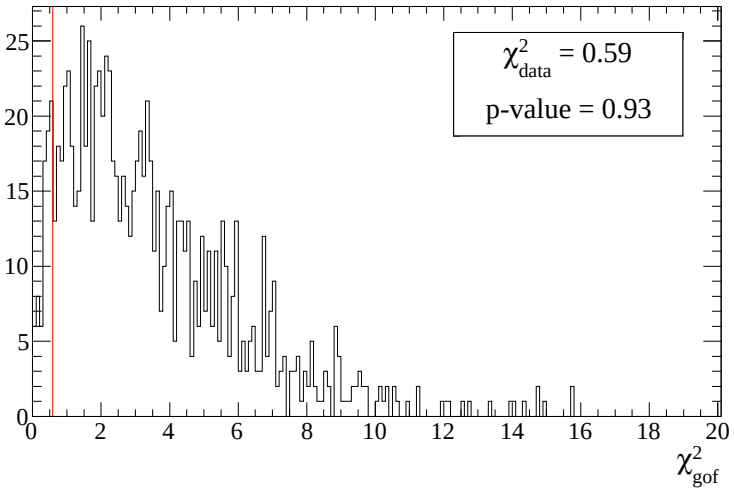


Figure 32: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments whose true input values were the Run 1+2+3+4 best-fit oscillation parameters. The χ^2_{gof} value from the fit of the Run 1+2+3+4 data (χ^2_{data}) is highlighted. The normal mass hierarchy is assumed.

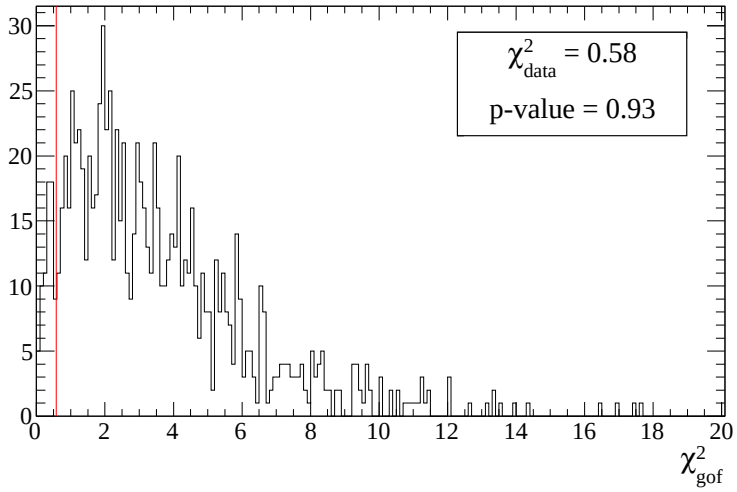


Figure 33: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments whose true input values were the Run 1+2+3+4 best-fit oscillation parameters. The χ^2_{gof} value for the fit of the Run 1+2+3+4 data (χ^2_{data}) is highlighted. The inverted mass hierarchy is assumed.

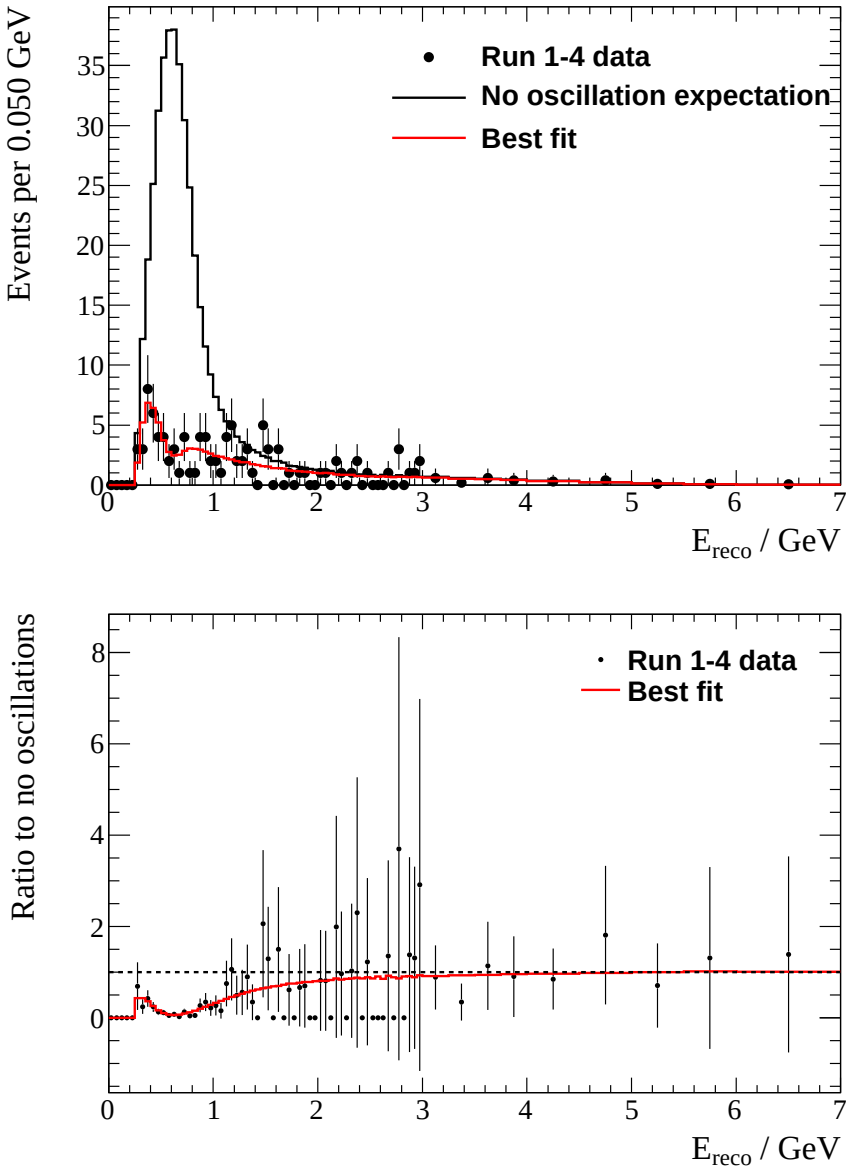


Figure 34: Best-fit reconstructed energy spectrum (top) obtained from the fit of the combined Run 1+2+3+4 dataset. The Run 1+2+3+4 dataset and the prediction obtained using the no-oscillation hypothesis are also shown. The spectra are shown using the 73 bins used in the fit (see Sec. 3.2.5). The ratios of the data and best-fit predicted spectrum to the no-oscillation hypothesis are shown in the lower plot. The characteristic energy-dependent deficit can be clearly seen. The normal mass hierarchy is assumed.

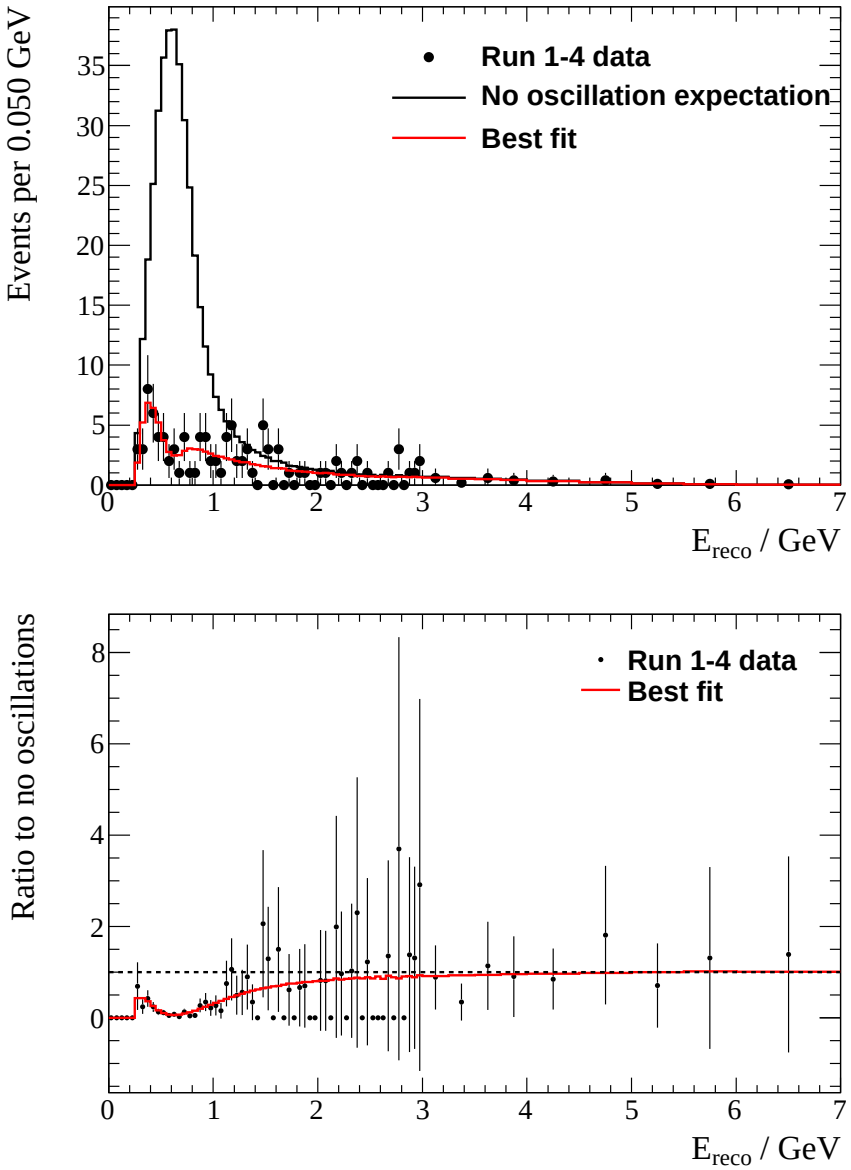


Figure 35: Best-fit reconstructed energy spectrum (top) obtained from the fit of the combined Run 1+2+3+4 dataset. The Run 1+2+3+4 dataset and the prediction obtained using the no-oscillation hypothesis are also shown. The spectra are shown using the 73 bins used in the fit (see Sec. 3.2.5). The ratios of the data and best-fit predicted spectrum to the no-oscillation hypothesis are shown in the lower plot. The characteristic energy-dependent deficit can be clearly seen. The inverted mass hierarchy is assumed.

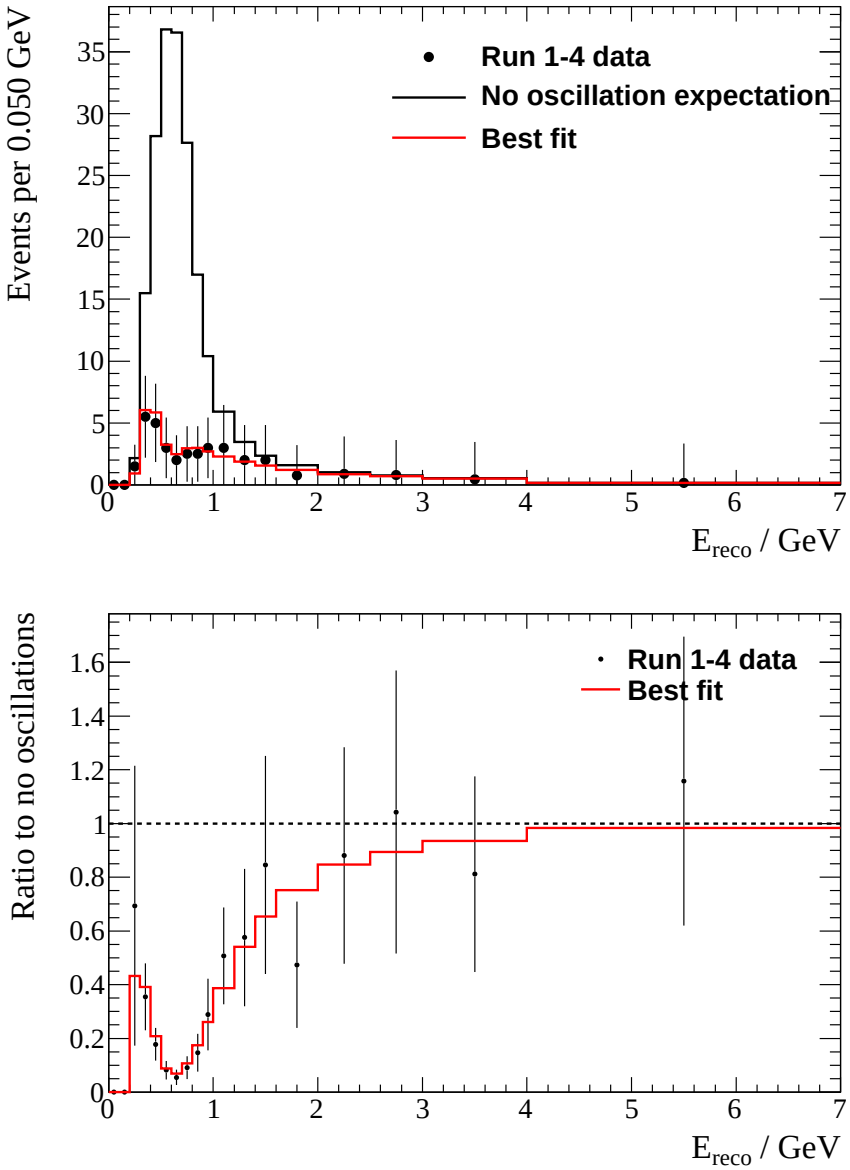


Figure 36: Best-fit reconstructed energy spectrum (top) obtained from the fit of the combined Run 1+2+3+4 dataset. The Run 1+2+3+4 dataset and the prediction obtained using the no-oscillation hypothesis are also shown. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). The characteristic energy-dependent deficit can be clearly seen. The normal mass hierarchy is assumed.

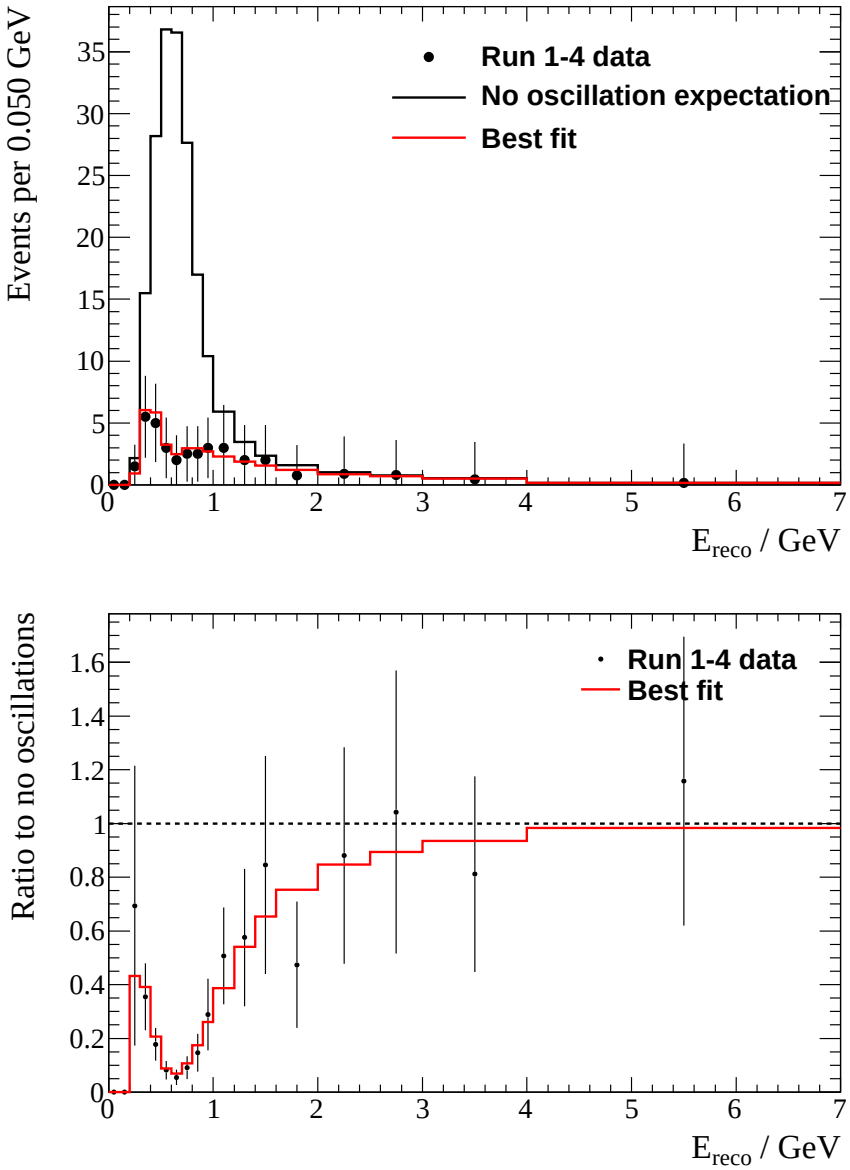


Figure 37: Best-fit reconstructed energy spectrum (top) obtained from the fit of the combined Run 1+2+3+4 dataset. The Run 1+2+3+4 dataset and the prediction obtained using the no-oscillation hypothesis are also shown. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). The characteristic energy-dependent deficit can be clearly seen. The inverted mass hierarchy is assumed.

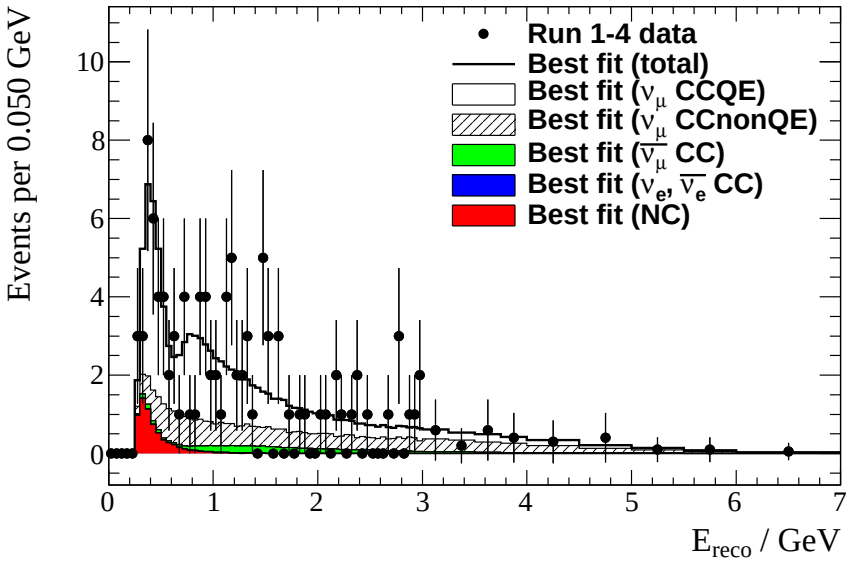


Figure 38: Best-fit reconstructed energy spectrum in the 73-bin scheme used in the fit (see Sec.3.2.5) and its components obtained from the fit of the Run 1+2+3+4 dataset. The 32 components, each of which is calculated separately in the analysis, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The normal mass hierarchy is assumed.

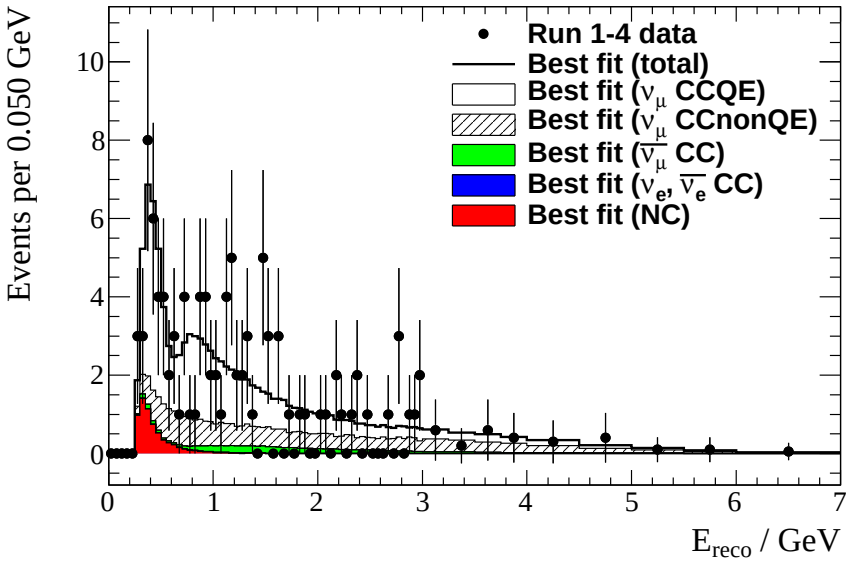


Figure 39: Best-fit reconstructed energy spectrum in the 73-bin scheme used in the fit (see Sec. 3.2.5)) and its components obtained from the fit of the Run 1+2+3+4 dataset. The 32 components, each of which is calculated separately in the analysis, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnQE, $\bar{\nu_\mu}$ CC, $\nu_e/\bar{\nu_e}$ CC and NC. The inverted mass hierarchy is assumed.

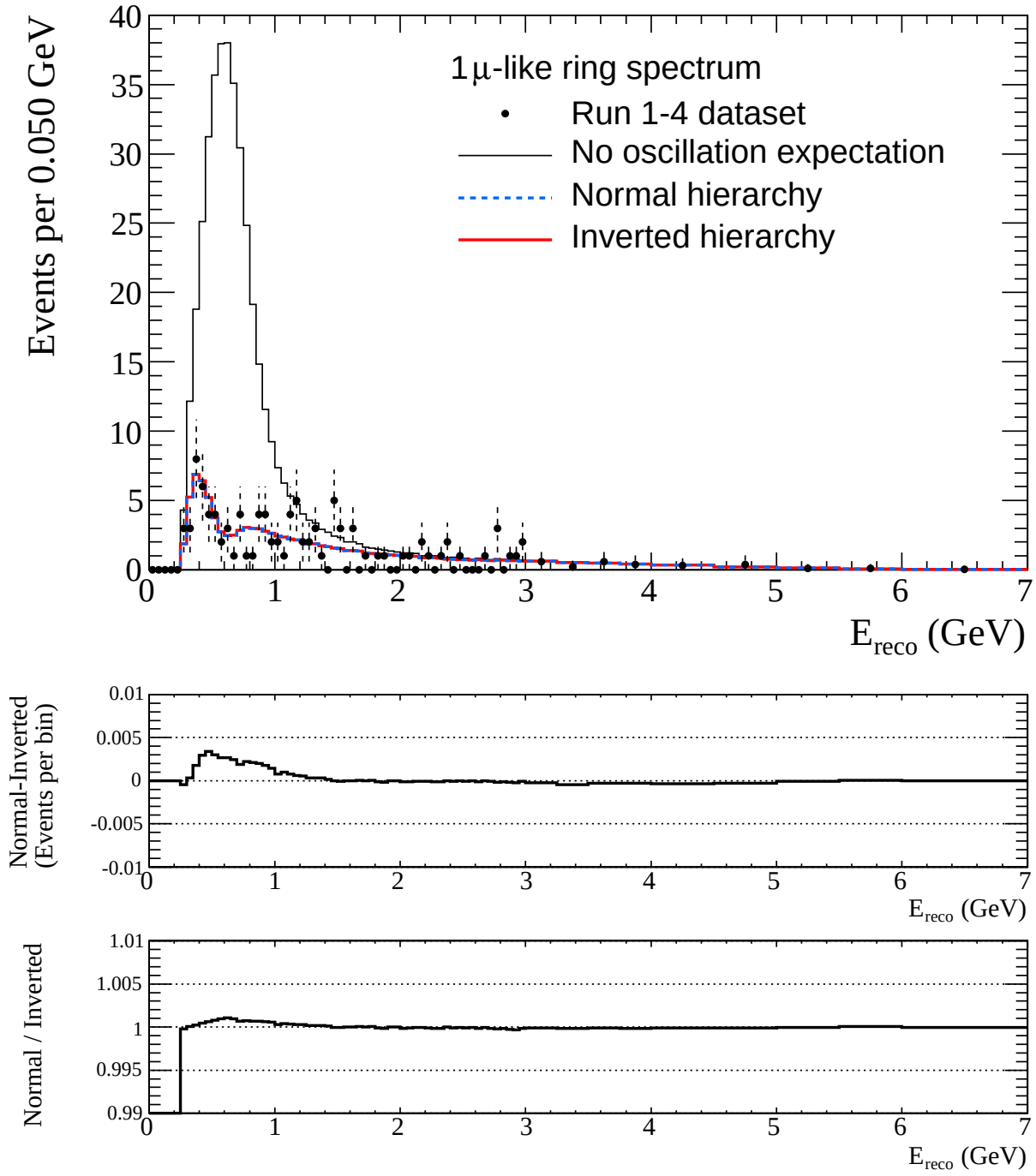


Figure 40: Comparison between best-fit reconstructed energy spectra obtained from the normal and inverted mass hierarchy fits of the Run 1+2+3+4 datasets. The spectra are shown using the 73 bins used in the fit (see Sec. 3.2.5). Also shown are the absolute difference between the normal and inverted mass hierarchy best-fit predicted spectra and the ratio between them.

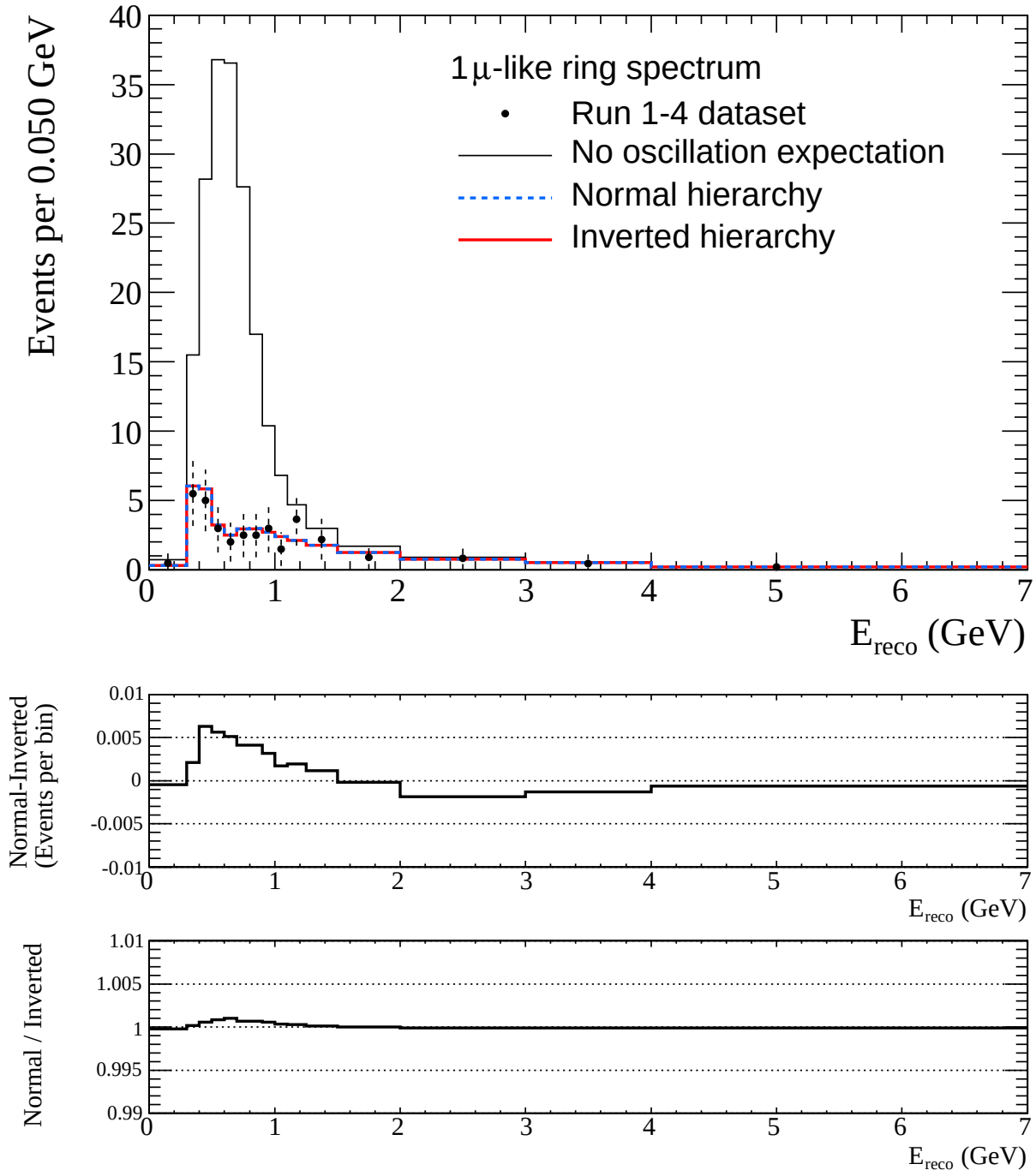


Figure 41: Comparison between best-fit reconstructed energy spectra obtained from the normal and inverted mass hierarchy fits of the Run 1+2+3+4 datasets. The spectra are shown using the coarse binning scheme discussed in the text (see Sec. 7.3). Also shown are the absolute difference between the normal and inverted mass hierarchy best-fit spectra and the ratio between them.

7.4. Confidence regions

The first step in constructing confidence contours in the 2-dimensional $\sin^2\theta_{23}$ - $|\Delta m^2_{32}|$ or $\sin^2\theta_{23}$ - $|\Delta m^2_{31}|$ oscillation parameter space is to construct the $\Delta\chi^2$ surface in that space. The oscillation parameter space is divided into a rectangular grid of width $\sin^2\theta_{23} = 0.0025$, $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}| = 0.025\times10^{-3}$ eV^2/c^4, with a grid point at the centre of each rectangle. For each mass hierarchy, two separate fits are made of the data at each grid point: in one fit, the values of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ are fixed to the true values of the grid point and the 41 systematic and four non-23 oscillation parameters are allowed to float, while, in the other fit, $\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$, and the systematic and non-23 oscillation parameters are allowed to float. $\Delta\chi^2$ is then calculated as the difference between the minimum values of χ^2 from the two fits (χ^2 is defined in Eq. 5 in Sec. 5.1).

The 2-dimensional $\Delta\chi^2$ surfaces from the fits of the Run 1+2+3+4 dataset is shown in Figs. 42 and 43 as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$, for the normal and inverted mass hierarchies respectively. The same $\Delta\chi^2$ range is used in both colour plots to allow comparison. Profile $\Delta\chi^2$ distributions are also shown as a function of $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ or $|\Delta m^2_{31}|$ from the fits of the Run 1+2+3+4 dataset in Figs. 44 and 45 respectively. Using these plots and the constant-$\Delta\chi^2$ method described in Sec. 5.3, 1-dimensional limits on the parameters can be obtained[9]. These limits are presented in Tab. 13. Confidence intervals were also constructed for each of $\sin^2\theta_{23}$ and Δm^2_{32} separately using the new procedure described in Sec. 5.3. These 1D intervals are being prepared, and will be shown in a later version of this note.

2-dimensional confidence regions were constructed using the constant-$\Delta\chi^2$ method described in Sec. 5.3 from the fits of the Run 1+2+3+4 dataset. The 68% and 90% CL regions are shown in Figs. 46 and 47 for the normal and inverted mass hierarchies respectively. All confidence regions presented in this note include the effect of systematic uncertainties.

To construct 2-dimensional confidence regions using the method of Feldman and Cousins [32], the $\Delta\chi^2_{critical}$ surfaces must be calculated in the $\sin^2\theta_{23}$ - $|\Delta m^2_{32}|$ or $\sin^2\theta_{23}$ - $|\Delta m^2_{31}|$ oscillation parameter space as described in Sec. 5.3. In Fig. **??**, we show a Monte Carlo estimation using the Feldman-Cousins method of the 2-dimensional $\Delta\chi^2_{critical}$ values used for calculation of the 68% and 90% CL confidence regions, assuming the normal mass hierarchy. The surfaces were generated using an ensemble of toy MC experiments which included both randomized systematic variations and statistical fluctuations. Slices of the 2-dimensional distributions shown in Fig. **??**, are shown in Figs. **??** and **??**.

Using the $\Delta\chi^2$ and $\Delta\chi^2_{critical}$ surfaces presented here, the 2-dimensional Feldman-Cousins 68% and 90% CL regions were constructed and they are shown in Fig. **??**.

In the next section, comparisons are presented of the confidence regions obtained using different construction methods. We also compare our contours from the fits of the combined Run 1+2+3+4 dataset with recent measurements from MINOS and SuperK.

[9] For 1 fit parameter, the critical values of $\Delta\chi^2$ used are 1.00 (68% CL), 2.71 (90% CL), 3.84 (95% CL), 6.63 (99% CL). For 2 parameters fitted simultaneously, the critical values of $\Delta\chi^2$ used are 2.30 (68% CL), 4.61 (90% CL), 5.99 (95% CL), 9.21 (99% CL).

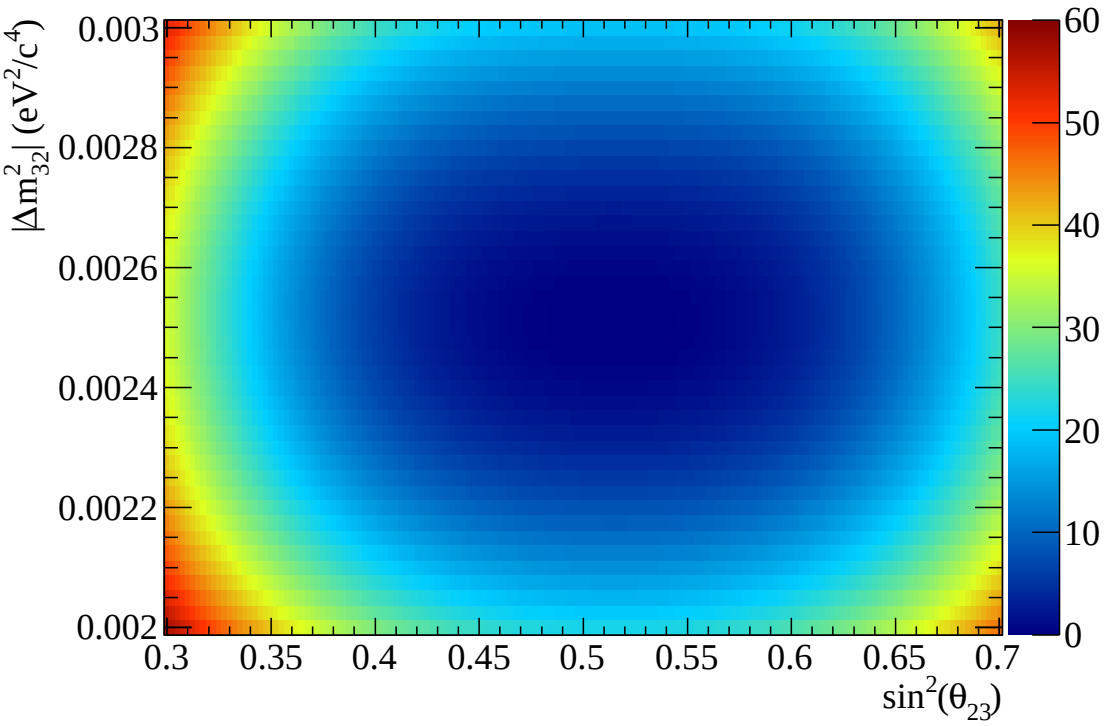


Figure 42: $\Delta\chi^2$ surface from the fit of the Run 1+2+3+4 dataset. The normal mass hierarchy is assumed.

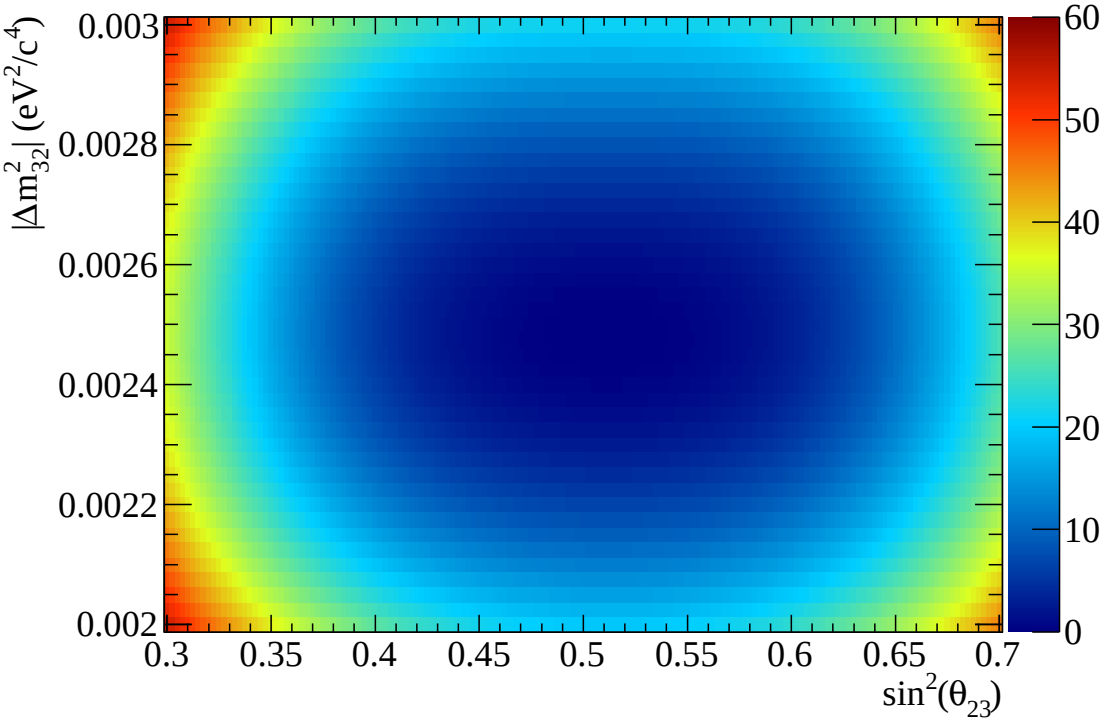


Figure 43: $\Delta\chi^2$ surface from the fit of the Run 1+2+3+4 dataset. The inverted mass hierarchy is assumed.

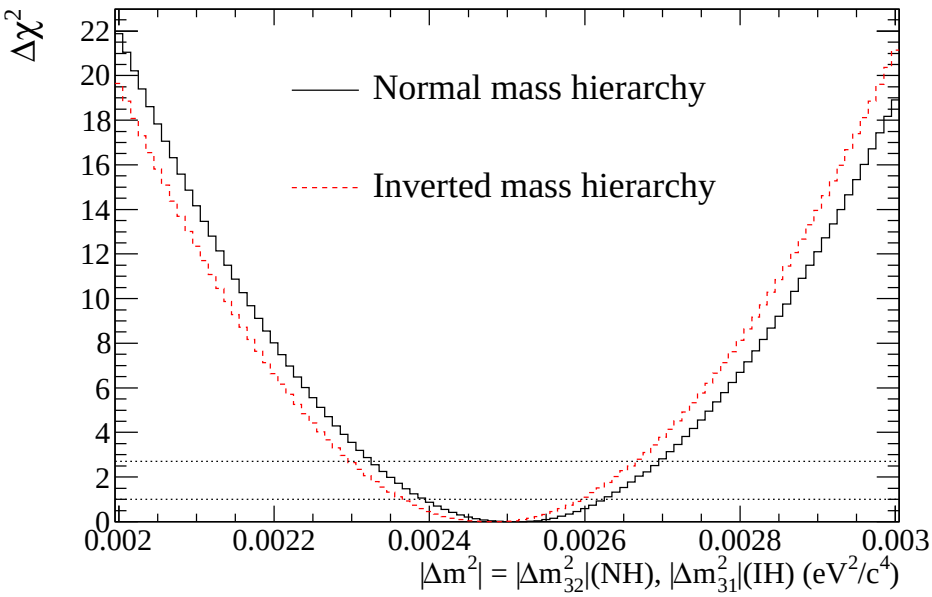


Figure 44: Profile $\Delta\chi^2$, as a function of $|\Delta m^2_{32}|$ for the normal mass hierarchy and of $|\Delta m^2_{31}|$ for the inverted mass hierarchy, from the fits of the Run 1+2+3+4 dataset. The 68% and 90% constant-$\Delta\chi^2$ values are shown as horizontal dotted lines. The 68% and 90% Feldman & Cousins values are shown as dashed curves.

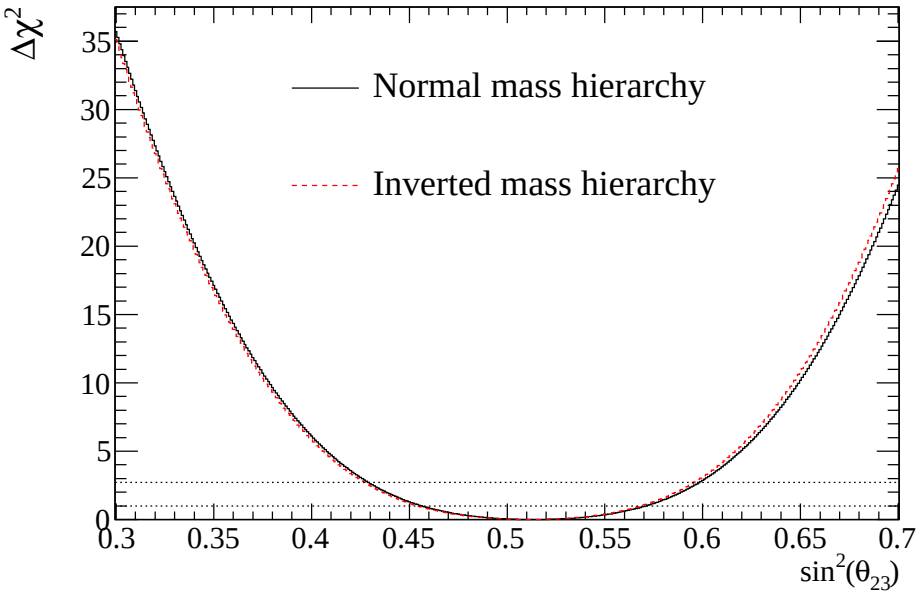


Figure 45: Profile $\Delta\chi^2$, as a function of $\sin^2\theta_{23}$, from the fits of the Run 1+2+3+4 dataset. The 68% and 90% constant-$\Delta\chi^2$ values are shown as horizontal dotted lines. The 68% and 90% Feldman & Cousins values are shown as dashed curves.

Method and mass hierarchy	Run period	Confidence limit	$\sin^2\theta_{23}$	$\lvert\Delta m^2_{32}\rvert$ or $\lvert\Delta m^2_{31}\rvert$ (10^{-3} eV^2/c^4)
Constant-$\Delta\chi^2$ (normal hierarchy)	Run 1+2+3+4	68%	0.514 ± 0.058	2.51 ± 0.12
		90%	> 0.429 < 0.599	2.32 2.70
Constant-$\Delta\chi^2$ (inverted hierarchy)	Run 1+2+3+4	68%	$0.511^{+0.058}_{-0.057}$	2.48 ± 0.12
		90%	> 0.427 < 0.596	2.29 2.67
Feldman & Cousins (normal hierarchy)	Run 1+2+3+4	68%	$0.514 \pm$???	$2.51 \pm$???
		90%	$>$??? $<$???	??? ???
Feldman & Cousins (inverted hierarchy)	Run 1+2+3+4	68%	$0.511 \pm$???	$2.48 \pm$???
		90%	$>$??? $<$???	??? ???

Table 13: Summary of one-dimensional limits on $\sin^2\theta_{23}$ and $\lvert\Delta m^2_{32}\rvert$ or $\lvert\Delta m^2_{31}\rvert$ from the fits of the Run 1+2+3+4 dataset. The limits are shown for both the constant-$\Delta\chi^2$ method and the new procedure.

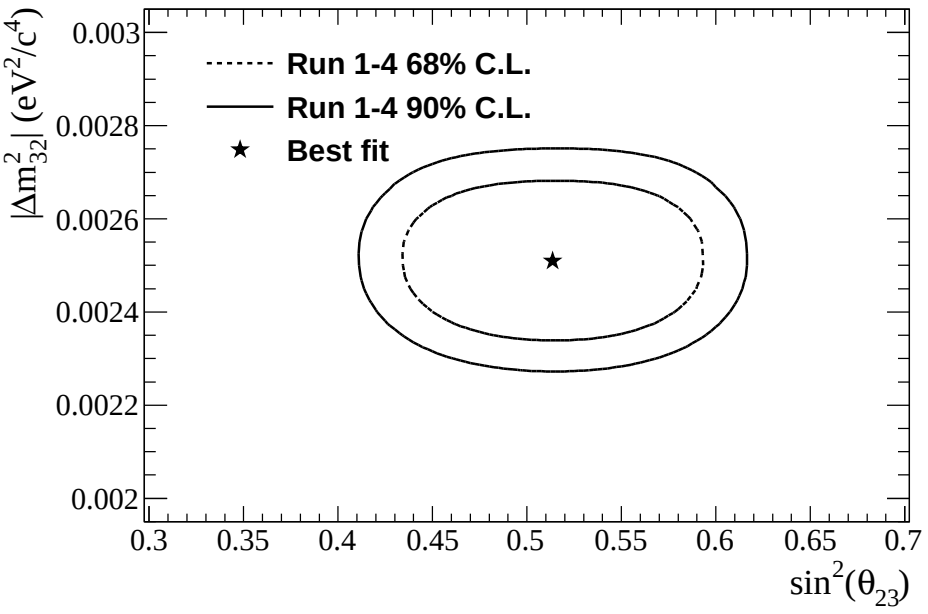


Figure 46: 68% CL and 90% CL allowed region (including systematics), constructed with the constant-$\Delta\chi^2$ method, from the fit of the Run 1+2+3+4 dataset. The best-fit point is also shown. The normal mass hierarchy is assumed.

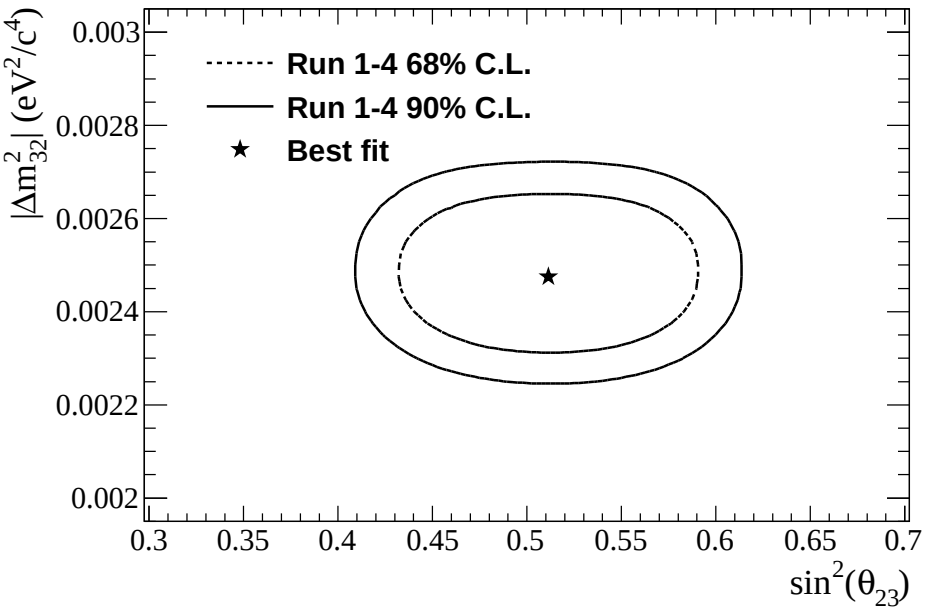


Figure 47: 68% CL and 90% CL allowed region (including systematics), constructed with the constant-$\Delta\chi^2$ method, from the fit of the Run 1+2+3+4 dataset. The best-fit point is also shown. The inverted mass hierarchy is assumed.

7.5. Comparisons of confidence regions

A comparison is shown in Fig. 48 of the Run 1+2+3+4 2-dimensional confidence regions obtained using the constant-$\Delta\chi^2$ method in $\sin^2\theta_{23}$ - Δm^2_{32} for the normal mass hierarchy and $\sin^2\theta_{23}$ - Δm^2_{31} for the inverted mass hierarchy. This figure shows that there is a significant difference between the two hierarchy assumptions.

The Run 1+2+3+4 2-dimensional confidence regions obtained using the constant-$\Delta\chi^2$ and Feldman-Cousins methods are compared in Fig. **??** for the normal mass hierarchy assumption.

In Fig. 49, the T2K Run 1+2+3+4 90% confidence region is compared with those from MINOS (2013) [39], SuperK atmospherics (2012 3-flavour zenith-angle analysis) [40], and T2K Run 1+2+3 [9]. A second comparison with external results is shown in Fig. 50: this figure compares the T2K Run 1+2+3+4 90% confidence region with those from MINOS (2013) [38], SuperK atmospherics (2012 2-flavour L/E and 3-flavour zenith-angle analyses) [40], and T2K Run 1+2+3 [9]. In this plot, the T2K Run 1+2+3+4 fit result has been converted from $\sin^2\theta_{23}$ to $\sin^2(2\theta_{23})$ by profiling the octant. This is the same method as that used by SuperK in their zenith angle contour, and, if this method had been applied in the T2K Run 1+2+3 result, it should have closely matched the T2K Run 1+2+3 $\theta_{23} \geq \pi/4$ contour. T2K now has the world's best limit on θ_{23}.

In Fig. **??** a comparison is shown between the Run 1+2+3+4 allowed region obtained with constant-$\Delta\chi^2$ method and the T2K 6.5700×10^{20} POT sensitivity contour produced at our best-fit point for the Run 1+2+3+4 dataset. The 'stat-only' sensitivity contours were computed by averaging the contours of 300 toy-MC experiments generated with statistical fluctuations and all systematic parameters fixed at their nominal values. The 'stat+syst' sensitivity contours were computed by averaging the contours of 300 toy-MC experiments generated with statistical fluctuations and all systematic parameters randomized.

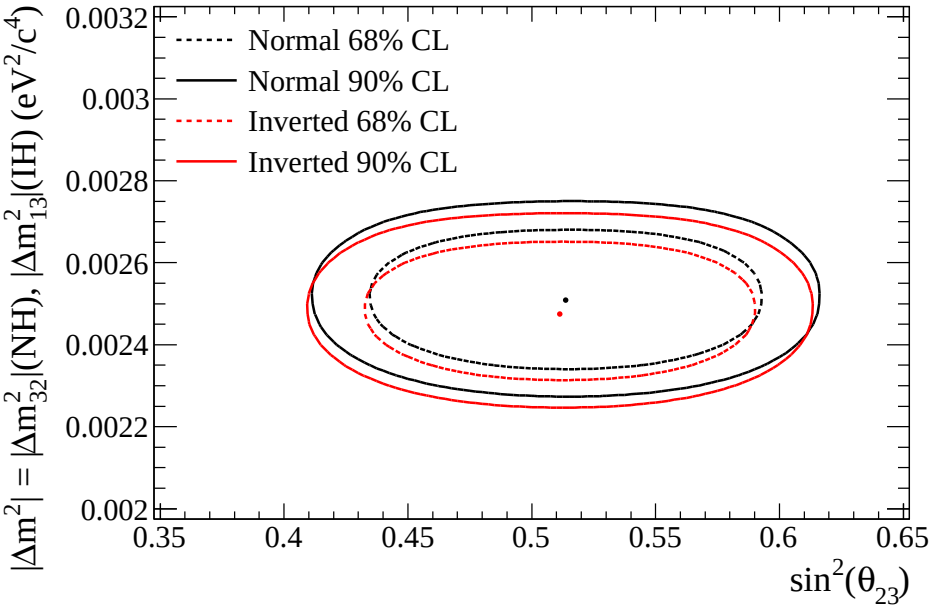


Figure 48: Comparison between the 68% CL and 90% CL allowed regions obtained using the constant-$\Delta\chi^2$ method from the fit of the Run 1+2+3+4 dataset for the normal and inverted hierarchy assumptions.

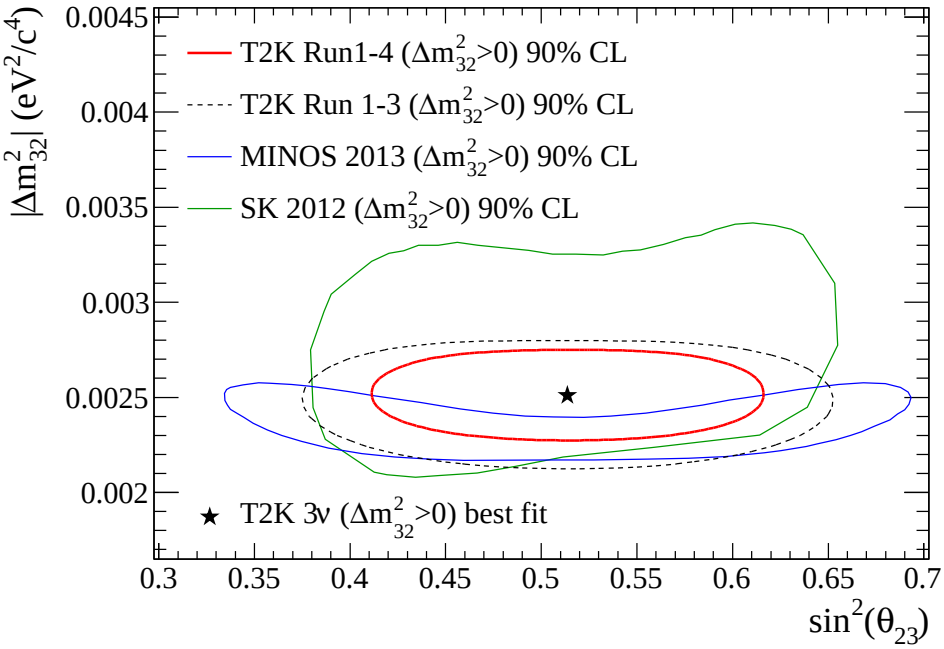


Figure 49: Comparison between 90% CL allowed regions from MINOS [39], SuperK atmospherics [40], T2K Run 1+2+3 [9], and the 90% CL region obtained from the VaLOR fit of the Run 1+2+3+4 dataset. Contours obtained using both the constant-$\Delta\chi^2$ and Feldman-Cousins methods are shown. The normal mass hierarchy is assumed for all results. An FC contour will be prepared and added to this plot.

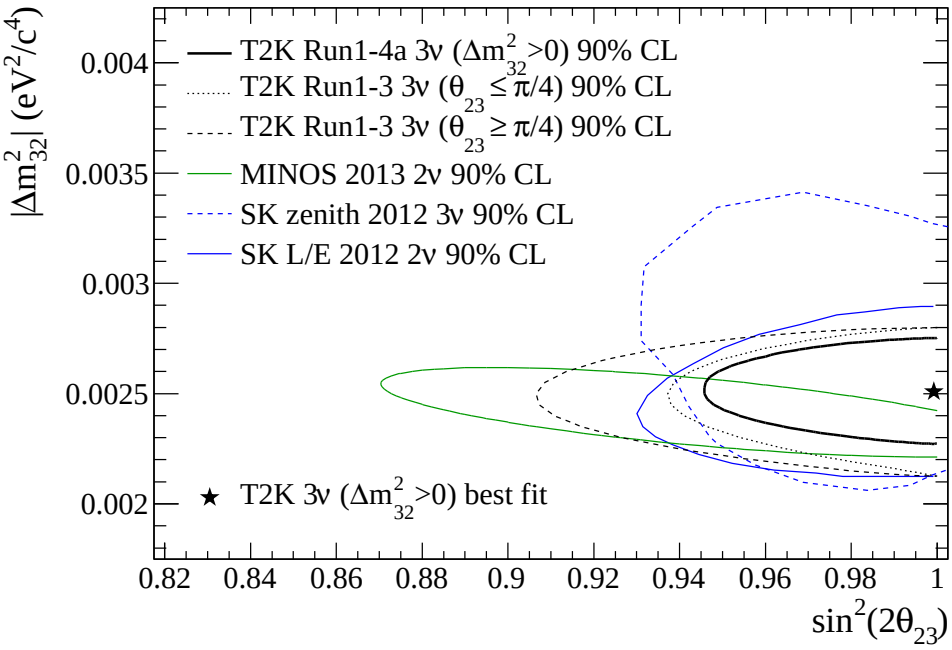


Figure 50: Comparison between 90% CL allowed regions from MINOS [38], SuperK atmospherics [40], T2K Run 1+2+3 [9], and the 90% CL region obtained from the VaLOR fit of the Run 1+2+3+4 dataset. Contours obtained using both the constant-$\Delta\chi^2$ and Feldman-Cousins methods are shown. The normal mass hierarchy is assumed for all results. An FC contour will be prepared and added to this plot.

8. Summary

In this technical note, we have presented the results of a 3-flavour ν_μ-disappearance analysis, performed by the Valencia-Lancaster-Oxford-Rutherford (VaLOR) group, on the combined Run1+2+3+4 (6.5700×10^{20} POT) dataset. This analysis is an update of the work presented in Refs. [1], [2], [3], [4] and [5] with improved methods and an enlarged dataset.

This analysis predicts 445.98± 23.46(syst) 1 μ-like ring events in SuperK in the absence of any oscillation, but only 120 were observed. The observed deficit has a strong energy dependence; the ratio of observed to expected, under the no-oscillation hypothesis, is ∼26% $<$ 0.5 GeV, ∼10% between 0.5 and 1 GeV and ∼72% $>$ 1 GeV.

A ν_μ-disappearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. In this analysis, 41 neutrino flux, neutrino cross section, efficiency, energy scale, initial-state nuclear environment and final-state rescattering, and oscillation parameter systematics were taken into account. The oscillation parameters $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, Δm^2_{21} and δ_{CP} were also treated as nuisance parameters. The observed reconstructed energy spectrum of 1 μ-like ring events was fitted, and separate fits were made for the normal and the inverted mass hierarchies. In these fits, $\sin^2\theta_{23}$ and either $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy) were allowed to float. The oscillation parameters $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, Δm^2_{21} and δ_{CP} and all 41 systematic parameters considered in this analysis were also allowed to float in the fits.

The 3-flavour ν_μ-disappearance fit of the combined Run 1+2+3+4 dataset, assuming the normal mass hierarchy, gives $|\Delta m^2_{32}| = 2.51\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.514$ as the best-fit values (p-value = 0.93). Assuming the inverted mass hierarchy, the fit yields $|\Delta m^2_{31}| = 2.48\times10^{-3}$ eV^2/c^4 and $\sin^2\theta_{23} = 0.511$ as the best-fit values (p-value = 0.93).

Confidence regions in the parameter spaces $\sin^2\theta_{23}$ - $|\Delta m^2_{32}|$ (normal hierarchy) and $\sin^2\theta_{23}$ - $|\Delta m^2_{31}|$ (inverted hierarchy) have been shown using both the constant-$\Delta\chi^2$ method and the method of Feldman and Cousins. Using the profile likelihoods obtained from the fit for $\sin^2\theta_{23}$ and either $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy) and the constant-$\Delta\chi^2$ method for one parameter, the 68% (90%) C.L. allowed regions for these parameters can be summarized as $0.456\ (0.429) < \sin^2\theta_{23} < 0.572\ (0.599)$ and 2.39×10^{-3} (2.32×10^{-3}) eV^2/c^4 $< |\Delta m^2_{32}| < 2.63\times10^{-3}$ (2.70×10^{-3}) eV^2/c^4 for the normal mass hierarchy, and $0.454\ (0.427) < \sin^2\theta_{23} < 0.569\ (0.596)$ and 2.36×10^{-3} (2.29×10^{-3}) eV^2/c^4 $< |\Delta m^2_{31}| < 2.60\times10^{-3}$ (2.67×10^{-3}) eV^2/c^4 for the inverted mass hierarchy.

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